A HANDBOOK FOR QUANT FINANCE

STOCHASTIC CALCULUS AND DERIVATIVE PRICING.

First Edition

PREFACE

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CHAPITRE 1

PROBABILITY AND MEASURE THEORY

Probability theory provides a mathematical framework for studying random phenomena. Many processes in nature, science, engineering, and finance involve outcomes that cannot be predicted with certainty. Examples include the outcome of a coin toss, the arrival time of customers in a queue, or the evolution of asset prices in financial markets. Probability theory allows us to model such situations and to reason quantitatively about uncertainty.

The modern formulation of probability theory is based on **measure theory**. In measure theory, the goal is to assign a non-negative number to certain subsets of a given set. This number is called a **measure** and represents a generalized notion of size, extending familiar ideas such as length, area, or volume. A fundamental property of a measure is **countable additivity**: if a collection of sets is disjoint, the measure of their union is the sum of their measures.

For mathematical reasons, it is not possible in general to assign a measure to every subset of a set. Instead, one restricts attention to a special collection of subsets called a σ-field . A set equipped with a σ-field is called a **measurable space**, and measures are defined only on these measurable sets.

Probability theory can be viewed as a special case of measure theory, where we are assigning a measure that represents our belief of occurrence of events. This measure is called a **Probability measure** and it has the additional property that the total measure of the entire space is equal to one. The triplet $(\Omega, \mathcal{A}, \mathbb{P})$ is called a **probability space** where :

- The set Ω is the **sample space**. It contains all possible outcomes of the experiment. Once an outcome $\omega \in \Omega$ is fixed, all random quantities associated with the experiment are completely determined.
- The collection $\mathcal{A}$ is a σ-field of subsets of Ω, called **events**. These are the sets to which probabilities can be assigned. From the point of view of measure theory, they are precisely the measurable sets. An event $A \in \mathcal{A}$ represents the occurrence of a given property of the outcome ω.
- The function $\mathbb{P} : \mathcal{A} \to [0, 1]$ is a **probability measure**. For each event $A \in \mathcal{A}$, the value $\mathbb{P}(A)$ represents the probability that the event A occurs.

This axiomatic formulation of probability theory was introduced by **Andrey Kolmogorov in 1933**. The Kolmogorov axioms provide a simple and powerful foundation

for probability and allow many results to be developed in a unified and rigorous way.

Although probability theory relies on measure theory, the goals of the two subjects are different. Measure theory studies general notions of size and integration, while probability theory focuses on modeling and analyzing randomness. In this chapter, we mainly develop probability theory while referring to measure-theoretic concepts whenever they are needed.

A prior knowledge of measure theory can be helpful but is not strictly required. The reader is assumed to have seen basic notions such as σ-fields , measurable functions, and measures. Whenever these ideas are used, we briefly recall their meaning and explain their role in probability.

The aim of this chapter is to present the fundamental concepts of probability theory in a clear and structured way, combining mathematical rigor with intuitive interpretations based on random experiments.

Important remark. In many situations, the explicit form of the probability measure $\mathbb{P}$ will not be specified. Instead, we focus on the properties of measurable functions defined on $(\Omega, \mathcal{A})$, which are called *random variables*.

Terminology. As in measure theory, sets of measure zero play an important role in probability theory. A property depending on $\omega \in \Omega$ is said to hold *almost surely* (a.s.) if it holds for all ω in an event of probability one. Thus, *almost surely* has the same meaning as "almost everywhere" in measure theory.

Notation. For a random variable $\mathbf{X}$, the notation $\mathbb{P}(\mathbf{X} \in A)$ means :

$$\mathbb{P}(\{\omega \in \Omega : \mathbf{X}(\omega) \in A\}).$$

1.1 General Definitions

In this section, we introduce the fundamental objects of probability theory. We begin with the definition of a probability space, which provides the mathematical framework for modeling random experiments. We then define random variables, which allow us to describe and analyze numerical quantities generated by randomness.

These definitions form the basis of all subsequent results in this chapter. They are stated in a general and rigorous setting, with close connections to measure theory whenever necessary.

1.1.1 Probability space

Definition 1.1.1: Probability space

A **probability space** is a triple $(\Omega, \mathcal{A}, \mathbb{P})$, where:

- Ω is a non-empty set called the **sample space**,
- $\mathcal{A}$ is a σ-field of subsets of Ω, and
- $\mathbb{P} : \mathcal{A} \to [0, 1]$ is a **probability measure**.

We remind the reader that a σ-field $\mathcal{A}$ is a collection of subsets of Ω such that:

- $\Omega \in \mathcal{A}$,
- if $A \in \mathcal{A}$ then $A^c = \Omega \setminus A \in \mathcal{A}$,
- if $A_1, A_2, \ldots \in \mathcal{A}$ then $\cup_{i=1}^{\infty} A_i \in \mathcal{A}$.

Definition 1.1.2: Probability measure

A **probability measure** $\mathbb{P}$ on a σ-field $\mathcal{A}$ of a set Ω is a function $\mathbb{P} : \mathcal{A} \to [0, 1]$ satisfying:

- $\mathbb{P}(\Omega) = 1$
- For any countable sequence of disjoint sets $A_1, A_2, \ldots \in \mathcal{A}$,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

Proposition 1.1.3

We have the following results :

1. If $A \subset B$ then :

$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$$

2. If $A, B \in \mathcal{A}$ then :

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

3. If $A_n \in \mathcal{A}$ and $A_n \subset A_{n+1}$ for every $n \in \mathbb{N}$ then :

$$\mathbb{P}(\bigcup_{n\in\mathbb{N}} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$$

4. If $A_n \in \mathcal{A}$ and $A_{n+1} \subset A_n$ for every $n \in \mathbb{N}$ then :

$$\mathbb{P}(\bigcap_{n\in\mathbb{N}} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n)$$

5. If $A_n \in \mathcal{A}$ for every $n \in \mathbb{N}$ then :

$$\mathbb{P}(\bigcup_{n\in\mathbb{N}} A_n) \leq \sum_{n\in\mathbb{N}} \mathbb{P}(A_n)$$

Proof for Proposition.

Left exercise for the reader.

1.1.2 Monotone class

In this section, we state and prove the monotone class theorem, which is a fundamental tool in probability theory.

Definition 1.1.4: Monotone class

Let Ω be a set and let $\mathcal{M} \subset \mathcal{P}(\Omega)$. The collection $\mathcal{M}$ is called a *monotone class* if:

1. $\Omega \in \mathcal{M}$;
2. if $A, B \in \mathcal{M}$ and $A \subset B$, then $B \setminus A \in \mathcal{M}$;
3. if $(A_n)_{n\in\mathbb{N}}$ is an increasing sequence in $\mathcal{M}$, i.e.

$$A_1 \subset A_2 \subset \cdots,$$

then

$$\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{M}.$$

Every σ-field is a monotone class. Conversely, a monotone class is a σ-field if and only if it is closed under finite intersections. Indeed, closure under finite intersections implies closure under complements and finite unions, and property (3) then yields closure under countable unions.

Arbitrary intersections of monotone classes are again monotone classes. Given a collection $\mathcal{C} \subset \mathcal{P}(\Omega)$, we define the *monotone class generated by* $\mathcal{C}$, denoted $\mathcal{M}(\mathcal{C})$,

as the intersection of all monotone classes containing $\mathcal{C}$.

Theorem 1.1.5: Monotone Class Theorem

Let $\mathcal{C} \subset \mathcal{P}(\Omega)$ be a collection of sets that is closed under finite intersections. Then

$$\mathcal{M}(\mathcal{C}) = \sigma(\mathcal{C}).$$

Consequently, if $\mathcal{M}$ is any monotone class such that $\mathcal{C} \subset \mathcal{M}$, then

$$\sigma(\mathcal{C}) \subset \mathcal{M}.$$

Proof for Theorem.

It is enough to prove the first assertion. Since any σ-field is a monotone class, it is clear that $\mathcal{M}(C) \subset \sigma(\mathcal{C})$. To prove the reverse inclusion, it is enough to verify that $\mathcal{M}(\mathcal{C})$ is a σ-field . As explained above, it suffices to verify that $\mathcal{M}(\mathcal{C})$ is closed under finite intersections.

For every $A \in \mathcal{P}(\Omega)$, set

$$\mathcal{M}_A = \{B \in \mathcal{M}(\mathcal{C}) : A \cap B \in \mathcal{M}(\mathcal{C})\}.$$

Fix $A \in \mathcal{C}$. Since $\mathcal{C}$ is closed under finite intersections, we have $\mathcal{C} \subset \mathcal{M}_A$. Let us verify that $\mathcal{M}_A$ is a monotone class:

- $\Omega \in \mathcal{M}_A$ is trivial.
- If $B, B' \in \mathcal{M}_A$ and $B \subset B'$, then $A \cap (B' \setminus B) = (A \cap B') \setminus (A \cap B) \in \mathcal{M}(\mathcal{C})$ and thus $B' \setminus B \in \mathcal{M}_A$.
- If $B_n \in \mathcal{M}_A$ for every $n \geq 0$ and the sequence $(B_n)_{n\geq 0}$ is increasing, we have $A \cap (\cup_{n\in\mathbb{N}} B_n) = \cup_{n\in\mathbb{N}}(A \cap B_n) \in \mathcal{M}(\mathcal{C})$ and therefore $\cup_{n\in\mathbb{N}} B_n \in \mathcal{M}_A$.

Since $\mathcal{M}_A$ is a monotone class that contains $\mathcal{C}$, $\mathcal{M}_A$ also contains $\mathcal{M}(\mathcal{C})$. We have thus obtained that

$$\forall A \in \mathcal{C},\ \forall B \in \mathcal{M}(\mathcal{C}),\ A \cap B \in \mathcal{M}(\mathcal{C}).$$

This is not yet the desired result, but we can use the same idea one more time. Precisely, we now fix $A \in \mathcal{M}(\mathcal{C})$. According to the first part of the proof, $\mathcal{C} \subset \mathcal{M}_A$. By exactly the same arguments as in the first part of the proof, we get that $\mathcal{M}_A$ is a monotone class. It follows that $\mathcal{M}(\mathcal{C}) \subset \mathcal{M}_A$, which shows that $\mathcal{M}(\mathcal{C})$ is closed under finite intersections, and completes the proof. ■

Proposition 1.1.6

Let $(\Omega, \mathcal{A})$ be a measurable space and let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Omega, \mathcal{A})$. Suppose there exists a collection $\mathcal{C} \subset \mathcal{A}$ such that:

- $\mathcal{C}$ is closed under finite intersections;
- $\sigma(\mathcal{C}) = \mathcal{A}$;

- $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \mathcal{C}$.

Then: We have $\mathbb{P} = \mathbb{Q}$;

Proof for Proposition.

Let $\mathcal{G} = \{A \in \mathcal{A} : \mathbb{P}(A) = \mathbb{Q}(A)\}$. By assumption, $\mathcal{C} \subset \mathcal{G}$. On the other hand, it is easy to verify that $\mathcal{G}$ is a monotone class. For instance, if $A, B \in \mathcal{G}$ and $A \subset B$, we have $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A) = \mathbb{Q}(B) - \mathbb{Q}(A) = \mathbb{Q}(B \setminus A)$, and hence $B \setminus A \in \Omega$ (note that we here use the fact that both $\mathbb{P}$ and $\mathbb{Q}$ are finite).

If follows that $\mathcal{G}$ contains $\mathcal{M}(\mathcal{C})$, which is equal to $\sigma(\mathcal{C})$ by the monotone class theorem. By our assumption $\sigma(\mathcal{C}) = \mathcal{A}$, we get $\mathcal{G} = \mathcal{A}$, which means that $\mathbb{P} = \mathbb{Q}$. ∎

1.1.3 Random variables

Definition 1.1.7: Random variable

A function from Ω to $\mathbb{R}^n$ is called a random variable (in measure theory is called measurable function), meaning an application $\mathbf{X}$ such that:

$$\mathbf{X}^{-1}(B) \in \mathcal{A}, \text{ for each set } B \in \mathcal{B}(\mathbb{R}^n)$$

where $\mathcal{B}(\mathbb{R}^n)$ denotes the Borel [a] σ-field .

a. The Borel σ-field on $\mathbb{R}^n$ is generated by the open subsets of $\mathbb{R}^n$. The Lebesgue measure on $\mathbb{R}^n$ can be first defined on rectangles of the form $[a_1, b_1] \times \cdots \times [a_n, b_n]$ by $(b_1 - a_1) \cdots (b_n - a_n)$, and then uniquely extended to a measure on the Borel σ-field . For more details, see the chapter [3, chap. 3].

Based on this definition, we can define a probability measure in the probabilistic space $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \mathbb{P}_{\mathbf{X}})$, where the probability measure (also called the distribution of $\mathbf{X}$) $\mathbb{P}_{\mathbf{X}}$ is defined as follows:

$$\mathbb{P}_{\mathbf{X}}(B) = \mathbb{P}(\mathbf{X}^{-1}(B)) = \mathbb{P}(\{\mathbf{X} \in B\}), \text{ for each set } B \in \mathcal{B}(\mathbb{R}^n)$$

We call $\mathbb{P}_{\mathbf{X}}$ the law (or the distribution) of the random variable $\mathbf{X}$. In order to characterize the distribution of a random variable, we use cumulative distribution functions and probability density functions.

Definition 1.1.8: Cumulative distribution function

Let $\mathbf{X}$ be a random variable with value space $\mathbb{R}^n$. For each $\mathbf{x} \in \mathbb{R}^n$, we define the **cumulative distribution function** as follows:

$$F_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(\mathbf{X} \leq \mathbf{x}) = \mathbb{P}(X_1 \leq x_1, \cdots, X_n \leq x_n)$$

Definition 1.1.9: Probability density function

Let $\mathbf{X}$ be a random variable with value space $\mathbb{R}^n$. For each $\mathbf{x} \in \mathbb{R}^n$, we define the **probability density function** as follows:

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\partial^n}{\partial \mathbf{x}} F_{\mathbf{X}}(\mathbf{x}) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{\mathbf{X}}(\mathbf{x})$$

if it exists on a non-negligible set of $\mathbb{R}^n$. This function is a density for the probability measure $\mathbb{P}_{\mathbf{X}}$ such that:

$$\mathbb{P}_{\mathbf{X}}(B) = \int_B d\mathbb{P}_{\mathbf{X}} = \int_B f_{\mathbf{X}}(\mathbf{x}) d\lambda(\mathbf{x})$$

Where λ is the Lebesgue measure on $\mathcal{B}(\mathbb{R}^n)$

Finally, we have the following theorem, which ensures that a constructed variable is indeed a random variable.

Theorem 1.1.10: Operations on random variables

1. Let X_1, X_2 be two random variables on $\mathbb{R}$, and $f : \mathbb{R}^2 \mapsto \mathbb{R}$ a measurable function (w.r.t Lebesgue measure and Borel σ-field), then (X_1, X_2), $f(X_1, X_2)$ are random variables.
2. Let (X_n) be a sequence of random variables that map into $\bar{\mathbb{R}}$ then :

$$\sup_{n\in\mathbb{N}} X_n, \quad \inf_{n\in\mathbb{N}} X_n, \quad \limsup_{n\to\infty} X_n, \quad \liminf_{n\to\infty} X_n$$

   are also random variables. In particular, if the sequence (X_n) converges pointwise, its limit is a random variable.
3. Let (X_n) be a sequence of random variables that map into $\bar{\mathbb{R}}$ then : The set A of all $x \in \Omega$ such that $X_n(\omega)$ converges in $\mathbb{R}$ as $n \to \infty$ is measurable ($A \in \mathcal{A}$). Furthermore, the function $h : \Omega \longrightarrow \mathbb{R}$ defined by

$$Y(\omega) := \begin{cases} \lim_{n\to\infty} X_n(\omega) & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A, \end{cases}$$

   is a random variable.

Proof for Theorem.

The proofs could be found in [3, sec. 1.3]. ■

Finally we have this two lemmas, that would be used frequently in most of probability-propositions proofs.

Lemma 1.1.11: Approximation by Simple Random Variables

Let X be a non-negative random variable with values in $\mathbb{R}$. Then there is a sequence (φ_n) of simple random variables with $\varphi_{n+1} \geq \varphi_n$ such that $X = \lim \varphi_n$ at each point of Ω. Where a simple random variable is a random variable having the form :

$$X = \sum_{i=1}^{m} c_i \mathbb{1}_{E_i}$$

For some integer m and real positive values c_i.

Proof for Lemma

The proof is left exercise for reader with the hint to use :

$$E_{n,k} = \{x : k2^{-n} \leq X(\omega) < (k+1)2^{-n}\}, \text{ and the sequence } \varphi_n = 2^{-n} \sum_{k=0}^{2^{2n}} k \mathbb{1}_{E_{n,k}}.$$

Lemma 1.1.12: Doob-Dynkin Lemma

If $\mathbf{Y}$ with values in $\mathbb{R}^m$ is $\sigma(\mathbf{X})$-measurable where $\mathbf{X}$ is a random variable with values in $\mathbb{R}^n$, there is a measurable function $f : \mathbb{R}^n \to \mathbb{R}^m$ such that $\mathbf{Y} = f(\mathbf{X})$.

Proof for Lemma

We'll proof the theorem for real valued random variables (with values in $\mathbb{R}$).

1. **Step 1: Indicator Functions** First, consider the case where $Y = \mathbb{1}_A$ for some $A \in \sigma(X)$. Since $A \in \sigma(X)$, by definition, there exists a Borel set $B \in \mathcal{B}(\mathbb{R})$ such that $A = X^{-1}(B)$. Define the function $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \mathbb{1}_B(x).$$

Then, for any $\omega \in \Omega$,

$$Y(\omega) = \mathbb{1}_A(\omega) = \mathbb{1}_{\mathbf{X}^{-1}(B)}(\omega) = \mathbb{1}_B(X(\omega)) = f(X(\omega)).$$

Hence, $Y = f(X)$ for this case.

2. **Step 2: Simple Functions** Now, consider Y as a simple function, i.e., $Y = \sum_{i=1}^{n} c_i \mathbb{1}_{A_i}$ where $c_i \in \mathbb{R}$ and $A_i \in \sigma(X)$. For each $A_i \in \sigma(X)$, there exists a Borel set $B_i \in \mathcal{B}(\mathbb{R})$ such that $A_i = X^{-1}(B_i)$. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \sum_{i=1}^{n} c_i \mathbb{1}_{B_i}(x).$$

Then, for any $\omega \in \Omega$,

$$Y(\omega) = \sum_{i=1}^{n} c_i \mathbb{1}_{A_i}(\omega) = \sum_{i=1}^{n} c_i \mathbb{1}_{\mathbf{X}^{-1}(B_i)}(\omega) = \sum_{i=1}^{n} c_i \mathbb{1}_{B_i}(X(\omega)) = f(X(\omega)).$$

Thus, $Y = f(X)$ for this case.

3. **Step 3: Positive Measurable Functions** Consider Y as a positive $\sigma(X)$-measurable function. There exists a sequence of simple functions Y_n such that $Y_n \nearrow Y$. (use of the lemma 1.1.11) From Step 2, for each n, there exists a Borel-measurable function $f_n : \mathbb{R} \to \mathbb{R}$ such that

$$Y_n = f_n(X).$$

Since $Y_n \nearrow Y$, we have $f_n(X) \to Y$ pointwise almost everywhere. Define $f(x) = \lim_{n\to\infty} f_n(x)$. Since each f_n is Borel-measurable, and the pointwise limit of Borel-measurable functions is Borel-measurable, f is Borel-measurable. Therefore, $Y = f(X)$.

4. **Step 4: General Measurable Functions** Finally, consider Y as an arbitrary $\sigma(X)$-measurable function. We can decompose Y into its positive and negative parts:

$$Y = Y^+ - Y^-,$$

where $Y^+ = \max(Y, 0)$ and $Y^- = -\min(Y, 0)$. Both Y^+ and Y^- are positive $\sigma(X)$-measurable functions. By Step 3, there exist Borel-measurable functions f^+ and f^- such that

$$Y^+ = f^+(X) \quad \text{and} \quad Y^- = f^-(X).$$

Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = f^+(x) - f^-(x).$$

Then,

$$Y = Y^+ - Y^- = f^+(X) - f^-(X) = f(X).$$

Thus, $Y = f(X)$ for some Borel-measurable function f.

■

A large number of proofs in probability theory follow the same general strategy. The idea is to first establish a result for very simple random variables and then extend it step by step to more general cases. This approach reflects the way many objects in probability theory are constructed from simpler ones.

More precisely, proofs often proceed through the following steps:

1. **Step 1: Indicator Functions.** The result is first verified for random variables of the form $\mathbb{1}_A$ where $A \in \mathcal{A}$
2. **Step 2: Simple Functions.** The result is then extended to simple random variables, which are finite linear combinations of indicator functions.
3. **Step 3: Positive Measurable Functions.** Using monotone approximation by simple functions, the result is extended to non-negative measurable random variables

via the approximation lemma 1.1.11.

4. **Step 4: General Measurable Functions.** Finally, the result is extended to general measurable random variables by decomposing them into their positive and negative parts.

In many situations, some of these steps may be omitted when they follow directly from previous arguments. Nevertheless, this four-step method provides a common structure that appears repeatedly in probability theory.

1.2 Expectation and Moments of Random Variables

1.2.1 Expectation

Definition 1.2.1: Expectation

Let $\mathbf{X}$ be a random variable with value space $\mathbb{R}^n$. The **expectation** or **expected value** of a random variable $\mathbf{X}$ with respect to a probability measure $\mathbb{P}$ is defined as

$$\mathbb{E}[\mathbf{X}] = \int_\Omega \mathbf{X}\, d\mathbb{P} = \int_{\mathbb{R}^n} \mathbf{x}\, d\mathbb{P}_{\mathbf{X}} = \int_{\mathbb{R}^n} \mathbf{x} f_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}$$

The integral here is a Lebesgue integral [a] with respect to the probability measure. We require the integral to exist, and the last equality holds only if a density exists.

a. The Lebesgue integral is the generalisation of integration on the measurable spaces, for the rigorous definition check the chapter [3, chap. 2]

We recall some properties of expectations as follow : Let X and Y be random variables, and let a and b be constants. The main properties of expectations are:

1. **Probability representation:**

$$\mathbb{P}[X \in A] = \mathbb{E}(\mathbb{1}_{X \in A})$$

2. **Linearity:**

$$\mathbb{E}[a\mathbf{X} + b\mathbf{Y}] = a\mathbb{E}[\mathbf{X}] + b\mathbb{E}[\mathbf{Y}]$$

3. **Expectation of a Constant:**

$$\mathbb{E}[a] = a$$

4. **Non-negativity:** If $\mathbf{X} \geq 0$ almost surely, then

$$\mathbb{E}[\mathbf{X}] \geq 0$$

5. **Finiteness:**

$$\mathbb{E}[X] < \infty, \Longrightarrow X < \infty \quad \mathbb{P}\text{-a.s.}$$

6. **Zero Expectation:** If X is a nonnegative random variable and

$$\mathbb{E}[X] = 0 \iff X = 0 \quad \mathbb{P}\text{-a.s.}$$

7. **Equality of Expectations:** If Y is another nonnegative random variable and

$$X = Y \quad \mathbb{P}\text{-a.s.}$$

then

$$\mathbb{E}[X] = \mathbb{E}[Y].$$

8. **Law of the Unconscious Statistician:** If g is a measurable function and $\mathbf{X}$ is a random variable, then

$$\mathbb{E}[g(\mathbf{X})] = \int g(\mathbf{X}) d\mathbb{P}$$

9. **Fatou's Lemma:** If (X_n) is a sequence of positive real random variables, then

$$\mathbb{E}[\liminf_{n\to\infty} X_n] \leq \liminf_{n\to\infty} \mathbb{E}[X_n]$$

10. **Monotone Convergence Theorem:** If (X_n) is a sequence of positive real random variables that converges to X almost surely and the sequence is a.s. increasing, then

$$\mathbb{E}[\lim_{n\to\infty} X_n] = \lim_{n\to\infty} \mathbb{E}[X_n]$$

11. **Dominated Convergence Theorem:** If (X_n) is a sequence of real random variables that converges to X almost surely and is dominated by an integrable random variable Y (i.e., $|X_n| \leq Y$ almost surely for all n), then

$$\mathbb{E}[\lim_{n\to\infty} X_n] = \lim_{n\to\infty} \mathbb{E}[X_n]$$

Before completing the section we highly invit the reader to consult the section of $\mathcal{L}^p$ spaces 1.3

1.2.2 Variance

Definition 1.2.2: Variance – Covariance – Correlation

Let $X \in L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$ be a real-valued random variable.

1. **Variance.** The **variance** of the random variable X measures how much X varies around its mean. It is defined by

$$\mathrm{Var}(X) = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big].$$

The **standard deviation** of X is defined as

$$\sigma(X) = \sqrt{\mathrm{Var}(X)}.$$

2. **Covariance.** Let X and Y be two random variables in $L^2(\Omega, \mathcal{A}, \mathbb{P})$. The **covariance** of X and Y measures how the two variables vary together and is defined by

$$\mathrm{Cov}(X, Y) = \mathbb{E}\Big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\Big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

3. **Correlation.** The **correlation** between X and Y is a normalized version of the covariance and is defined by

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma(X)\sigma(Y)}.$$

It takes values between -1 and 1.

4. **Covariance matrix.** Let $\mathbf{X} = (X_1, \ldots, X_n)$ be a random vector in $\mathbb{R}^n$ with $\mathbf{X} \in L^2_{\mathbb{R}^n}(\Omega, \mathcal{A}, \mathbb{P})$. The **covariance matrix** of $\mathbf{X}$ is defined by

$$\mathrm{Var}(\mathbf{X}) = \mathbb{E}\Big[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^T\Big] = \Big(\mathrm{Cov}(X_i, X_j)\Big)_{1 \le i,j \le n}.$$

This matrix belongs to $M_n(\mathbb{R})$.

Theorem 1.2.3: Markov Inequalities

Markov's Inequality If X is a nonnegative random variable and $a > 0$,

$$\mathbb{P}(X \ge a) \le \frac{1}{a}\mathbb{E}[X]$$

Bienaymé-Chebyshev Inequality If $X \in L^2(\Omega, \mathcal{A}, \mathbb{P})$ and $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \ge a) \le \frac{1}{a^2}\,\mathrm{var}(X)$$

Proof for Theorem.

A classical proof, left exercise for the reader.
Hint : Use $\mathbb{E}[\mathbb{1}_{X \ge a}] \le \mathbb{E}[X/a]$ for Markov's inequality, and apply Markov's inequality to the nonnegative random variable $(X - \mathbb{E}[X])^2$ for Chebyshev's inequality. ■

1.2.3 Characteristic Function

We start with a basic definition. We use the notation $x \cdot y$ for the standard Euclidean scalar product in $\mathbb{R}^d$.

Definition 1.2.4: Characteristic function

Let X be a random variable with values in $\mathbb{R}^d$. The characteristic function of X is the function $\Phi_X : \mathbb{R}^d \longrightarrow \mathbb{C}$ defined by

$$\Phi_X(\xi) := \mathbb{E}[\exp(i\xi \cdot X)], \qquad \xi \in \mathbb{R}^d.$$

The definition of expectation 1.2.1 allows us to write

$$\Phi_X(\xi) = \int_{\mathbb{R}^d} e^{i\xi \cdot x} \, d\mathbb{P}_X(x)$$

and we can also view Φ_X as the Fourier transform of the law $\mathbb{P}_X$ of X. For this reason, we sometimes write $\Phi_X(\xi) = \widehat{\mathbb{P}_X}(\xi)$. It's easy to notice (in the case $d = 1$), the dominated convergence theorem ensures that the function Φ_X is (bounded and) continuous on $\mathbb{R}^d$.

Our first goal is to show that the characteristic function determines the law $\mathbb{P}_X$ of X (as the name suggests!). This is equivalent to showing that the Fourier transform acting on probability measures on $\mathbb{R}^d$ is injective. We begin with an important calculation for a special case of the Gaussian distribution. We will study this case in more detail in Section 1.6, where we will derive the following formula:

$$\Phi_X(\xi) = \exp\left(im\xi - \frac{\sigma^2\xi^2}{2}\right), \quad \xi \in \mathbb{R}$$

Theorem 1.2.5: Characteristic function determines the law

The characteristic function of a random variable X with values in $\mathbb{R}^d$ determines the law of this random variable. In other words, the Fourier transform acting on probability measures on $\mathbb{R}^d$ is injective.

Proof for Theorem.

Let us consider the case $d = 1$. For every $\sigma > 0$, let g_σ be the density of the Gaussian $\mathcal{N}(0, \sigma^2)$ distribution:

$$g_\sigma(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}.$$

Let μ be a probability measure on $\mathbb{R}$, and set

$$\begin{cases} f_\sigma(x) & = \int_{\mathbb{R}} g_\sigma(x-y)\,\mu(dy) \overset{(\text{def})}{=} g_\sigma * \mu(x), \\ d\mu_\sigma(x) & = f_\sigma(x)\,dx. \end{cases}$$

Let $C_b(\mathbb{R})$ denote the space of all bounded continuous functions $\varphi : \mathbb{R} \longrightarrow \mathbb{R}$, and recall that a probability measure ν on $\mathbb{R}$ is characterized by the values of $\int \varphi(x)\,d\nu(x)$ for $\varphi \in C_b(\mathbb{R})$. The result of the theorem (when $d = 1$) will follow if we are able to prove that:

1. μ_σ is determined by $\widehat{\mu}$.
2. For every $\varphi \in C_b(\mathbb{R})$, $\int \varphi(x)\mu_\sigma(dx) \longrightarrow \int \varphi(x)\mu(dx)$ as $\sigma \to 0$.

In order to establish (1), we use the formula of the characteristic function for gaussian distribution to write, for every $x \in \mathbb{R}$,

$$\sigma\sqrt{2\pi}\, g_\sigma(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right) = \int_{\mathbb{R}} e^{i\xi x} g_{1/\sigma}(\xi)\, d\xi.$$

It follows that

$$\begin{aligned} f_\sigma(x) &= \int_{\mathbb{R}} g_\sigma(x-y)\, \mu(dy) \\ &= (\sigma\sqrt{2\pi})^{-1} \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{i\xi(x-y)} g_{1/\sigma}(\xi)\, d\xi \right) \mu(dy) \\ &= (\sigma\sqrt{2\pi})^{-1} \int_{\mathbb{R}} e^{i\xi x} g_{1/\sigma}(\xi) \left(\int_{\mathbb{R}} e^{-i\xi y} \mu(dy) \right) d\xi \\ &= (\sigma\sqrt{2\pi})^{-1} \int_{\mathbb{R}} e^{i\xi x} g_{1/\sigma}(\xi)\, \widehat{\mu}(-\xi)\, d\xi. \end{aligned}$$

In the penultimate equality, we used the Fubini-Lebesgue theorem 1.4.4. The application of this theorem is easily justified since μ is a probability measure and the function $g_{1/\sigma}$ is integrable with respect to Lebesgue measure. The formula of the last display shows that f_σ (and hence μ_σ) is determined by $\widehat{\mu}$. As for (2), we start by writing, for every $\varphi \in C_b(\mathbb{R})$,

$$\begin{aligned} \int \varphi(x)\mu_\sigma(dx) &= \int \varphi(x) \left(\int g_\sigma(y-x)\mu(dy) \right) dx \\ &= \int \left(\int g_\sigma(y-x)\varphi(x)\, dx \right) \mu(dy) \\ &= \int g_\sigma * \varphi(y)\, \mu(dy). \end{aligned}$$

where we have again applied the Fubini-Lebesgue theorem, with the same justification as above, also the last equality use the convolution that we will see in 1.4. Then, we note that the properties

$$\begin{cases} \int g_\sigma(x)\, dx = 1, \\ \lim_{\sigma\to 0} \int_{\{|x|>\varepsilon\}} g_\sigma(x)\, dx = 0, \quad \forall \varepsilon > 0. \end{cases}$$

imply that, for every $y \in \mathbb{R}$,

$$\lim_{\sigma\to 0} g_\sigma * \varphi(y) = \varphi(y)$$

thanks to Proposition **??** (1) that we will see in the section measure product and independence 1.4. By dominated convergence, using the fact that $|g_\sigma * \varphi| \leq \sup\{|\varphi(x)| : x \in \mathbb{R}\}$, we get

$$\lim_{\sigma\to 0} \int \varphi(x)\mu_\sigma(dx) = \lim_{\sigma\to 0} \int g_\sigma * \varphi(y)\mu(dy) = \int \varphi(x)\mu(dx),$$

which completes the proof of (2) and of the case $d = 1$ of the theorem.
The proof in the general case is similar. We now use the auxiliary functions

$$g_\sigma^{(d)}(x_1, \dots, x_d) = \prod_{j=1}^{d} g_\sigma(x_j)$$

noting that, for $\xi = (\xi_1, \dots, \xi_d) \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d} g_\sigma^{(d)}(x) e^{i\xi\cdot x}\, dx = \prod_{j=1}^{d} \int_{\mathbb{R}} g_\sigma(x_j) e^{i\xi_j x_j}\, dx_j = (2\pi/\sigma^2)^{d/2} g_{1/\sigma}^{(d)}(\xi).$$

The arguments of the case $d = 1$ then apply with obvious changes. ■

Proposition 1.2.6

Let $X = (X_1, \dots, X_d)$ be a random variable with values in $\mathbb{R}^d$. Assume that $\mathbb{E}[(X_j)^2] < \infty$ for every $j \in \{1, \dots, d\}$. Then, Φ_X is twice continuously differentiable and

$$\Phi_X(\xi) = 1 + i \sum_{j=1}^{d} \xi_j \mathbb{E}[X_j] - \frac{1}{2} \sum_{j=1}^{d} \sum_{k=1}^{d} \xi_j \xi_k \mathbb{E}[X_j X_k] + o(|\xi|^2)$$

when $\xi = (\xi_1, \dots, \xi_d)$ tends to 0.

Proof for Proposition.

By differentiating under the integral sign (easy application of dominance convergence theorem), we get, for every $1 \le j \le d$,

$$\frac{\partial \Phi_X}{\partial \xi_j}(\xi) = i\, \mathbb{E}[X_j e^{i\xi\cdot X}].$$

The justification is easy since $|iX_j e^{i\xi\cdot X}| = |X_j|$ and $X_j \in L^2(\Omega, \mathcal{A}, \mathbb{P}) \subset L^1(\Omega, \mathcal{A}, \mathbb{P})$. Similarly, for every $j, k \in \{1, \dots, d\}$, the fact that $X_j X_k \in L^1(\Omega, \mathcal{A}, \mathbb{P})$ allows us to differentiate once more and to get

$$\frac{\partial^2 \Phi_X}{\partial \xi_j \partial \xi_k}(\xi) = -\mathbb{E}[X_j X_k e^{i\xi\cdot X}].$$

Moreover, the dominated convergence theorem ensures that the quantity $\mathbb{E}[X_j X_k e^{i\xi\cdot X}]$ is a continuous function of ξ. We have thus proved that Φ_X is twice continuously differentiable.

Finally, the last assertion of the proposition is just Taylor's expansion of Φ_X at order two at the origin. ■

1.2.4 Laplace transform

For a random variable with values in $\mathbb{R}_+$, one often uses the Laplace transform instead of the characteristic function.

Definition 1.2.7: Laplace transform

Let X be a nonnegative random variable. The Laplace transform of (the law of) X is the function L_X defined on $\mathbb{R}_+$ by

$$L_X(\lambda) := \mathbb{E}[e^{-\lambda X}] = \int_{\mathbb{R}_+} e^{-\lambda x}\, \mathbb{P}_X(dx).$$

The dominated convergence theorem shows that L_X is continuous on $[0, \infty)$. An application of the theorem of differentiation under the integral [which is just easy result of convergence theorem] sign shows that L_X is infinitely differentiable on $(0, \infty)$. The function L_X is also convex (as a consequence of the fact that the functions $\lambda \mapsto e^{-\lambda x}$ are convex, for every $x \geq 0$) and thus has a (possibly infinite) right derivative at 0, which is given by

$$L'_X(0) = -\lim_{\lambda \downarrow 0} \mathbb{E}\left[\frac{1 - e^{-\lambda X}}{\lambda}\right] = -\mathbb{E}[X]$$

where the second equality follows from the monotone convergence theorem.

Theorem 1.2.8: Laplace transform determines the law

If X is a nonnegative random variable, the Laplace transform of X determines the law $\mathbb{P}_X$ of X.

Proof for Theorem.

Suppose X and X' are two nonnegative random variables, and that $L_X = L_{X'}$. For every $\lambda \geq 0$ and every $x \geq 0$, set $\psi_\lambda(x) = e^{-\lambda x}$. We extend the functions ψ_λ to $[0, \infty]$ by continuity, setting $\psi_\lambda(\infty) = 0$ if $\lambda > 0$, and $\psi_0(\infty) = 1$.

Let H be the vector space spanned by the functions ψ_λ, $\lambda \geq 0$. Then H is a subspace of the space $C([0, \infty], \mathbb{R})$ of all real continuous functions on $[0, \infty]$, which is equipped with the topology of uniform convergence. Moreover, the Stone-Weierstrass theorem ensures that H is dense in $C([0, \infty], \mathbb{R})$.

The equality $L_X = L_{X'}$ implies that $\mathbb{E}[\phi(X)] = \mathbb{E}[\phi(X')]$ for every $\phi \in H$. On the other hand, the mapping $\phi \mapsto \mathbb{E}[\phi(X)] = \int \phi\, d\mathbb{P}_X$ is clearly continuous on $C([0, \infty], \mathbb{R})$. Since H is dense, it follows that we have $\mathbb{E}[\psi(X)] = \mathbb{E}[\psi(X')]$ for every $\psi \in C([0, \infty], \mathbb{R})$, and in particular $\mathbb{E}[\psi(X)] = \mathbb{E}[\psi(X')]$ for every continuous function $\psi : \mathbb{R}_+ \to \mathbb{R}$ with compact support, which suffices to say that $\mathbb{P}_X = \mathbb{P}_{X'}$. ■

Finally, in the case of integer-valued random variables, one often uses generating functions.

Definition 1.2.9: Generating function

Let X be a random variable with values in $\mathbb{Z}_+$. The generating function of X is the

function $g_X : [0,1] \to [0,1]$ defined by

$$g_X(r) := \mathbb{E}[r^X] = \sum_{n=0}^{\infty} \mathbb{P}(X = n)\, r^n.$$

Note that $g_X(0) = \mathbb{P}(X = 0)$ and $g_X(1) = 1$. The function g_X is continuous on $[0,1]$. Since the radius of convergence of the series in the definition is at least 1, the function g_X is infinitely differentiable on $[0,1)$. Moreover, it is obvious that g_X determines $\mathbb{P}_X$, since the quantities $\mathbb{P}(X = n)$ appear in the Taylor expansion of g_X at the origin.

The derivative of g_X is

$$g'_X(r) = \sum_{n=1}^{\infty} n\, \mathbb{P}(X = n)\, r^{n-1}$$

for $r \in [0,1)$. It follows that g_X has a (possibly infinite) left-derivative at $r = 1$, and

$$g'_X(1) = \sum_{n=1}^{\infty} n\, \mathbb{P}(X = n) = \mathbb{E}[X].$$

More generally, for every integer $p \geq 1$, if $g_X^{(p)}$ denotes the p-th derivative of g_X,

$$\lim_{r \uparrow 1} g_X^{(p)}(r) = \mathbb{E}[X(X-1)\cdots(X-p+1)]$$

which shows how to recover all moments of X from the knowledge of its generating function.

1.3 Lp Spaces

This section is mainly devoted to the study of the Lebesgue space $\mathcal{L}^p$ of random variables whose p-th power of the absolute value is having expectaiton (integrable) on a given probability measure. The fundamental inequalities of Hölder, Minkowski and Jensen provide essential ingredients for this study. We investigate the Banach space structure of $\mathcal{L}^p$, and in the special case $p = 2$, the Hilbert space structure of $\mathcal{L}^2$, which has important applications in probability theory.

Density theorems showing that any function of $\mathcal{L}^p$ can be well approximated by *smoother* functions play an important role in many analytic developments. As an application of the Hilbert space structure of L^2 , we establish the Radon-Nikodym theorem, which allows one to decompose an arbitrary measure as the sum of a measure having a density with respect to the reference measure and a singular measure. The Radon-Nikodym theorem will be a crucial ingredient of the theory of conditional expectations.

1.3.1 Definition of the space

Throughout this section, we consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. For a real $p \geq 1$, we let $\mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ denote the space of all random variables $\mathbf{X} : \Omega \mapsto \mathbb{R}^d$ such that:

$$\int |\mathbf{X}|^p \, \mathrm{d}\mathbb{P} < \infty$$

We also introduce the space $\mathcal{L}^\infty_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ of all random variables $\mathbf{X} : \Omega \to \mathbb{R}$ such that there exists a constant $C \in \mathbb{R}_+$with

$$|\mathbf{X}| \leq C, \mathbb{P} \text{ a.e.}$$

For every $p \in [1, \infty]$, we define an equivalence relation on the set of p-integrable random variables by setting :

$$\mathbf{X} \equiv \mathbf{Y} \quad \text{if and only if} \quad \mathbf{X} = \mathbf{Y}, \ \mathbb{P}\text{-a.s.}$$

We then consider the quotient space

$$L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}) = \mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}) / \equiv,$$

An element of $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is thus an equivalence class of random variables that are equal almost surely. In fact, $\langle X \rangle_c \in \mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is defined as :

$$\langle X \rangle_c = \left\{ Y \in \mathcal{L}^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}) \quad \text{such that } \mathbf{X} = \mathbf{Y} \quad \mathbb{P} - a.s. \right\}$$

In fact we can define operation

In what follows, we will almost always abuse notation and identify an element of $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ with one of its representatives: when we speak of a random variable in $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$, we mean that a particular representative has been chosen, and all subsequent considerations do not depend on this choice.

We will often write $L^p(\Omega, \mathcal{A}, \mathbb{P})$, $L^p(\mathbb{P})$, $L^p_{\mathbb{R}^d}$ or simply L^p, instead of $L^p_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ when there is no risk of confusion. Notice that the space L^0 is the space of all random variables, the space L^1 is precisely the space of integrable random variables, and that the space L^2 is the space of square-integrable random variables, which is of particular importance in probability theory.

For every random variable $\mathbf{X}$ on $(\Omega, \mathcal{A}, \mathbb{P})$ and every $p \in [1, \infty)$, we set :

$$\|\mathbf{X}\|_p = \left(\mathbb{E}[|\mathbf{X}|^p]\right)^{1/p},$$

with the convention $\infty^{1/p} = \infty$, and

$$\|\mathbf{X}\|_\infty = \inf\{C \in [0, \infty] : |\mathbf{X}| \leq C, \ \mathbb{P}\text{-a.s.}\}.$$

These definitions are such that $|\mathbf{X}| \leq \|\mathbf{X}\|_\infty$ $\mathbb{P}$-a.s., and $\|\mathbf{X}\|_\infty$ is the smallest number in $[0, \infty]$ with this property. We observe that, if X and Y are two random variables such

that $X = Y$ $\mathbb{P}$-a.s., then $\|\mathbf{X}\|_p = \|\mathbf{Y}\|_p$. Hence $\|\mathbf{X}\|_p$ is well defined for $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$.

It is worth noting that when dealing with random variables taking values in $\mathbb{R}^d$, the absolute value can be replaced by the Euclidean norm, indeed for a for a vector $|\mathbf{x}|$ the euclidian norm is $\sqrt{x_1^1 + ... + x_d^2}$.

1.3.2 Principale inequalities

Definition 1.3.1: Conjugate exponents

Let $p, q \in [1, \infty]$. We say that p and q are *conjugate exponents* if

$$\frac{1}{p} + \frac{1}{q} = 1.$$

In particular, $p = 1$ and $q = \infty$

Theorem 1.3.2: Holder Inequality

Let p and q be conjugate exponents. Then, if $\mathbf{X}$ and $\mathbf{Y}$ are two random variables on $(\Omega, \mathcal{A}, \mathbb{P})$,

$$\mathbb{E}[|\mathbf{XY}|] \leq \|\mathbf{X}\|_p \, \|\mathbf{Y}\|_q.$$

In particular, $\mathbf{XY} \in L^1(\Omega, \mathcal{A}, \mathbb{P})$ whenever $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$ and $\mathbf{Y} \in L^q(\Omega, \mathcal{A}, \mathbb{P})$.

Proof for Theorem.

If $\|\mathbf{X}\|_p = 0$, then $\mathbf{X} = 0$ $\mathbb{P}$-a.s., which implies $\mathbb{E}[|\mathbf{XY}|] = 0$, and the inequality is trivial. We may therefore assume that $\|\mathbf{X}\|_p > 0$ and $\|\mathbf{Y}\|_q > 0$. Without loss of generality, we may also assume that $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$ and $\mathbf{Y} \in L^q(\Omega, \mathcal{A}, \mathbb{P})$ (otherwise $\|\mathbf{X}\|_p \|\mathbf{Y}\|_q = \infty$ and there is nothing to prove).

The case $p = 1$ and $q = \infty$ is immediate, since $|\mathbf{XY}| \leq \|\mathbf{Y}\|_\infty |\mathbf{X}|$ $\mathbb{P}$-a.s., which implies :

$$\mathbb{E}[|\mathbf{XY}|] \leq \|\mathbf{Y}\|_\infty \, \mathbb{E}[|\mathbf{X}|] = \|Y\|_\infty \, \|\mathbf{X}\|_1.$$

In what follows, we assume that $1 < p < \infty$ (and thus $1 < q < \infty$). Let $\alpha \in (0, 1)$. Then, for every $x \in \mathbb{R}_+$,

$$x^\alpha - \alpha x \leq 1 - \alpha.$$

Indeed, define $\varphi_\alpha(x) = x^\alpha - \alpha x$ for $x \geq 0$. For $x > 0$,

$$\varphi'_\alpha(x) = \alpha(x^{\alpha-1} - 1),$$

so that $\varphi'_\alpha(x) > 0$ for $x \in (0, 1)$ and $\varphi'_\alpha(x) < 0$ for $x \in (1, \infty)$. Hence φ_α attains its maximum at $x = 1$, which yields the inequality above.

Applying this inequality to $x = u/v$, where $u \geq 0$ and $v > 0$, we obtain

$$u^\alpha v^{1-\alpha} \leq \alpha u + (1 - \alpha) v,$$

and the inequality still holds when $v = 0$. We now take $\alpha = 1/p$ (so that $1 - \alpha = 1/q$) and set :

$$u = \frac{|\mathbf{X}(\omega)|^p}{\|\mathbf{X}\|_p^p}, \qquad v = \frac{|\mathbf{Y}(\omega)|^q}{\|\mathbf{Y}\|_q^q}.$$

This yields

$$\frac{|\mathbf{X}(\omega)Y(\omega)|}{\|\mathbf{X}\|_p\|\mathbf{Y}\|_q} \leq \frac{1}{p}\frac{|\mathbf{X}(\omega)|^p}{\|\mathbf{X}\|_p^p} + \frac{1}{q}\frac{|Y(\omega)|^q}{\|\mathbf{Y}\|_q^q}, \quad \mathbb{P}\text{-a.s.}$$

Taking expectations on both sides gives

$$\frac{1}{\|\mathbf{X}\|_p\|\mathbf{Y}\|_q}\,\mathbb{E}[|\mathbf{XY}|] \leq \frac{1}{p} + \frac{1}{q} = 1,$$

which completes the proof. ■

Theorem 1.3.3: Cauchy-Schwartz Inequality

If $\mathbf{X}$ and $\mathbf{Y}$ are two random variables on $(\Omega, \mathcal{A}, \mathbb{P})$, then

$$\mathbb{E}[|\mathbf{XY}|] \leq \left(\mathbb{E}\left[|\mathbf{X}|^2\right]\right)^{1/2}\left(\mathbb{E}\left[|\mathbf{Y}|^2\right]\right)^{1/2} = \|\mathbf{X}\|_2\,\|\mathbf{Y}\|_2.$$

Proof for Theorem.

Just a special case of Holder Inequality with $p = 2$ ■

Proposition 1.3.4

Suppose that $\mathbb{P}$ is a probability measure and let $p > 1$. Then, for any random variable $\mathbf{X} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$,

$$\|\mathbf{X}\|_1 \leq \|\mathbf{X}\|_p.$$

Consequently, $L^p(\Omega, \mathcal{A}, \mathbb{P}) \subset L^1(\Omega, \mathcal{A}, \mathbb{P})$ for every $p \in (1, \infty]$. More generally, for any $1 \leq r < r' \leq \infty$ and any random variable $\mathbf{X} \in L^{r'}(\Omega, \mathcal{A}, \mathbb{P})$,

$$\|\mathbf{X}\|_r \leq \|\mathbf{X}\|_{r'},$$

and thus $L^{r'}(\Omega, \mathcal{A}, \mathbb{P}) \subset L^r(\Omega, \mathcal{A}, \mathbb{P})$.

Proof for Proposition.

The inequality $\|\mathbf{X}\|_1 \leq \|\mathbf{X}\|_p$ follows directly from Hölder's inequality by taking $\mathbf{Y} = 1$. For the second bound in the corollary, it suffices to replace $\mathbf{X}$ by $|\mathbf{X}|^r$ and to take $p = r'/r$. ■

Theorem 1.3.5: Jensen Inequality

Suppose that $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{A})$, and let $\varphi : \mathbb{R} \to \mathbb{R}_+$ be a convex

function. Then, for every integrable real-valued random variable $X \in L^1_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$,

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}[X]) .$$

remark : The expectation $\mathbb{E}[\varphi(X)]$ is well defined, since $\varphi(X)$ is a nonnegative measurable random variable.

Proof for Theorem.

Set

$$\mathcal{E}_\varphi = \{(a, b) \in \mathbb{R}^2 : \forall x \in \mathbb{R},\ \varphi(x) \geq ax + b\}.$$

Elementary properties of convex functions show that

$$\forall x \in \mathbb{R}, \quad \varphi(x) = \sup_{(a,b)\in\mathcal{E}_\varphi} (ax + b).$$

Since $\varphi(X) \geq aX + b$ almost surely for every $(a, b) \in \mathcal{E}_\varphi$, we obtain

$$\mathbb{E}[\varphi(X)] \geq \sup_{(a,b)\in\mathcal{E}_\varphi} \mathbb{E}[aX + b] = \sup_{(a,b)\in\mathcal{E}_\varphi} \Big(a\,\mathbb{E}[X] + b\Big) = \varphi(\mathbb{E}[X]) .$$

Theorem 1.3.6: Minkowski's Inequality

Let $p \in [1, \infty]$, and let $\mathbf{X}, \mathbf{Y} \in L^p(\Omega, \mathcal{A}, \mathbb{P})$. Then :

$$\mathbf{X} + \mathbf{Y} \in L^p(\Omega, \mathcal{A}, \mathbb{P}) \quad \text{and} \quad \|\mathbf{X} + \mathbf{Y}\|_p \leq \|\mathbf{X}\|_p + \|\mathbf{Y}\|_p.$$

Proof for Theorem.

The cases $p = 1$ and $p = \infty$ are immediate, using the inequality $|\mathbf{X} + \mathbf{Y}| \leq |\mathbf{X}| + |\mathbf{Y}|$. We therefore assume that $1 < p < \infty$.

Writing

$$|\mathbf{X} + \mathbf{Y}|^p \leq (|\mathbf{X}| + |\mathbf{Y}|)^p \leq \Big(2\max(|\mathbf{X}|, |\mathbf{Y}|)\Big)^p \leq 2^p\Big(|\mathbf{X}|^p + |\mathbf{Y}|^p\Big),$$

we see that $\mathbb{E}[|\mathbf{X} + \mathbf{Y}|^p] < \infty$, and hence $\mathbf{X} + \mathbf{Y} \in L^p$.

Next, observe that

$$|\mathbf{X} + \mathbf{Y}|^p = |\mathbf{X} + \mathbf{Y}| \cdot |\mathbf{X} + \mathbf{Y}|^{p-1} \leq |\mathbf{X}|\,|\mathbf{X} + \mathbf{Y}|^{p-1} + |\mathbf{Y}|\,|\mathbf{X} + \mathbf{Y}|^{p-1}.$$

Taking expectations yields

$$\mathbb{E}[|\mathbf{X} + \mathbf{Y}|^p] \leq \mathbb{E}\Big[|\mathbf{X}|\,|\mathbf{X} + \mathbf{Y}|^{p-1}\Big] + \mathbb{E}\Big[|\mathbf{Y}|\,|\mathbf{X} + \mathbf{Y}|^{p-1}\Big] .$$

Applying Hölder's inequality with conjugate exponents p and $q = p/(p - 1)$, we

obtain

$$\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p] \leq \|\mathbf{X}\|_p \left(\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p]\right)^{(p-1)/p} + \|\mathbf{Y}\|_p \left(\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p]\right)^{(p-1)/p}.$$

If $\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p] = 0$, the inequality is trivial. Otherwise, dividing both sides by $\left(\mathbb{E}[|\mathbf{X}+\mathbf{Y}|^p]\right)^{(p-1)/p}$ yields

$$\|\mathbf{X}+\mathbf{Y}\|_p \leq \|\mathbf{X}\|_p + \|\mathbf{Y}\|_p,$$

which completes the proof. ■

1.3.3 Topological structure of Lp-space

Proposition 1.3.7

The formula

$$d(X,Y) = \mathbb{E}\Big[\,|X-Y| \wedge 1\Big]$$

defines a distance on $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$, and the space $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is complete for the distance d.

Proof for Proposition.

Let's prove that the space $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is complete for the distance d. So let $(X_n)_{n\in\mathbb{N}}$ be a Cauchy sequence for the distance d. We can then find a subsequence $Y_k = X_{n_k}$, $k \in \mathbb{N}$, such that for every $k \geq 1$,

$$d(Y_k, Y_{k+1}) \leq 2^{-k}.$$

Then,

$$\mathbb{E}\left[\sum_{k=1}^{\infty}\Big(|Y_{k+1}-Y_k| \wedge 1\Big)\right] = \sum_{k=1}^{\infty} d(Y_k, Y_{k+1}) < \infty,$$

which implies that $\sum_{k=1}^{\infty}(|Y_{k+1}-Y_k| \wedge 1) < \infty$ almost surely, and therefore also $\sum_{k=1}^{\infty}|Y_{k+1}-Y_k| < \infty$ almost surely (there are almost surely only finitely many values of k such that $|Y_{k+1}-Y_k| \geq 1$). We then define a random variable in $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ by setting

$$X = Y_1 + \sum_{k=1}^{\infty}(Y_{k+1}-Y_k),$$

on the event where the series converges absolutely, and $X = 0$ on the complementary event, which has zero probability. By construction, the sequence $(Y_k)_{k\in\mathbb{N}}$ converges almost surely to X, and this implies

$$d(Y_k, X) = \mathbb{E}\Big[\,|Y_k - X| \wedge 1\Big] \longrightarrow 0 \quad \text{as } k \to \infty,$$

by an application of the dominated convergence theorem. Hence the sequence $(Y_k)_{k\in\mathbb{N}}$ converges to X for the distance d, and since the sequence $(X_n)_{n\in\mathbb{N}}$ is Cauchy and has a

convergent subsequence, it must also converge.

Theorem 1.3.8: Riesz Theorem

For every $p \in [1, \infty]$, the space $L^p(\Omega, \mathcal{A}, \mathbb{P})$ equipped with the norm $\mathbf{X} \mapsto \|\mathbf{X}\|_p$ is a Banach space, that is, a complete normed linear space.

Proof for Theorem.

Can be found in [3, p. 68]

Theorem 1.3.9: The Hilbert Space L2

The space $L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P})$ equipped with the scalar product

$$\langle X, Y \rangle = \int XY \, d\mathbb{P} = \mathbb{E}(XY)$$

is a real Hilbert space.

Proof for Theorem.

The Cauchy-Schwarz inequality shows that, if $X, Y \in L^2(\Omega, \mathcal{A}, \mathbb{P})$, then XY is integrable and hence $\langle X, Y \rangle$ is well defined. It is then clear that the mapping $(X, Y) \mapsto \langle X, Y \rangle$ defines an inner product on $L^2(\Omega, \mathcal{A}, \mathbb{P})$, and that $\langle X, X \rangle = \|X\|_2^2$. Finally, the fact that $\left(L^2(\Omega, \mathcal{A}, \mathbb{P}), \|\cdot\|_2\right)$ is complete is a special case of the Riesz theorem 1.3.8.

Classical results from the theory of Hilbert spaces [see appendix **??**] can therefore be applied to $L^2(\Omega, \mathcal{A}, \mathbb{P})$. In particular, if $\Phi : L^2_{\mathbb{R}}(\Omega, \mathcal{A}, \mathbb{P}) \to \mathbb{R}$ is a continuous linear functional, then there exists a unique element $Y \in L^2_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ such that

$$\Phi(X) = \langle X, Y \rangle \quad \text{for every } X \in L^2_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P}),$$

where $\langle X, Y \rangle = \mathbb{E}[XY]$; .

1.3.4 The Radon-Nikodym Theorem

Definition 1.3.10: Measures continuity

Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\Omega, \mathcal{A})$. We say that:

1. $\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$ (notation $\mathbb{Q} \ll \mathbb{P}$) if

$$\forall A \in \mathcal{A}, \quad \mathbb{P}(A) = 0 \;\Rightarrow\; \mathbb{Q}(A) = 0.$$

2. $\mathbb{Q}$ is *singular* with respect to $\mathbb{P}$ (notation $\mathbb{Q} \perp \mathbb{P}$) if there exists $\Omega_0 \in \mathcal{A}$ such that

$$\mathbb{P}(\Omega_0) = 0 \quad \text{and} \quad \mathbb{Q}(\Omega_0^c) = 0.$$

3. $\mathbb{P}$ and $\mathbb{Q}$ are said to be *equivalent* (notation $\mathbb{P} \sim \mathbb{Q}$) if

$$\mathbb{P} \ll \mathbb{Q} \quad \text{and} \quad \mathbb{Q} \ll \mathbb{P}.$$

Now we present the theorem of Radon-Nikodym :

Theorem 1.3.11: Radon-Nikodym Theorem

Suppose that $\mathbb{P}$ is a σ-finite measure [a] on $(\Omega, \mathcal{A})$, and let $\mathbb{Q}$ be another σ-finite measure on $(\Omega, \mathcal{A})$. Then there exists a unique pair $(\mathbb{Q}_a, \mathbb{Q}_s)$ of σ-finite measures on $(\Omega, \mathcal{A})$ such that:

1. $\mathbb{Q} = \mathbb{Q}_a + \mathbb{Q}_s$,
2. $\mathbb{Q}_a \ll \mathbb{P}$ and $\mathbb{Q}_s \perp \mathbb{P}$.

Moreover, there exists a unique [b] nonnegative measurable function $g : \Omega \to \mathbb{R}_+$ such that

$$d\mathbb{Q}_a = g \cdot d\mathbb{P},$$

meaning that

$$\forall A \in \mathcal{A}, \qquad \mathbb{Q}_a(A) = \int_A g \, d\mathbb{P}.$$

a. A measure $\mathbb{P}$ on $(\Omega, \mathcal{A})$ is called *finite* if $\mathbb{P}(\Omega) < \infty$. It is called *$\sigma$-finite* if there exists a sequence of measurable sets $(A_n)_{n\geq 1}$ such that $\Omega = \bigcup_{n=1}^{\infty} A_n$ and $\mathbb{P}(A_n) < \infty$ for all n.

b. Uniqueness is in the sense that if $\tilde{g}$ is another measurable function satisfying the same properties, then $\tilde{g} = g$ $\mathbb{P}$-a.s.

Proof for Theorem.

Could be found in [3, p. 76] proven in the cadre of measure theory. ■

Remark: A useful formula one can use in *change of measures* in Radon-Nikodym calculations is this :

$$d\mathbb{Q} = g \cdot d\mathbb{P} \qquad \textit{then} \qquad \int_A f d\mathbb{Q} = \int_A f \cdot g \cdot d\mathbb{P}$$

Theorem 1.3.12: Radon-Nikodym Theorem in Probability

Suppose that $\mathbb{P}$ and $\mathbb{Q}$ are two probability measures such that $\mathbb{Q} \ll \mathbb{P}$ then :
There exists a unique [a] nonnegative random variable $X : \Omega \to \mathbb{R}_+$ such that

$$d\mathbb{Q} = X \cdot d\mathbb{P},$$

Furthermore, if $\mathbb{P} \ll \mathbb{Q}$ then $1/X$ satisfies :

$$d\mathbb{P} = \frac{1}{X} \cdot d\mathbb{Q},$$

a. Uniqueness is in the sense that if $\tilde{X}$ is another random variable satisfying the same properties,

then $\tilde{g} = g$ $\mathbb{P}$-a.s.

Proposition 1.3.13

Let $\mathbb{P}$ be a σ-finite measure on $(\Omega, \mathcal{A})$, and let $\mathbb{Q}$ be a finite measure on $(\Omega, \mathcal{A})$. The following conditions are equivalent:

1. $\mathbb{Q} \ll \mathbb{P}$;
2. For every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $A \in \mathcal{A}$,

$$\mathbb{P}(A) < \delta \implies \mathbb{Q}(A) < \varepsilon.$$

Proof for Proposition.

The implication (ii) $\Rightarrow$ (i) is immediate. Conversely, suppose that $\mathbb{Q} \ll \mathbb{P}$. By the Radon-Nikodym theorem 1.3.11, there exists a nonnegative measurable function $g : \Omega \to \mathbb{R}_+$ such that

$$d\mathbb{Q} = g \cdot d\mathbb{P}.$$

Note that

$$\int_\Omega g \, d\mathbb{P} = \mathbb{Q}(\Omega) < \infty.$$

The dominated convergence theorem then yields

$$\int_\Omega \mathbb{1}_{\{g>n\}} \, g \, d\mathbb{P} \xrightarrow{n\to\infty} 0.$$

Given $\varepsilon > 0$, we may therefore choose $M \in \mathbb{N}$ large enough such that

$$\int_\Omega \mathbb{1}_{\{g>M\}} \, g \, d\mathbb{P} < \frac{\varepsilon}{2}.$$

Setting $\delta = \varepsilon/(2M)$, we obtain, for every $A \in \mathcal{A}$ such that $\mathbb{P}(A) < \delta$,

$$\mathbb{Q}(A) = \int_A g \, d\mathbb{P} \leq \int_\Omega \mathbb{1}_{\{g>M\}} \, g \, d\mathbb{P} + \int_{A\cap\{g\leq M\}} g \, d\mathbb{P} < \frac{\varepsilon}{2} + M \, \mathbb{P}(A) \leq \varepsilon.$$

This proves (i) $\Rightarrow$ (ii) and completes the proof. ∎

1.4 Measure product and Independence

The notion of independence is one of the first genuinely new concepts introduced in probability theory, in the sense that it does not arise as a straightforward adaptation of a corresponding notion from measure theory. While the independence of two events or two random variables admits a natural intuitive interpretation, the most fundamental and structurally relevant notion is that of independence between two or more σ-fields .

A central result shows that two random variables are independent if and only if the joint distribution of the pair coincides with the product of their marginal distributions. Combined with Fubini's theorem, which we present first, this characterization yields

several equivalent and practical formulations of independence.

We then establish the classical Borel-Cantelli lemma and illustrate its strength through an application to the dyadic expansion of a randomly chosen real number, leading to several striking probabilistic properties.

1.4.1 Probabilities measures product

Let $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$ be two probabilistic spaces. The product σ-field $\mathcal{A}_1 \otimes \mathcal{A}_2$ is defined as the σ-field on $\Omega_1 \times \Omega_2$ given by

$$\mathcal{A}_1 \otimes \mathcal{A}_2 = \sigma\Big(A \times B : A \in \mathcal{A}_1,\ B \in \mathcal{A}_2\Big).$$

Sets of the form $A \times B$, with $A \in \mathcal{A}_1$ and $B \in \mathcal{A}_2$, are called *measurable rectangles.*

remark : The definition of the product σ-field extends naturally to a finite collection of probabilistic spaces $(\Omega_1, \mathcal{A}_1), \ldots, (\Omega_n, \mathcal{A}_n)$ by setting

$$\mathcal{A}_1 \otimes \cdots \otimes \mathcal{A}_n = \sigma(\mathcal{A}_1 \times \cdots \times \mathcal{A}_n).$$

The expected associativity properties hold; for instance, if $n = 3$,

$$(\mathcal{A}_1 \otimes \mathcal{A}_2) \otimes \mathcal{A}_3 = \mathcal{A}_1 \otimes (\mathcal{A}_2 \otimes \mathcal{A}_3) = \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \mathcal{A}_3.$$

Let $C \subset \Omega_1 \times \Omega_2$ and let $\omega_1 \in \Omega_1$. We define the *section* of C at ω_1 by

$$C_{\omega_1} = \{\omega_2 \in \Omega_2 : (\omega_1, \omega_2) \in C\}.$$

Similarly, for $\omega_2 \in \Omega_2$, define

$$C^{\omega_2} = \{\omega_1 \in \Omega_1 : (\omega_1, \omega_2) \in C\}.$$

If X is a random variable on $\Omega_1 \times \Omega_2$ and $\omega_1 \in \Omega_1$, we denote by X_{ω_1} the random variable on Ω_2 defined by

$$X_{\omega_1}(\omega_2) = X(\omega_1, \omega_2).$$

Similarly, for $\omega_2 \in \Omega_2$, we define $X^{\omega_2}(\omega_1) = X(\omega_1, \omega_2)$.

Proposition 1.4.1

1. If $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$, then for every $\omega_1 \in \Omega_1$, the section C_{ω_1} belongs to $\mathcal{A}_2$, and for every $\omega_2 \in \Omega_2$, the section C^{ω_2} belongs to $\mathcal{A}_1$.
2. Let X be a random variable on $\Omega_1 \times \Omega_2$ with respect to $\mathcal{A}_1 \otimes \mathcal{A}_2$. Then, for every $\omega_1 \in \Omega_1$, X_{ω_1} is $\mathcal{A}_2$-measurable, and for every $\omega_2 \in \Omega_2$, X^{ω_2} is $\mathcal{A}_1$-measurable.

Proof for Proposition.

1. Fix $\omega_1 \in \Omega_1$ and set

$$\mathcal{C} = \{C \in \mathcal{A}_1 \otimes \mathcal{A}_2 : C_{\omega_1} \in \mathcal{A}_2\}.$$

Then $\mathcal{C}$ contains all measurable rectangles (if $C = A \times B$, we have $C_{\omega_1} = B$ or $C_{\omega_1} = \mathcal{O}$ according as $\omega_1 \in A$ or $\omega_1 \notin A$). On the other hand, it is easy to verify that $\mathcal{C}$ is a σ-field, and it follows that $\mathcal{C} = \mathcal{A}_1 \otimes \mathcal{A}_2$. This gives the property $C_{\omega_1} \in \mathcal{A}_2$ for every $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$. The property $C^{\omega_2} \in \mathcal{A}_1$ is derived in a similar manner.

2. For every measurable subset D of G,

$$\begin{aligned} X_{\omega_1}^{-1}(D) &= \{\omega_2 \in \Omega_2 : X_{\omega_1}(\omega_2) \in D\} \\ &= \{\omega_2 \in \Omega_2 : (\omega_1, \omega_2) \in X^{-1}(D)\} \\ &= (X^{-1}(D))_{\omega_1}. \end{aligned}$$

which belongs to $\mathcal{A}_2$ by part (1).

Theorem 1.4.2: Existence of product measure

Let $\mathbb{P}_1$ and $\mathbb{P}_2$ be two probability measures on $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$, respectively.

1. There exists a unique measure $\mathbb{P}_1 \otimes \mathbb{P}_2$ on $(\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2)$ such that

$$(\mathbb{P}_1 \otimes \mathbb{P}_2)(A \times B) = \mathbb{P}_1(A)\,\mathbb{P}_2(B), \qquad A \in \mathcal{A}_1,\ B \in \mathcal{A}_2,$$

2. For every $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$, the mappings

$$\omega_1 \mapsto \mathbb{P}_2(C_{\omega_1}) \quad \text{and} \quad \omega_2 \mapsto \mathbb{P}_1(C^{\omega_2})$$

are measurable (ie random variables), and

$$(\mathbb{P}_1 \otimes \mathbb{P}_2)(C) = \int_{\Omega_1} \mathbb{P}_2(C_{\omega_1})\, d\mathbb{P}_1(\omega_1) = \int_{\Omega_2} \mathbb{P}_1(C^{\omega_2})\, d\mathbb{P}_2(\omega_2).$$

Remarks:

1. Suppose now that we consider an arbitrary finite number of σ-finite measures $\mathbb{P}_1, \ldots, \mathbb{P}_n$, defined on $(E_1, \mathcal{A}_1), \ldots, (E_n, \mathcal{A}_n)$ respectively. We can then define the product measure

$$\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n$$

on

$$(E_1 \times \cdots \times E_n,\ \mathcal{A}_1 \otimes \cdots \otimes \mathcal{A}_n)$$

by setting

$$\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n = \mathbb{P}_1 \otimes (\mathbb{P}_2 \otimes (\cdots \otimes \mathbb{P}_n)).$$

The way we insert parentheses is in fact unimportant since $\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n$ is characterized by its values on (measurable) rectangles,

$$\mathbb{P}_1 \otimes \cdots \otimes \mathbb{P}_n(A_1 \times \cdots \times A_n) = \mathbb{P}_1(A_1) \cdots \mathbb{P}_n(A_n).$$

Theorem 1.4.3: Fubini-Tonelli Theorem

Let $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$ be probabilistic spaces, and equip $\Omega_1 \times \Omega_2$ with the product σ-field $\mathcal{A}_1 \otimes \mathcal{A}_2$ and Let $\mathbb{P}_1$ and $\mathbb{P}_2$ be two probability measures on $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$, respectively, and let $X : \Omega_1 \times \Omega_2 \to [0, \infty]$ be a positive random variable.

1. The functions

$$\Omega_1 \ni \omega_1 \longmapsto \int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2), \qquad \Omega_2 \ni \omega_2 \longmapsto \int_{\Omega_1} X(\omega_1, \omega_2)\, d\mathbb{P}_1(\omega_1),$$

   with values in $[0, \infty]$, are $\mathcal{A}_1$-measurable and $\mathcal{A}_2$-measurable, respectively (ie. they are random variables).

2. We have :

$$\begin{aligned}\int_{\Omega_1 \times \Omega_2} X\, d(\mathbb{P}_1 \otimes \mathbb{P}_2) &= \int_{\Omega_1} \left(\int_{\Omega_2} X(\omega_1, \omega_2)\, d\mathbb{P}_2(\omega_2) \right) d\mathbb{P}_1(\omega_1) \\ &= \int_{\Omega_2} \left(\int_{\Omega_1} X(\omega_1, \omega_2)\, d\mathbb{P}_1(\omega_1) \right) d\mathbb{P}_2(\omega_2).\end{aligned}$$

   Or more simply :

$$\mathbb{E}_{\mathbb{P}_1 \otimes \mathbb{P}_2}[X] = \mathbb{E}_{\mathbb{P}_1}\Big[\mathbb{E}_{\mathbb{P}_2}\big[X(\omega_1, \cdot)\big]\Big] = \mathbb{E}_{\mathbb{P}_2}\Big[\mathbb{E}_{\mathbb{P}_1}\big[X(\cdot, \omega_2)\big]\Big].$$

Proof for Theorem.

We give the proof for the special case $X = \mathbb{1}_C$ for some $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$. The general case can be deduced from this special case by applying the monotone convergence theorem to the sequence of random variables $X_n \xrightarrow{n \to \infty} X$.
In fact, let $X = \mathbb{1}_C$ for some $C \in \mathcal{A}_1 \otimes \mathcal{A}_2$ the functions in the theorem are reduced to :

$$\Omega_1 \ni \omega_1 \longmapsto \int_{\Omega_2} \mathbb{1}_{C_{\omega_1}}(\omega_2), d\mathbb{P}_2(\omega_2), \qquad \Omega_2 \ni \omega_2 \longmapsto \int_{\Omega_1} \mathbb{1}_{C^{\omega_2}}(\omega_1)\, d\mathbb{P}_1(\omega_1),$$

Which are exactly :

$$\omega_1 \mapsto \mathbb{P}_2(C_{\omega_1}) \quad \text{and} \quad \omega_2 \mapsto \mathbb{P}_1(C^{\omega_2})$$

By the second point of Theorem 1.4.2, these functions are measurable, and we have

$$(\mathbb{P}_1 \otimes \mathbb{P}_2)(C) = \int_{\Omega_1} \mathbb{P}_2(C_{\omega_1})\, d\mathbb{P}_1(\omega_1) = \int_{\Omega_2} \mathbb{P}_1(C^{\omega_2})\, d\mathbb{P}_2(\omega_2).$$

Which is :

$$\mathbb{E}_{\mathbb{P}_1\otimes\mathbb{P}_2}[X] = \mathbb{E}_{\mathbb{P}_1}\Big[\mathbb{E}_{\mathbb{P}_2}\big[X(\omega_1,\cdot)\big]\Big] = \mathbb{E}_{\mathbb{P}_2}\Big[\mathbb{E}_{\mathbb{P}_1}\big[X(\cdot,\omega_2)\big]\Big].$$

■

Theorem 1.4.4: Fubini-Lebesgue Theorem

Let $X \in L^1(\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mathbb{P}_1 \otimes \mathbb{P}_2)$. Then:

1. For $\mathbb{P}_1$-almost every $\omega_1 \in \Omega_1$, the random variable $\omega_2 \mapsto X(\omega_1,\omega_2)$ belongs to $L^1(\Omega_2, \mathcal{A}_2, \mathbb{P}_2)$, and for $\mathbb{P}_2$-almost every $\omega_2 \in \Omega_2$, the random variable $\omega_1 \mapsto X(\omega_1,\omega_2)$ belongs to $L^1(\Omega_1, \mathcal{A}_1, \mathbb{P}_1)$.
2. The functions

$$\omega_1 \longmapsto \int_{\Omega_2} X(\omega_1,\omega_2)\, d\mathbb{P}_2(\omega_2), \qquad \omega_2 \longmapsto \int_{\Omega_1} X(\omega_1,\omega_2)\, d\mathbb{P}_1(\omega_1),$$

   belong to $L^1(\Omega_1, \mathcal{A}_1, \mathbb{P}_1)$ and $L^1(\Omega_2, \mathcal{A}_2, \mathbb{P}_2)$, respectively.
3. We have :

$$\begin{aligned}\int_{\Omega_1\times\Omega_2} X\, d(\mathbb{P}_1\otimes\mathbb{P}_2) &= \int_{\Omega_1}\left(\int_{\Omega_2} X(\omega_1,\omega_2)\, d\mathbb{P}_2(\omega_2)\right) d\mathbb{P}_1(\omega_1)\\ &= \int_{\Omega_2}\left(\int_{\Omega_1} X(\omega_1,\omega_2)\, d\mathbb{P}_1(\omega_1)\right) d\mathbb{P}_2(\omega_2).\end{aligned}$$

   Or more simply :

$$\mathbb{E}_{\mathbb{P}_1\otimes\mathbb{P}_2}[X] = \mathbb{E}_{\mathbb{P}_1}\Big[\mathbb{E}_{\mathbb{P}_2}\big[X(\omega_1,\cdot)\big]\Big] = \mathbb{E}_{\mathbb{P}_2}\Big[\mathbb{E}_{\mathbb{P}_1}\big[X(\cdot,\omega_2)\big]\Big].$$

Proof for Theorem.

1. By theorem 1.4.3 applied to $|X|$, we have

$$\int_{\Omega_1}\left(\int_{\Omega_2} |X(\omega_1,\omega_2)|\, d\mathbb{P}_2(\omega_2)\right) d\mathbb{P}_1(\omega_1) = \int |X|\, d\mathbb{P}_1\otimes\mathbb{P}_2 < \infty.$$

   Thanks to propertry (5) following definition of expectation 1.2.1, this implies that

$$\int_{\Omega_2} |X(\omega_1,\omega_2)|\, d\mathbb{P}_2(\omega_2) < \infty \quad \text{for } \mathbb{P}_1\text{-a.e. } \omega_1.$$

   Thus, the function $\omega_1 \mapsto X(\omega_1,\omega_2)$ (which is measurable by proposition 1.4.1) belongs to $L^1(\Omega_2, \mathcal{A}_2, \mathbb{P}_2)$ except on the $\mathcal{A}_1$-measurable set

$$N := \left\{\omega_1 \in \Omega_1 : \int_{\Omega_2} |X(\omega_1,\omega_2)|\, d\mathbb{P}_2(\omega_2) = \infty\right\},$$

   which has zero $\mathbb{P}_1$-measure. The same argument applies to the function $\omega_2 \mapsto X(\omega_1,\omega_2)$.

2. If $\omega_1 \in N^c$, $\int_{\Omega_2} X(\omega_1,\omega_2)\, d\mathbb{P}_2(\omega_2)$ is well defined since the function is only defined

(thanks to (1)) outside a measurable set of values of ω_1 of zero $\mathbb{P}_1$-measure. We can always agree that that $\int_{\Omega_2} X(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2) = 0$ if $\omega_1 \in N$. Then the function

$$\begin{aligned}\omega_1 \mapsto \int_{\Omega_2} X(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2) &= \mathbb{1}_{N^c}(\omega_1)\int_{\Omega_2} X^+(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2)\\ &\quad - \mathbb{1}_{N^c}(\omega_1)\int_{\Omega_2} X^-(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2).\end{aligned}$$

is $\mathcal{A}_1$-measurable by 1.4.3 (1), and we have also, thanks to part (2) of the same theorem,

$$\begin{aligned}\int_{\Omega_1}\left|\int_{\Omega_2} X(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2)\right| d\mathbb{P}_1(\omega_1) &\le \int_{\Omega_1}\left(\int_{\Omega_2} |X(\omega_1,\omega_2)|\,d\mathbb{P}_2(\omega_2)\right) d\mathbb{P}_1(\omega_1)\\ &= \int |X|\,d\mathbb{P}_1\otimes\mathbb{P}_2\\ &< \infty.\end{aligned}$$

This proves that the function $\omega_1 \mapsto \int_{\Omega_2} X(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2)$ is in $L^1(\Omega_1,\mathcal{A}_1,\mathbb{P}_1)$. The same argument applies to the function $\omega_2 \mapsto \int_{\Omega_1} X(\omega_1,\omega_2)\,d\mathbb{P}_1(\omega_1)$.

3. Theorem 1.4.3 gives the two equalities

$$\begin{aligned}\int_{\Omega_1}\left(\int_{\Omega_2} X^+(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2)\right) d\mathbb{P}_1(\omega_1) &= \int_{\Omega_1\times\Omega_2} X^+\,d\mathbb{P}_1\otimes\mathbb{P}_2,\\ \int_{\Omega_1}\left(\int_{\Omega_2} X^-(\omega_1,\omega_2)\,d\mathbb{P}_2(\omega_2)\right) d\mathbb{P}_1(\omega_1) &= \int_{\Omega_1\times\Omega_2} X^-\,d\mathbb{P}_1\otimes\mathbb{P}_2.\end{aligned}$$

We can replace the integral over Ω_1 by an integral over N^c in the left-hand sides of these formulas. Then we just have to subtract the second equality from the first one (using part (2)), in order to arrive at the first equality in (3), and the second equality is derived in a similar manner.

■

1.4.2 Convolution

Throughout this section, we consider the measure space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$. Let f and g be two real measurable functions on $\mathbb{R}^d$. For $x \in \mathbb{R}^d$, the convolution

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x-y)g(y)\,dy$$

is well defined provided

$$\int_{\mathbb{R}^d} |f(x-y)g(y)|\,dy < \infty.$$

In that case, the fact that Lebesgue measure on $\mathbb{R}^d$ is invariant under translations and under the symmetry $y \mapsto -y$ shows that $(g * f)(x)$ is also well defined and $(g * f)(x) = (f * g)(x)$.

Proposition 1.4.5

Let $f \in L^1(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ and $g \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$, where $p \in [1, \infty)$. Then, for λ a.e. $x \in \mathbb{R}^d$ [a], the convolution $(f * g)(x)$ is well defined. Moreover $f * g \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ and :

$$\|f * g\|_p \leq \|f\|_1 \|g\|_p$$

a. for all x except a set of null measure

Proof for Proposition.

Again, we may agree that, on the set of zero Lebesgue measure where $f * g$ is not well defined, we take $f * g(x) = 0$.

We leave aside the case $p = \infty$, which is easier (and where stronger results are proved in Proposition 1.4.6 below). We assume that $\|f\|_1 > 0$, since otherwise the results are trivial. Since the measure with density $\|f\|_1^{-1}|f(x)|$ with respect to λ is a probability measure, Jensen's inequality gives, for every $x \in \mathbb{R}^d$,

$$\begin{aligned}\left(\int_{\mathbb{R}^d} |g(x-y)||f(y)|\, dy\right)^p &= \|f\|_1^p \left(\int_{\mathbb{R}^d} |g(x-y)| \frac{|f(y)|}{\|f\|_1} dy\right)^p \\ &\leq \|f\|_1^{p-1} \int_{\mathbb{R}^d} |g(x-y)|^p |f(y)|\, dy.\end{aligned}$$

Using Fubini-Tonelli Theorem 1.4.3, we have then :

$$\begin{aligned}\int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(x-y)||f(y)|\, dy\right)^p dx &\leq \|f\|_1^{p-1} \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(x-y)|^p |f(y)|\, dy\right) dx \\ &= \|f\|_1^{p-1} \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(y)|^p |f(x-y)|\, dy\right) dx \\ &= \|f\|_1^{p-1} \int_{\mathbb{R}^d} |g(y)|^p \left(\int_{\mathbb{R}^d} |f(x-y)| dx\right) dy \\ &= \|f\|_1^p \|g\|_p^p < \infty,\end{aligned}$$

which shows that

$$\int_{\mathbb{R}^d} |f(x-t)||g(t)| dt < \infty, \qquad \lambda(dx) \text{ a.e.}$$

and gives the first assertion. The second one also follows since the previous calculation gives

$$\int_{\mathbb{R}^d} |f * g(x)|^p dx \leq \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |g(x-y)||f(y)|\, dy\right)^p dx \leq \|f\|_1^p \|g\|_p^p.$$

The next proposition gives another important case where the convolution $f * g$ is well defined and enjoys nice continuity properties.

Proposition 1.4.6

Let $p \in [1, \infty)$ and $q \in (1, \infty)$ be conjugate exponents $\left(\frac{1}{p} + \frac{1}{q} = 1\right)$. Let $f \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ and $g \in L^q(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$. Then, the convolution $f * g(x)$ is well defined and $|f * g(x)| \leq \|f\|_p \|g\|_q$, for every $x \in \mathbb{R}^d$. Furthermore, the function $f * g$ is uniformly continuous on $\mathbb{R}^d$.

Proof for Proposition.

By the Hölder inequality,

$$\int_{\mathbb{R}^d} |f(x-y)g(y)|dy \leq \left(\int |f(x-y)|^p dy\right)^{1/p} \|g\|_q = \|f\|_p \|g\|_q.$$

This gives the first assertion and also proves that $f * g$ is bounded above by $\|f\|_p \|g\|_q$. To get the property of uniform continuity, we use the Lemma 1.4.7 it is easy to complete the proof of the proposition. Indeed, for every $x, x' \in \mathbb{R}^d$,

$$\begin{aligned} |f * g(x) - f * g(x')| &\leq \int |f(x-y) - f(x'-y)||g(y)|dy \\ &\leq \|g\|_q \left(\int |f(x-y) - f(x'-y)|^p dy\right)^{1/p} \\ &= \|g\|_q \|f \circ \sigma_{-x} - f \circ \sigma_{-x'}\|_p \end{aligned}$$

and we use Lemma 1.4.7 to say that $\|f \circ \sigma_{-x} - f \circ \sigma_{-x'}\|_p$ tends to 0 when $x - x' \to 0$.

Lemma 1.4.7: Continuity of translation

Set $\sigma_x(y) = y - x$ for every $x, y \in \mathbb{R}^d$. Let $p \in [1, \infty)$ and $f \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$. Then the mapping $x \mapsto f \circ \sigma_x$ is uniformly continuous from $\mathbb{R}^d$ into $L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$.

Proof for Lemma

The proof of this lemma is omitted, but could be found in the book [3, lem 5.7]

Approximations of the Dirac Measure

Definition 1.4.8: Approximation of the Dirac Measure

Let us say that a sequence $(\varphi_n)_{n \in \mathbb{N}}$ of nonnegative continuous functions on $\mathbb{R}^d$ is an approximation of the Dirac measure δ_0 if:

1. For every $n \in \mathbb{N}$,
$$\int_{\mathbb{R}^d} \varphi_n(x)\, dx = 1.$$

2. For every $\delta > 0$,
$$\lim_{n\to\infty} \int_{\{|x|>\delta\}} \varphi_n(x)\, dx = 0.$$

It is easy to construct approximations of δ_0. If φ is any nonnegative continuous function on $\mathbb{R}^d$ such that $\int \varphi(x)\, dx = 1$, we can set $\varphi_n(x) = n^d \varphi(nx)$ for every $x \in \mathbb{R}^d$ and every $n \in \mathbb{N}$. Note that, if we start from a function φ with compact support, all functions φ_n will vanish outside the same closed ball.

Proposition 1.4.9

Let $(\varphi_n)_{n\in\mathbb{N}}$ be an approximation of the Dirac measure δ_0.

1. For any bounded continuous function $f : \mathbb{R}^d \longrightarrow \mathbb{R}$, the sequence $\varphi_n * f$ converges to f as $n \to \infty$, uniformly on every compact subset of $\mathbb{R}^d$.
2. Let $f \in L^p(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$, where $p \in [1, \infty)$. Then $\varphi_n * f \to f$ in L^p.

Proof for Proposition.

1. First note that $\varphi_n * f$ is well defined since φ_n is integrable and f is bounded. Then, we have, for every $\delta > 0$,
$$\begin{aligned}\varphi_n * f(x) - f(x) = &\int_{|y|\le\delta} (f(x-y) - f(x))\varphi_n(y)\, dy \\ &+ \int_{|y|>\delta} (f(x-y) - f(x))\varphi_n(y)\, dy. \qquad \text{(i)}\end{aligned}$$
Fix $\varepsilon > 0$ and a compact subset H of $\mathbb{R}^d$. Since f is uniformly continuous on any compact subset of $\mathbb{R}^d$, we can choose $\delta > 0$ such that $|f(x-y) - f(x)| \le \varepsilon$ for every $x \in H$ and $y \in \mathbb{R}^d$ such that $|y| \le \delta$. Recalling (1) in Definition 1.4.8, we get that the first term in the right-hand side of (i) is smaller than ε in absolute value, for every $x \in H$ and $n \in \mathbb{N}$. Furthermore, (2) in Definition 1.4.8 and the boundedness of f show that the second term in the right-hand side of (i) tends to 0 when $n \to \infty$, uniformly when x varies in $\mathbb{R}^d$. It follows that we have $|\varphi_n * f(x) - f(x)| < 2\varepsilon$ for every $x \in H$, as soon as n is sufficiently large.
2. The fact that $\varphi_n * f$ is well defined a.e. and belongs to L^p follows from Proposition 1.4.5 since φ_n is integrable. We then observe that
$$\begin{aligned}\int |\varphi_n * f(x) - f(x)|^p dx &\le \int \left(\int \varphi_n(y) |f(x-y) - f(x)|\, dy \right)^p dx \\ &\le \int \left(\int \varphi_n(y) |f(x-y) - f(x)|^p dy \right) dx \\ &= \int \left(\int \varphi_n(y) |f(x-y) - f(x)|^p dx \right) dy \\ &= \int \varphi_n(y) \left(\int |f(x-y) - f(x)|^p dx \right) dy\end{aligned}$$
where the second inequality is a consequence of Jensen's inequality (note that

$\varphi_n(y)dy$ is a probability measure), and the next equality uses Fubini-Tonelli Theorem 1.4.3. For every $y \in \mathbb{R}^d$, set

$$h(y) = \int |f(x-y) - f(x)|^p dx = \|f \circ \sigma_y - f\|_p^p,$$

with the notation of Lemma 1.4.7. The function h is bounded and $h(y) \to 0$ as $y \to 0$ by Lemma 1.4.7. It then immediately follows from (1) and (2) in Definition 1.4.8 that

$$\lim_{n\to\infty} \int \varphi_n(y)h(y)\, dy = 0.$$

This completes the proof.

Application In dimension $d = 1$, we can take

$$\varphi_n(x) = c_n(1-x^2)^n \mathbb{1}_{\{|x|\leq 1\}}$$

where the constant c_n is chosen so that $\int \varphi_n(x)dx = 1$. Then let $[a, b]$ be a compact subinterval of $(0, 1)$, and let f be a continuous function on $[a, b]$. It is easy to extend f to a continuous function on $\mathbb{R}$ that vanishes outside $[0, 1]$ (for instance, take f to be affine on both intervals $[0, a]$ and $[b, 1]$). Then the preceding proposition shows that

$$\varphi_n * f(x) = c_n \int (1-(x-y)^2)^n \mathbb{1}_{\{|x-y|\leq 1\}} f(y)\, dy \longrightarrow f(x)$$

uniformly on $[a, b]$. If $x \in [a, b]$, we can remove the indicator function $\mathbb{1}_{\{|x-y|\leq 1\}}$ in the last display, since the conditions $x \in [a, b]$ and $|x - y| > 1$ force $f(y) = 0$. It now follows that f is a uniform limit of polynomials on the interval $[a, b]$ (this is a special case of the Stone-Weierstrass theorem).

1.4.3 Independent Events

Definition 1.4.10: Independence of events

1. Two events A and B are said to be *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

2. Let $A_1, \ldots, A_n$ be events. We say that the events $A_1, \ldots, A_n$ are *independent* if, for every nonempty subset $\{i_1, \ldots, i_p\} \subset \{1, \ldots, n\}$, we have

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_p}) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_p}).$$

remark: In particular, for $n \geq 3$, pairwise independence does not imply independence.

Proposition 1.4.11

The events $A_1, \ldots, A_n$ are independent if and only if

$$\mathbb{P}(B_1 \cap \cdots \cap B_n) = \mathbb{P}(B_1) \cdots \mathbb{P}(B_n)$$

for every choice of events $B_i \in \sigma(A_i) = \{\varnothing, A_i, A_i^c, \Omega\}$, $i = 1, \ldots, n$.

Proof for Proposition.

Left exercise for the reader.

1.4.4 Independence for σ-fields and Random Variables

We say that $\mathcal{G}$ is a *sub-σ-field* of $\mathcal{A}$ if $\mathcal{G}$ is a σ-field and $\mathcal{G} \subset \mathcal{A}$. Roughly speaking, a sub-σ-field represents partial information in the probability space, namely the information obtained by knowing which events in $\mathcal{G}$ have occurred. For instance, if X is a random variable, the σ-field $\sigma(X)$ corresponds to the information provided by the value of X.

This suggests that the most general notion of independence is that of independent sub-σ-fields : informally, two sub-σ-fields $\mathcal{G}$ and $\mathcal{G}'$ are independent if knowing which events of $\mathcal{G}$ have occurred gives no information about the occurrence of events in $\mathcal{G}'$, and conversely. We make this precise in the following definition.

Definition 1.4.12: Independence of [finite] σ-fields

1. Let $\mathcal{G}_1, \ldots, \mathcal{G}_n$ be sub-σ-fields of $\mathcal{A}$. We say that $\mathcal{G}_1, \ldots, \mathcal{G}_n$ are *independent* if, for every choice of events

$$A_1 \in \mathcal{G}_1, \ A_2 \in \mathcal{G}_2, \ \ldots, \ A_n \in \mathcal{G}_n,$$

we have

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2) \cdots \mathbb{P}(A_n).$$

2. Let $X_1, \ldots, X_n$ be random variables taking values in the measurable spaces $(E_1, \mathcal{E}_1), \ldots, (E_n, \mathcal{E}_n)$, respectively. We say that the random variables $X_1, \ldots, X_n$ are *independent* if the σ-fields $\sigma(X_1), \ldots, \sigma(X_n)$ are independent.
3. Equivalently, $X_1, \ldots, X_n$ are independent if and only if, for every choice of sets $F_1 \in \mathcal{E}_1, \ldots, F_n \in \mathcal{E}_n$,

$$\mathbb{P}\Big(X_1 \in F_1, \ldots, X_n \in F_n\Big) = \mathbb{P}(X_1 \in F_1) \cdots \mathbb{P}(X_n \in F_n).$$

Firstly, we introduce a suffisant condition for the independence of σ-fields .

Theorem 1.4.13: A suffisant condition for independence of σ-fields

Let $\mathcal{G}_1, \ldots, \mathcal{G}_n$ be sub-σ-fields of $\mathcal{A}$. For each $i \in \{1, \ldots, n\}$, let $\mathcal{C}_i \subset \mathcal{G}_i$ be a class of sets such that:

- $\mathcal{C}_i$ is closed under finite intersections,
- $\Omega \in \mathcal{C}_i$,
- $\sigma(\mathcal{C}_i) = \mathcal{G}_i$.

Assume that

$$\forall C_1 \in \mathcal{C}_1, \ldots, C_n \in \mathcal{C}_n, \qquad \mathbb{P}(C_1 \cap \cdots \cap C_n) = \prod_{i=1}^{n} \mathbb{P}(C_i).$$

Then the σ-fields $\mathcal{G}_1, \ldots, \mathcal{G}_n$ are independent.

Proof for Theorem.

We first fix $C_2 \in \mathcal{C}_2, \ldots, C_n \in \mathcal{C}_n$ and define

$$\mathcal{M}_1 = \left\{ B_1 \in \mathcal{G}_1 : \mathbb{P}(B_1 \cap C_2 \cap \cdots \cap C_n) = \mathbb{P}(B_1) \prod_{i=2}^{n} \mathbb{P}(C_i) \right\}.$$

By assumption, $\mathcal{C}_1 \subset \mathcal{M}_1$. It is straightforward to check that $\mathcal{M}_1$ is a monotone class. By the monotone class theorem,

$$\mathcal{M}_1 \supset \sigma(\mathcal{C}_1) = \mathcal{G}_1.$$

Hence, for all $B_1 \in \mathcal{G}_1$ and all $C_2 \in \mathcal{C}_2, \ldots, C_n \in \mathcal{C}_n$,

$$\mathbb{P}(B_1 \cap C_2 \cap \cdots \cap C_n) = \mathbb{P}(B_1) \prod_{i=2}^{n} \mathbb{P}(C_i).$$

We now fix $B_1 \in \mathcal{G}_1$, $C_3 \in \mathcal{C}_3, \ldots, C_n \in \mathcal{C}_n$ and define

$$\mathcal{M}_2 = \left\{ B_2 \in \mathcal{G}_2 : \mathbb{P}(B_1 \cap B_2 \cap C_3 \cap \cdots \cap C_n) = \mathbb{P}(B_1)\mathbb{P}(B_2) \prod_{i=3}^{n} \mathbb{P}(C_i) \right\}.$$

Again, $\mathcal{C}_2 \subset \mathcal{M}_2$, and $\mathcal{M}_2$ is a monotone class, so $\mathcal{M}_2 = \mathcal{G}_2$.

Proceeding inductively, we obtain that, for all $B_1 \in \mathcal{G}_1, \ldots, B_n \in \mathcal{G}_n$,

$$\mathbb{P}(B_1 \cap \cdots \cap B_n) = \prod_{i=1}^{n} \mathbb{P}(B_i).$$

This is precisely the definition of independence of the σ-fields $\mathcal{G}_1, \ldots, \mathcal{G}_n$. ■

The next theorem provides a fundamental characterization of the independence of n random variables.

Theorem 1.4.14: Characterization of Random Variable Independence

Let $X_1, \dots, X_n$ be random variables taking values in $\mathbb{R}$, the following statements are equivalent:

1. The random variables $X_1, \dots, X_n$ are independent.
2. the law of the random vector $(X_1, \dots, X_n)$ is the product of the respective laws of $X_1, \dots, X_n$, that is,
$$\mathbb{P}_{(X_1,\dots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}.$$
3. For every $a_1, \dots, a_n \in \mathbb{R}$,
$$\mathbb{P}(X_1 \leq a_1, \dots, X_n \leq a_n) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq a_i).$$
4. For every choice of continuous functions $f_1, \dots, f_n : \mathbb{R} \to \mathbb{R}_+$ with compact support,
$$\mathbb{E}\left[\prod_{i=1}^{n} f_i(X_i)\right] = \prod_{i=1}^{n} \mathbb{E}[f_i(X_i)].$$
5. The characteristic function of the random vector $X = (X_1, \dots, X_n)$ factorizes as
$$\Phi_X(\xi_1, \dots, \xi_n) = \prod_{i=1}^{n} \Phi_{X_i}(\xi_i), \qquad (\xi_1, \dots, \xi_n) \in \mathbb{R}^n.$$

Proof for Theorem.

- $(1) \iff (2)$:
  Let $F_i \in \mathcal{E}_i$, for every $i \in \{1, \dots, n\}$. On one hand,
$$\mathbb{P}_{(X_1,\dots,X_n)}(F_1 \times \cdots \times F_n) = \mathbb{P}\Big((X_1 \in F_1) \cap \cdots \cap (X_n \in F_n)\Big).$$
  On the other hand,
$$\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}(F_1 \times \cdots \times F_n) = \prod_{i=1}^{n} \mathbb{P}_{X_i}(F_i) = \prod_{i=1}^{n} \mathbb{P}(X_i \in F_i).$$
  Comparing with the definition of independence, we see that $X_1, \dots, X_n$ are independent if and only if the two probability measures $\mathbb{P}_{(X_1,\dots,X_n)}$ and $\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$ take the same values on boxes $F_1 \times \cdots \times F_n$. This is equivalent to saying that :
$$\mathbb{P}_{(X_1,\dots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n},$$
  since we know that a probability measure on a product space is characterized by its values on boxes (theorem 1.4.2) we conclude the result.

- (2) $\Longleftrightarrow$ (3) :
  Just by taking $F_i =]-\infty, a_i]$ we can prove the (2) $\Longleftarrow$ (3), conversely, Just by taking $C_i = \left\{]-\infty, a] \;\middle|\; a \in \mathbb{R} \right\}$, and apply the theorem 1.4.13 we obtain the independence (1) and therefore (2).
- (1) $\Longleftrightarrow$ (4) :
  We have (4) $\Longrightarrow$ (3) $\Longrightarrow$ (1) by taking $f_i(x) = \mathbb{1}_{x \leq a_i}$. The other implication is imediate from the next proposition.
- (2) $\Longleftrightarrow$ (5) :
  To show this equivalence, we note that the Fourier transform of the product measure $\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$ is

$$\begin{aligned}(\xi_1, \ldots, \xi_n) \longmapsto & \int \exp\left(i \sum_{j=1}^{n} \xi_j x_j \right) \mathbb{P}_{X_1}(dx_1) \cdots \mathbb{P}_{X_n}(dx_n) \\ &= \prod_{j=1}^{n} \int e^{i\xi_j x_j} \, \mathbb{P}_{X_j}(dx_j) \\ &= \prod_{j=1}^{n} \Phi_{X_j}(\xi_j).\end{aligned}$$

  using the Fubini theorem in the first equality. Hence the property (5) is equivalent to the fact that the Fourier transform of $\mathbb{P}_{(X_1,\ldots,X_n)}$ coincides with the Fourier transform of $\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$. By pervious property, this holds if and only if

$$\mathbb{P}_{(X_1,\ldots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n},$$

From this theorem follow this interesting results :

Proposition 1.4.15

Let $X_1, \ldots, X_n$ be independent real-valued random variables.

1. Let $f_1, \ldots, f_n$ be non negative measurable functions we have :

$$\mathbb{E}\left[\prod_{i=1}^{n} f_i(X_i) \right] = \prod_{i=1}^{n} \mathbb{E}[f_i(X_i)] \,,$$

2. Let $f_1, \ldots, f_n$ be measurable functions such that

$$\mathbb{E}[|f_i(X_i)|] < \infty \quad \text{for every } i \in \{1, \ldots, n\}.$$

   Then

$$\mathbb{E}\left[\prod_{i=1}^{n} f_i(X_i) \right] = \prod_{i=1}^{n} \mathbb{E}[f_i(X_i)],$$

   and the expectation on the left-hand side is well defined. The same statement

holds for complex-valued functions f_i under the same integrability assumption.

3. As a special case of (1), if $X_1, \dots, X_n$ are independent random variables in $\mathcal{L}^1$, then $X_1 \cdots X_n \in \mathcal{L}^1$ and

$$\mathbb{E}[X_1 \cdots X_n] = \prod_{i=1}^{n} \mathbb{E}[X_i].$$

In general, the product of random variables in $\mathcal{L}^1$ need not belong to $\mathcal{L}^1$ without independence.

4. If X_1 and X_2 are independent real-valued random variables in L^2, then

$$\operatorname{cov}(X_1, X_2) = 0.$$

5. Suppose that, for every $i \in \{1, \dots, n\}$, the random variable X_i admits a density p_i with respect to Lebesgue measure and that $X_1, \dots, X_n$ are independent. Then the random vector $(X_1, \dots, X_n)$ admits a density p given by

$$p(x_1, \dots, x_n) = \prod_{i=1}^{n} p_i(x_i).$$

6. Conversely, suppose that $(X_1, \dots, X_n)$ admits a density p of the form

$$p(x_1, \dots, x_n) = \prod_{i=1}^{n} q_i(x_i),$$

where $q_1, \dots, q_n$ are nonnegative measurable functions on $\mathbb{R}$. Then $X_1, \dots, X_n$ are independent, and for every $i \in \{1, \dots, n\}$, X_i admits a density p_i of the form $p_i = C_i q_i$, where $C_i > 0$ is a normalization constant.

Proof for Proposition.

1. Using the Fubini-Tonelli theorem 1.4.3 for the case of two independent variables :

$$\begin{aligned}
\mathbb{E}\Big(f_1(X_1)f_2(X_2)\Big) &= \int_{\mathbb{R}^2} f_1(x_1)f_2(x_2), d\mathbb{P}_{(X_1,X_2)}(x_1, x_2) \\
&= \int_{\mathbb{R}^2} f_1(x_1)f_2(x_2), d(\mathbb{P}_{X_1} \otimes \mathbb{P}_{X_2})(x_1, x_2) \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} f_1(x_1)f_2(x_2), d\mathbb{P}_{X_2}(x_2), d\mathbb{P}_{X_1}(x_1) \\
&= \int_{\mathbb{R}} f_1(x_1), d\mathbb{P}_{X_1}(x_1) \int_{\mathbb{R}} f_2(x_2), d\mathbb{P}_{X_2}(x_2) \\
&= \mathbb{E}\Big(f_1(X_1)\Big) \times \mathbb{E}\Big(f_2(X_2)\Big).
\end{aligned}$$

The general case can be achived by a simple reccurence.

2. Similar to (1), but using the Fubini-Lebesgue theorem 1.4.4 and ensuring the

$f_i(X_i) \in \mathcal{L}^1$ which is the condition :

$$\mathbb{E}[|f_i(X_i)|] < \infty \quad \text{for every } i \in \{1, \dots, n\}.$$

3. Clear conclusion from (1).
4. This follows from the preceding remarks since :

$$\mathrm{Cov}(X_1, X_2) = \mathbb{E}(X_1 X_2) - \mathbb{E}(X_1)\mathbb{E}(X_2) = 0$$

5. This is an immediate consequence of the caracterisation of independece $\mathbb{P}_{(X_1,\dots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}$. Indeed, by the Fubini theorem, under the assumption

$$\mathbb{P}_{X_i}(dx_i) = p_i(x_i)\, dx_i \quad \text{for } i \in \{1, \dots, n\},$$

we have

$$\mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n}(dx_1 \cdots dx_n) = \left(\prod_{i=1}^n p_i(x_i) \right) dx_1 \cdots dx_n.$$

6. We first observe, again thanks to the Fubini theorem, that

$$\prod_{i=1}^n \left(\int q_i(x)\, dx \right) = \int_{\mathbb{R}^n} p(x_1, \dots, x_n)\, dx_1 \cdots dx_n = 1.$$

In particular, the quantity

$$K_i = \int q_i(x)\, dx$$

is positive and finite for every $i \in \{1, \dots, n\}$. Then, by Proposition 8.6, the density of X_i is

$$\begin{aligned} p_i(x_i) &= \int_{\mathbb{R}^{n-1}} p(x_1, \dots, x_n)\, dx_1 \cdots dx_{i-1}\, dx_{i+1} \cdots dx_n \\ &= \left(\prod_{j \neq i} K_j \right) q_i(x_i) = \frac{1}{K_i}\, q_i(x_i). \end{aligned}$$

Thus $p_i = C_i q_i$ with $C_i = 1/K_i$. This also allows us to rewrite the density of $(X_1, \dots, X_n)$ as

$$p(x_1, \dots, x_n) = \prod_{i=1}^n q_i(x_i) = \prod_{i=1}^n p_i(x_i),$$

and therefore

$$\mathbb{P}_{(X_1,\dots,X_n)} = \mathbb{P}_{X_1} \otimes \cdots \otimes \mathbb{P}_{X_n},$$

which shows that $X_1, \dots, X_n$ are independent.

■

Definition 1.4.16: Independence of [infinite] σ-fields

Let $(\mathcal{G}_i)_{i\in I}$ be an arbitrary collection of sub-σ-fields of $\mathcal{A}$. We say that the σ-fields $\mathcal{G}_i$, $i \in I$, are *independent* if, for every finite subset $\{i_1, \dots, i_p\} \subset I$, the σ-fields $\mathcal{G}_{i_1}, \dots, \mathcal{G}_{i_p}$ are independent.
If $(X_i)_{i\in I}$ is an arbitrary collection of random variables, we say that the random variables X_i, $i \in I$, are *independent* if the σ-fields $\sigma(X_i)$, $i \in I$, are independent.
Similarly, for an arbitrary collection $(A_i)_{i\in I}$ of events, we say that the events A_i, $i \in I$, are independent if the σ-fields $\sigma(A_i)$, $i \in I$, are independent.

Proposition 1.4.17

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables. Then, for every $p \in \mathbb{N}$, the σ-fields

$$\mathcal{G}_1 = \sigma(X_1, \dots, X_p), \qquad \mathcal{G}_2 = \sigma(X_{p+1}, X_{p+2}, \dots)$$

are independent.

Proof for Proposition.

We apply the suffisant condition of σ-fields independence given in 1.4.13. Set

$$\mathcal{C}_1 = \sigma(X_1, \dots, X_p) = \mathcal{G}_1$$

and

$$\mathcal{C}_2 = \bigcup_{k=p+1}^{\infty} \sigma(X_{p+1}, \dots, X_k) \subset \mathcal{G}_2.$$

The class $\mathcal{C}_2$ is closed under finite intersections, contains Ω, and satisfies $\sigma(\mathcal{C}_2) = \mathcal{G}_2$.

By independence of the random variables $(X_n)_{n\in\mathbb{N}}$, and by the grouping-by-blocks principle, we have for every $C_1 \in \mathcal{C}_1$ and $C_2 \in \mathcal{C}_2$,

$$\mathbb{P}(C_1 \cap C_2) = \mathbb{P}(C_1)\mathbb{P}(C_2).$$

Therefore, the assumptions of theorem 1.4.13 are satisfied, and it follows that the σ-fields $\mathcal{G}_1$ and $\mathcal{G}_2$ are independent. ∎

1.4.5 Sums of Independent Random Variables

Sums of independent random variables play a central role in probability theory. Their distributional and analytic properties are conveniently described using the notion of convolution of probability measures. Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $\mathbb{R}^d$. The *convolution* $\mathbb{P} * \mathbb{Q}$ is defined as the pushforward of the product measure $\mathbb{P} \otimes \mathbb{Q}$ under the mapping $(x, y) \mapsto x + y$. Equivalently, for every nonnegative measurable function φ on $\mathbb{R}^d$,

$$\int_{\mathbb{R}^d} \varphi(z)\,(\mathbb{P} * \mathbb{Q})(dz) = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \varphi(x + y)\,d\mathbb{P}(x)\,d\mathbb{Q}(y).$$

If $\mathbb{P}$ and $\mathbb{Q}$ admit densities f and g with respect to Lebesgue measure, then $\mathbb{P} * \mathbb{Q}$ admits the density $f * g$ given by

$$(f * g)(z) = \int_{\mathbb{R}^d} f(x)\, g(z - x)\, dx.$$

Proposition 1.4.18

Let X and Y be two independent random variables with values in $\mathbb{R}^d$.

1. The law of $X + Y$ is the convolution of the laws of X and Y:

$$\mathbb{P}_{X+Y} = \mathbb{P}_X * \mathbb{P}_Y.$$

   In particular, if X and Y admit densities p_X and p_Y, then $X + Y$ admits the density $p_X * p_Y$.

2. The characteristic function of $X + Y$ satisfies

$$\Phi_{X+Y}(\xi) = \Phi_X(\xi)\, \Phi_Y(\xi), \qquad \xi \in \mathbb{R}^d.$$

3. If $\mathbb{E}[\|X\|^2] < \infty$ and $\mathbb{E}[\|Y\|^2] < \infty$, then the covariance matrices satisfy

$$K_{X+Y} = K_X + K_Y.$$

   In particular, if $d = 1$ and $X, Y \in L^2$, then

$$\mathrm{var}(X + Y) = \mathrm{var}(X) + \mathrm{var}(Y).$$

Proof for Proposition.

1. We know that $\mathbb{P}_{(X,Y)} = \mathbb{P}_X \otimes \mathbb{P}_Y$, and thus, for every nonnegative measurable function φ on $\mathbb{R}^d$,

$$\mathbb{E}[\varphi(X + Y)] = \int\int \varphi(x + y)\, \mathbb{P}_X(dx)\, \mathbb{P}_Y(dy) = \int \varphi(z)\, (\mathbb{P}_X * \mathbb{P}_Y)(dz),$$

   by the definition of $\mathbb{P}_X * \mathbb{P}_Y$. The case with density follows from the remarks before the proposition.

2. This follows from the equality

$$\mathbb{P}_{X+Y} = \mathbb{P}_X * \mathbb{P}_Y$$

   and the last observation before the proposition.

3. If $X = (X_1, \ldots, X_d)$ and $Y = (Y_1, \ldots, Y_d)$, proposition 1.4.15 implies that

$$\mathrm{cov}(X_i, Y_j) = 0 \quad \text{for every } i, j \in \{1, \ldots, d\}.$$

Consequently, using bilinearity,

$$\operatorname{cov}(X_i + Y_i,\ X_j + Y_j) = \operatorname{cov}(X_i, X_j) + \operatorname{cov}(Y_i, Y_j),$$

which gives

$$K_{X+Y} = K_X + K_Y.$$

1.4.6 The Borel–Cantelli Lemma

Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events. Recall that

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k.$$

The event $\limsup_{n\to\infty} A_n$ consists of those $\omega \in \Omega$ that belong to infinitely many of the events A_n. Note that Lemma 1.7 implies

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) \geq \limsup_{n\to\infty} \mathbb{P}(A_n).$$

Theorem 1.4.19: Borel–Cantelli Lemma

Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events.

1. If $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) < \infty$, then

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 0.$$

   Equivalently,

$$\Big\{n \in \mathbb{N} :\ \omega \in A_n\Big\} \quad \text{is finite almost surely.}$$

2. If $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) = \infty$ and the events $(A_n)_{n\in\mathbb{N}}$ are independent, then

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 1.$$

   Equivalently,

$$\Big\{n \in \mathbb{N} :\ \omega \in A_n\Big\} \quad \text{is infinite almost surely.}$$

Proof for Theorem.

1. If $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) < \infty$, then

$$\mathbb{E}\left[\sum_{n\in\mathbb{N}} \mathbb{1}_{A_n}\right] = \sum_{n\in\mathbb{N}} \mathbb{P}(A_n) < \infty.$$

It follows that

$$\sum_{n\in\mathbb{N}} \mathbb{1}_{A_n} < \infty \quad \text{almost surely,}$$

which means that ω belongs to only finitely many of the events A_n. Hence,

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 0.$$

2. Assume now that $\sum_{n\in\mathbb{N}} \mathbb{P}(A_n) = \infty$ and that the events A_n are independent. Fix $n_0 \in \mathbb{N}$. For $n \geq n_0$, we have

$$\mathbb{P}\left(\bigcap_{k=n_0}^{n} A_k^c\right) = \prod_{k=n_0}^{n} \mathbb{P}(A_k^c) = \prod_{k=n_0}^{n} \Big(1 - \mathbb{P}(A_k)\Big).$$

Since the series $\sum_{k\in\mathbb{N}} \mathbb{P}(A_k)$ diverges, the right-hand side converges to 0 as $n \to \infty$. Consequently,

$$\mathbb{P}\left(\bigcap_{k=n_0}^{\infty} A_k^c\right) = 0.$$

As this holds for every $n_0 \in \mathbb{N}$, we obtain

$$\mathbb{P}\left(\bigcup_{n_0=1}^{\infty} \bigcap_{k=n_0}^{\infty} A_k^c\right) = 0.$$

Taking complements yields

$$\mathbb{P}\left(\bigcap_{n_0=1}^{\infty} \bigcup_{k=n_0}^{\infty} A_k\right) = 1,$$

that is,

$$\mathbb{P}\Big(\limsup_{n\to\infty} A_n\Big) = 1.$$

■

1.5 Conditioning

1.5.1 Definition of conditional expectaiton

Discret case

As in the previous chapters, we consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. If $B \in \mathcal{A}$ is an event of positive probability, we can define a new probability measure on $(\Omega, \mathcal{A})$ by setting, for every $A \in \mathcal{A}$,

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This probability measure $A \mapsto \mathbb{P}(A \mid B)$ is called **the conditional probability given** B. Similarly, for every nonnegative random variable X, or for $X \in L^1(\Omega, \mathcal{A}, \mathbb{P})$, the

conditional expectation of X given B is defined by

$$\mathbb{E}[X \mid B] := \frac{\mathbb{E}[X\mathbb{1}_B]}{\mathbb{P}(B)}.$$

One verifies that $\mathbb{E}[X \mid B]$ is also the expected value of X under $\mathbb{P}(\cdot \mid B)$. We can interpret $\mathbb{E}[X \mid B]$ as the mean value of X knowing that the event B occurs.

Let us now define the conditional expectation knowing a discrete random variable. We consider a random variable Y taking values in a countable space E equipped as usual with the σ-field of all its subsets. Let

$$E' = \{y \in E : \mathbb{P}(Y = y) > 0\}.$$

If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P})$, we may consider, for every $y \in E'$,

$$\mathbb{E}[X \mid Y = y] = \frac{\mathbb{E}[X\mathbb{1}_{\{Y=y\}}]}{\mathbb{P}(Y = y)},$$

as a special case of the conditional expectation $\mathbb{E}[X \mid B]$ when $\mathbb{P}(B) > 0$.

The conditional expectation of X knowing Y is the real random variable

$$\mathbb{E}[X \mid Y] = \varphi(Y),$$

where the function $\varphi : E \to \mathbb{R}$ is given by

$$\varphi(y) = \begin{cases} \mathbb{E}[X \mid Y = y], & \text{if } y \in E', \\ 0, & \text{if } y \in E \setminus E'. \end{cases}$$

General case

Now we turn to the definition of conditional expectation with respect to a sub-σ-field . For that purpose we need the next theorem that provides the existance and uniqueness of conditional expectation.

Theorem 1.5.1: Existance of Conditional Expectation

Let $\mathcal{B}$ be a sub-σ-field of $\mathcal{A}$, and let $X \in L^1(\Omega, \mathcal{A}, \mathbb{P})$. There exists a unique element of $L^1(\Omega, \mathcal{B}, \mathbb{P})$, which is denoted by $\mathbb{E}[X \mid \mathcal{B}]$, such that

$$\forall B \in \mathcal{B}, \quad \mathbb{E}[X\mathbb{1}_B] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]\mathbb{1}_B]. \tag{i}$$

We have more generally, for every bounded $\mathcal{B}$-measurable real random variable Z,

$$\mathbb{E}[XZ] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]Z]. \tag{ii}$$

If $X \geq 0$, we have $\mathbb{E}[X \mid \mathcal{B}] \geq 0$ a.s.

Proof for Theorem.

Let us start by proving uniqueness. Let X' and X'' be two random variables in $L^1(\Omega, \mathcal{B}, \mathbb{P})$ such that

$$\forall B \in \mathcal{B}, \quad \mathbb{E}[X'\mathbb{1}_B] = \mathbb{E}[X\mathbb{1}_B] = \mathbb{E}[X''\mathbb{1}_B].$$

Taking $B = \{X' > X''\}$ (which is in $\mathcal{B}$ since both X' and X'' are $\mathcal{B}$-measurable), we get

$$\mathbb{E}[(X' - X'')\mathbb{1}_{\{X'>X''\}}] = 0,$$

which implies that $X' \leq X''$ a.s., and we have similarly $X' \geq X''$ a.s. Thus $X' = X''$ a.s., which means that X' and X'' are equal as elements of $L^1(\Omega, \mathcal{B}, \mathbb{P})$.

Let us now turn to existence. We first assume that $X \geq 0$ and $\mathbb{E}(X) \neq 0$, then we let $\mathbb{Q}$ be the finite measure on $(\Omega, \mathcal{B})$ defined by

$$\forall B \in \mathcal{B}, \quad \mathbb{Q}(B) := \frac{1}{\mathbb{E}(X)}\mathbb{E}[X\mathbb{1}_B].$$

Let us emphasize that we define $\mathbb{Q}(B)$ only for $B \in \mathcal{B}$. We may also view $\mathbb{P}$ as a probability measure on $(\Omega, \mathcal{B})$, by restricting the mapping $B \mapsto \mathbb{P}(B)$ to $B \in \mathcal{B}$, and it is immediate that $\mathbb{Q} \ll \mathbb{P}$. The Radon–Nikodym theorem 1.3.11 applied to the probability measures $\mathbb{P}$ and $\mathbb{Q}$ on the measurable space $(\Omega, \mathcal{B})$ yields the existence of a nonnegative $\mathcal{B}$-measurable random variable $\tilde{X}$ such that

$$\forall B \in \mathcal{B}, \quad \mathbb{E}[X\mathbb{1}_B] = \mathbb{Q}(B) = \mathbb{E}[\tilde{X}\mathbb{1}_B].$$

Taking $B = \Omega$, we have $\mathbb{E}[\tilde{X}] = \mathbb{E}[X] < \infty$, and thus $\tilde{X} \in L^1(\Omega, \mathcal{B}, \mathbb{P})$. The random variable $\mathbb{E}[X \mid \mathcal{B}] = \tilde{X}$ satisfies (1). (for the case of $\mathbb{E}(X) = 0$ we have $X = 0$ a.s. and we can take $\tilde{X} = 0$)

When X is of arbitrary sign, we just have to take

$$\mathbb{E}[X \mid \mathcal{B}] = \mathbb{E}[X^+ \mid \mathcal{B}] - \mathbb{E}[X^- \mid \mathcal{B}],$$

and it is clear that (1) also holds in that case.

Finally, to see that (1) implies (2), we rely on the usual measure-theoretic arguments. (2) follows from (1) when Z is a simple random variable (taking only finitely many values), and in the general case lemma 1.1.11 allows us to write Z as the pointwise limit of a sequence $(Z_n)_{n\in\mathbb{N}}$ of simple $\mathcal{B}$-measurable random variables that are uniformly bounded by the same constant K (such that $|Z| \leq K$) and the dominated convergence theorem yields the desired result. ∎

The crucial point is the fact that $\mathbb{E}[X \mid \mathcal{B}]$ is $\mathcal{B}$-measurable. Either of properties (1) and (2) characterizes the conditional expectation $\mathbb{E}[X \mid \mathcal{B}]$ among random variables of $L^1(\Omega, \mathcal{B}, \mathbb{P})$. In what follows, we will refer to (1) or (2) as the *characteristic property* of $\mathbb{E}[X \mid \mathcal{B}]$.

In the special case where $\mathcal{B} = \sigma(Y)$ is generated by a random variable Y, we will write

indifferently

$$\mathbb{E}[X \mid \mathcal{B}] = \mathbb{E}[X \mid \sigma(Y)] = \mathbb{E}[X \mid Y].$$

We now turn to the definition of $\mathbb{E}[X \mid \mathcal{B}]$ for a nonnegative random variable X which allows X to take the value $+\infty$.

Theorem 1.5.2: Existence of Conditional Expectation for nonnegative random variables

Let X be a random variable with values in $[0, \infty]$. There exists a $\mathcal{B}$-measurable random variable with values in $[0, \infty]$, which is denoted by $\mathbb{E}[X \mid \mathcal{B}]$ and is such that, for every nonnegative $\mathcal{B}$-measurable random variable Z,

$$\mathbb{E}[XZ] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]Z]. \tag{iii}$$

Furthermore $\mathbb{E}[X \mid \mathcal{B}]$ is unique up to a $\mathcal{B}$-measurable set of probability zero.

Proof for Theorem.

We define $\mathbb{E}[X \mid \mathcal{B}]$ by setting

$$\mathbb{E}[X \mid \mathcal{B}] := \lim_{n \uparrow \infty} \mathbb{E}[X \wedge n \mid \mathcal{B}] \quad \text{a.s.}$$

This definition makes sense because, for every $n \in \mathbb{N}$, $X \wedge n$ is bounded hence integrable and thus $\mathbb{E}[X \wedge n \mid \mathcal{B}]$ is well defined by the previous section. Furthermore, the fact that the sequence $(\mathbb{E}[X \wedge n \mid \mathcal{B}])_{n \in \mathbb{N}}$ is (a.s.) increasing follows from the fact that the conditional expectation of a.s. non-negative random variable is also an a.s. non-negative random variable. Then, if Z is nonnegative and $\mathcal{B}$-measurable, the monotone convergence theorem implies that

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]Z] = \lim_{n \to \infty} \mathbb{E}[\mathbb{E}[X \wedge n \mid \mathcal{B}]Z\mathbb{1}_{\{Z<\infty\}}] = \lim_{n \to \infty} \mathbb{E}[(X \wedge n)Z\mathbb{1}_{\{Z<\infty\}}] = \mathbb{E}[XZ].$$

It remains to establish uniqueness. Let X' and X'' be two $\mathcal{B}$-measurable random variables with values in $[0, \infty]$, such that

$$\mathbb{E}[X'Z] = \mathbb{E}[X''Z]$$

for every nonnegative $\mathcal{B}$-measurable random variable Z. Let us fix two nonnegative rationals $a < b$, and take

$$Z = \mathbb{1}_{\{X' \leq a < b \leq X''\}}.$$

It follows that

$$a\,\mathbb{P}(X' \leq a < b \leq X'') \leq \mathbb{E}[\mathbb{1}_{\{X' \leq a < b \leq X''\}}X'] = \mathbb{E}[\mathbb{1}_{\{X' \leq a < b \leq X''\}}X''] \geq b\,\mathbb{P}(X' \leq a < b \leq X''),$$

which is only possible if $\mathbb{P}(X' \leq a < b \leq X'') = 0$. Hence,

$$\mathbb{P}\left(\bigcup_{\substack{a,b\in\mathbb{Q}_+\\ a<b}} \{X' \leq a < b \leq X''\}\right) = 0,$$

which implies $X' \geq X''$ a.s., and interchanging the roles of X' and X'' also gives $X'' \geq X'$ a.s. ■

Remark: The uniqueness part of the theorem means that we should view $\mathbb{E}[X \mid \mathcal{B}]$ as an equivalence class of $\mathcal{B}$-measurable random variables that are equal a.s. But of course it will be more convenient to speak about "the random variable" $\mathbb{E}[X \mid \mathcal{B}]$.

In the case where X is also integrable, the definition of the preceding theorem is consistent with that of Theorem 1.5.1. Indeed (3) with $Z = 1$ shows that $\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]] = \mathbb{E}[X]$, and in particular $\mathbb{E}[X \mid \mathcal{B}] < \infty$ a.s. so that we can take a representative with values in $[0, \infty)$ and $\mathbb{E}[X \mid \mathcal{B}] \in L^1$. We then just have to note that (1) (in the theorem we are citing) is the special case of (3) where $Z = \mathbb{1}_B$, $B \in \mathcal{B}$.

Similarly as in the case of integrable variables, we will refer to (3) (or to its variant when $Z = \mathbb{1}_B$, $B \in \mathcal{B}$) as the characteristic property of $\mathbb{E}[X \mid \mathcal{B}]$.

1.5.2 Properties of conditional Expectations

We now list the properties of conditional expectaitons :

Proposition 1.5.3

1. For a $X, Y \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ and $\alpha, \beta \in \mathbb{R}$ we have :

$$\mathbb{E}(\alpha X + \beta Y \mid \mathcal{B}) = \alpha\mathbb{E}(X \mid \mathcal{B}) + \beta\mathbb{E}(Y \mid \mathcal{B})$$

2. For a X, Y non negative random variables and $\alpha, \beta \in \mathbb{R}^+$ we have :

$$\mathbb{E}(\alpha X + \beta Y \mid \mathcal{B}) = \alpha\mathbb{E}(X \mid \mathcal{B}) + \beta\mathbb{E}(Y \mid \mathcal{B})$$

3. If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ (or if X is just non negative) and X is $\mathcal{B}$-measurable, then :

$$\mathbb{E}[X \mid \mathcal{B}] = X$$

4. If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ (or if X is just non negative), then :

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]] = \mathbb{E}[X]$$

5. If $X, X' \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$ and $X \geq X'$, then :

$$\mathbb{E}[X \mid \mathcal{B}] \geq \mathbb{E}[X' \mid \mathcal{B}] \quad \text{a.s}$$

6. If $X \in L^1(\Omega, \mathcal{A}, \mathbb{P}))$, then :

$$|\mathbb{E}[X \mid \mathcal{B}]| \leq \mathbb{E}[|X| \mid \mathcal{B}] \quad \text{a.s.}$$

and, consequently,

$$\mathbb{E}[|\mathbb{E}[X \mid \mathcal{B}]|] \leq \mathbb{E}[|X|].$$

Therefore the mapping $X \mapsto \mathbb{E}[X \mid \mathcal{B}]$ is a contraction of $L^1(\Omega, \mathcal{A}, \mathbb{P}))$.

Proof for Proposition.

The proofs for these properties are left exercise for the reader.

Now the properties allowing intereaction between limits and conditional expectations :

Proposition 1.5.4

1. If $(X_n)_{n\in\mathbb{N}}$ is an increasing sequence of nonnegative random variables, and $X = \lim \uparrow X_n$, then

$$\mathbb{E}[X \mid \mathcal{B}] = \lim_{n\to\infty} \uparrow \mathbb{E}[X_n \mid \mathcal{B}], \quad \text{a.s.}$$

As consequence : if $(Y_n)_{n\in\mathbb{N}}$ is a sequence of nonnegative random variables, we have

$$\mathbb{E}\left[\sum_{n\in\mathbb{N}} Y_n \,\middle|\, \mathcal{B}\right] = \sum_{n\in\mathbb{N}} \mathbb{E}[Y_n \mid \mathcal{B}].$$

2. If $(X_n)_{n\in\mathbb{N}}$ is any sequence of nonnegative random variables, then

$$\mathbb{E}[\liminf X_n \mid \mathcal{B}] \leq \liminf \mathbb{E}[X_n \mid \mathcal{B}], \quad \text{a.s.}$$

3. Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of integrable random variables that converges a.s. to X. Assume that there exists a nonnegative random variable Z such that $|X_n| \leq Z$ a.s. for every $n \in \mathbb{N}$, and $\mathbb{E}[Z] < \infty$. Then,

$$\mathbb{E}[X \mid \mathcal{B}] = \lim_{n\to\infty} \mathbb{E}[X_n \mid \mathcal{B}], \quad \text{a.s.}$$

The convergence is also in $\mathcal{L}^1$ that is :

$$\mathbb{E}\left[\left|\mathbb{E}[X \mid \mathcal{B}] - \mathbb{E}[X_n \mid \mathcal{B}]\right|\right] \longrightarrow 0 \quad \textit{as } n \to +\infty$$

Proof for Proposition.

1. It follows from last property that we have $\mathbb{E}[X \mid \mathcal{B}] \geq \mathbb{E}[Y \mid \mathcal{B}]$ if $X \geq Y \geq 0$. Under the assumptions of (1), we can therefore set

$$X' = \lim \uparrow \mathbb{E}[X_n \mid \mathcal{B}],$$

which is a $\mathcal{B}$-measurable random variable with values in $[0, \infty)$. Then, for every nonnegative $\mathcal{B}$-measurable random variable Z, the monotone convergence theorem gives

$$\mathbb{E}[ZX'] = \lim \uparrow \mathbb{E}[Z\mathbb{E}[X_n \mid \mathcal{B}]] = \lim \uparrow \mathbb{E}[ZX_n] = \mathbb{E}[ZX],$$

which implies $X' = \mathbb{E}[X \mid \mathcal{B}]$ thanks to the characteristic property (3) from theorem 1.5.1. The second assertion in (1) follows by applying the first one to $X_n = Y_1 + \cdots + Y_n$.

2. Using (1), we have

$$\mathbb{E}[\liminf X_n \mid \mathcal{B}] = \mathbb{E}\left[\lim_{k\uparrow\infty}\left(\inf_{n\geq k} X_n\right) \middle| \mathcal{B}\right] = \lim_{k\uparrow\infty} \mathbb{E}\left[\inf_{n\geq k} X_n \middle| \mathcal{B}\right].$$

3. It suffices to apply (2) twice:

$$\mathbb{E}[Z - X \mid \mathcal{B}] = \mathbb{E}[\liminf(Z - X_n) \mid \mathcal{B}] \leq \mathbb{E}[Z \mid \mathcal{B}] - \limsup \mathbb{E}[X_n \mid \mathcal{B}],$$

$$\mathbb{E}[Z + X \mid \mathcal{B}] = \mathbb{E}[\liminf(Z + X_n) \mid \mathcal{B}] \leq \mathbb{E}[Z \mid \mathcal{B}] + \liminf \mathbb{E}[X_n \mid \mathcal{B}],$$

which leads to

$$\mathbb{E}[X \mid \mathcal{B}] \leq \liminf \mathbb{E}[X_n \mid \mathcal{B}] \leq \limsup \mathbb{E}[X_n \mid \mathcal{B}] \leq \mathbb{E}[X \mid \mathcal{B}],$$

giving the desired almost sure convergence. The convergence in L^1 is now a consequence of the dominated convergence theorem, since we have

$$|\mathbb{E}[X_n \mid \mathcal{B}]| \leq \mathbb{E}[|X_n| \mid \mathcal{B}] \leq \mathbb{E}[Z \mid \mathcal{B}]$$

and $\mathbb{E}[\mathbb{E}[Z \mid \mathcal{B}]] = \mathbb{E}[Z] < \infty$.

Theorem 1.5.5: Jensen Inequality conditional Expectation

If $f : \mathbb{R} \to \mathbb{R}_+$ is convex, and if $X \in L^1$, then

$$\mathbb{E}[f(X) \mid \mathcal{B}] \geq f(\mathbb{E}[X \mid \mathcal{B}]).$$

Proof for Theorem.

Set

$$E_f = \{(a, b) \in \mathbb{R}^2 : \forall x \in \mathbb{R},\ f(x) \geq ax + b\}.$$

Then,

$$\forall x \in \mathbb{R}, \quad f(x) = \sup_{(a,b)\in E_f} (ax + b) = \sup_{(a,b)\in E_f\cap\mathbb{Q}^2} (ax + b).$$

We can take advantage of the fact that $\mathbb{Q}^2$ is countable to discard a countable

collection of sets of probability zero and to get that, a.s.,

$$\mathbb{E}[f(X) \mid \mathcal{B}] = \mathbb{E}\left[\sup_{(a,b)\in E_f\cap\mathbb{Q}^2}(aX+b)\,\middle|\,\mathcal{B}\right] \geq \sup_{(a,b)\in E_f\cap\mathbb{Q}^2}\mathbb{E}[aX+b \mid \mathcal{B}] = f(\mathbb{E}[X \mid \mathcal{B}]).$$

■

As a consequence of Jensen's inequality, we obtain that, for every $p \geq 1$, the mapping $X \mapsto \mathbb{E}[X \mid \mathcal{B}]$ maps $L^p(\Omega, \mathcal{A}, \mathbb{P})$ into itself, and is a contraction of $L^p(\Omega, \mathcal{A}, \mathbb{P})$. Indeed, for every $X \in L^p(\Omega, \mathcal{A}, \mathbb{P})$, we have using property (d) in Section 11.2.1,

$$\mathbb{E}[|\mathbb{E}[X \mid \mathcal{B}]|^p] \leq \mathbb{E}[\mathbb{E}[|X| \mid \mathcal{B}]^p] \leq \mathbb{E}[\mathbb{E}[|X|^p \mid \mathcal{B}]] = \mathbb{E}[|X|^p].$$

This contraction property also holds for $p = \infty$.

Theorem 1.5.6: The orthogonal projection property

If $X \in L^2(\Omega, \mathcal{A}, \mathbb{P})$, then $\mathbb{E}[X \mid \mathcal{B}]$ is the orthogonal projection of X on $L^2(\Omega, \mathcal{B}, \mathbb{P})$.

Proof for Theorem.

Jensen's inequality shows that $\mathbb{E}[X \mid \mathcal{B}]^2 \leq \mathbb{E}[X^2 \mid \mathcal{B}]$, a.s. This implies that

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}]^2] \leq \mathbb{E}[\mathbb{E}[X^2 \mid \mathcal{B}]] = \mathbb{E}[X^2] < \infty,$$

and thus the random variable $\mathbb{E}[X \mid \mathcal{B}]$ belongs to $L^2(\Omega, \mathcal{B}, \mathbb{P})$.

On the other hand, for every bounded $\mathcal{B}$-measurable random variable Z,

$$\mathbb{E}[Z(X - \mathbb{E}[X \mid \mathcal{B}])] = \mathbb{E}[ZX] - \mathbb{E}[Z\mathbb{E}[X \mid \mathcal{B}]] = 0,$$

by the characteristic property (11.2). Hence $X - \mathbb{E}[X \mid \mathcal{B}]$ is orthogonal to the space of bounded $\mathcal{B}$-measurable random variables, and the latter space is dense in $L^2(\Omega, \mathcal{B}, \mathbb{P})$ (for instance by Theorem 4.8). It follows that $X - \mathbb{E}[X \mid \mathcal{B}]$ is orthogonal to $L^2(\Omega, \mathcal{B}, \mathbb{P})$, which gives the desired result. ■

Theorem 1.5.7: Pull-out property of conditional expectation

Let X and Y be real random variables and assume that Y is $\mathcal{B}$-measurable. Then

$$\mathbb{E}[YX \mid \mathcal{B}] = Y\,\mathbb{E}[X \mid \mathcal{B}],$$

provided X and Y are both nonnegative, or X and YX are both integrable.

Proof for Theorem.

Suppose that $X \geq 0$ and $Y \geq 0$. Then, for every nonnegative $\mathcal{B}$-measurable random variable Z, we have

$$\mathbb{E}[Z(Y\mathbb{E}[X \mid \mathcal{B}])] = \mathbb{E}[(ZY)\mathbb{E}[X \mid \mathcal{B}]] = \mathbb{E}[ZYX],$$

using *characteristic property* (3) from theorem 1.5.1 and the fact that ZY is $\mathcal{B}$-measurable. Since $Y\mathbb{E}[X \mid \mathcal{B}]$ is a nonnegative $\mathcal{B}$-measurable random variable, the characteristic property from theorem 1.5.1 implies that

$$Y\mathbb{E}[X \mid \mathcal{B}] = \mathbb{E}[YX \mid \mathcal{B}].$$

In the case when X and YX are integrable, we get the desired result by decomposing

$$X = X^+ - X^-, \qquad Y = Y^+ - Y^-.$$

■

Theorem 1.5.8: Nested σ-Fields

Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two sub-σ-fields of $\mathcal{A}$ such that $\mathcal{B}_1 \subset \mathcal{B}_2$. Then, for every nonnegative (or integrable) random variable X, we have

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_2] \mid \mathcal{B}_1] = \mathbb{E}[X \mid \mathcal{B}_1].$$

Proof for Theorem.

Consider the case where $X \geq 0$. Let Z be a nonnegative $\mathcal{B}_1$-measurable random variable. Then, since Z is also $\mathcal{B}_2$-measurable,

$$\mathbb{E}[Z\,\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_2] \mid \mathcal{B}_1]] = \mathbb{E}[Z\,\mathbb{E}[X \mid \mathcal{B}_2]] = \mathbb{E}[ZX].$$

Hence $\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_2] \mid \mathcal{B}_1]$ satisfies the characteristic property of $\mathbb{E}[X \mid \mathcal{B}_1]$, and the desired result follows. ■

Remark: We have also $\mathbb{E}[\mathbb{E}[X \mid \mathcal{B}_1] \mid \mathcal{B}_2] = \mathbb{E}[X \mid \mathcal{B}_1]$ under the same assumptions, but this is trivial since $\mathbb{E}[X \mid \mathcal{B}_1]$ is $\mathcal{B}_2$-measurable.

Theorem 1.5.9: Independence and conditional expectaiton

Two sub-σ-fields $\mathcal{B}_1$ and $\mathcal{B}_2$ of $\mathcal{A}$ are independent if and only if, for every $B \in \mathcal{B}_2$, we have

$$\mathbb{E}[\mathbb{1}_B \mid \mathcal{B}_1] = \mathbb{P}(B).$$

Furthermore, if $\mathcal{B}_1$ and $\mathcal{B}_2$ are independent, we have also $\mathbb{E}[X \mid \mathcal{B}_1] = \mathbb{E}[X]$ for every nonnegative $\mathcal{B}_2$-measurable random variable X and for every $X \in L^1(\Omega, \mathcal{B}_2, \mathbb{P})$.

Proof for Theorem.

Left exercise for the reader. ■

Theorem 1.5.10: Factorization under independence

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be two measurable spaces, and let X and Y be random variables taking values in E and F respectively. Assume that X is independent

of $\mathcal{B}$ and that Y is $\mathcal{B}$-measurable. Then, for every $\mathcal{E} \otimes \mathcal{F}$-measurable function $g : E \times F \to \mathbb{R}_+$,

$$\mathbb{E}[g(X,Y) \mid \mathcal{B}] = \int g(x,Y)\,\mathbb{P}_X(dx),$$

where we recall that $\mathbb{P}_X$ denotes the law of X. The right-hand side is the composition of the random variable Y with the mapping $\Psi : F \to \mathbb{R}_+$ defined by

$$\Psi(y) = \int g(x,y)\,\mathbb{P}_X(dx).$$

Proof for Theorem.

We have to prove that, for any nonnegative $\mathcal{B}$-measurable random variable Z, we have

$$\mathbb{E}[g(X,Y)Z] = \mathbb{E}[\Psi(Y)Z].$$

Write $\mathbb{P}_{X,Y,Z}$ for the law of the triple (X,Y,Z), which is a probability measure on $E \times F \times \mathbb{R}_+$. Since X is independent of $\mathcal{B}$, X is independent of the pair (Y,Z), and therefore

$$\mathbb{P}_{X,Y,Z} = \mathbb{P}_X \otimes \mathbb{P}_{Y,Z}.$$

Then, using the Fubini theorem, we have

$$\begin{aligned}
\mathbb{E}[g(X,Y)Z] &= \int g(x,y)z\,\mathbb{P}_{X,Y,Z}(dx\,dy\,dz) \\
&= \int g(x,y)z\,\mathbb{P}_X(dx)\,\mathbb{P}_{Y,Z}(dy\,dz) \\
&= \int_{F\times\mathbb{R}_+} z\left(\int_E g(x,y)\,\mathbb{P}_X(dx)\right)\mathbb{P}_{Y,Z}(dy\,dz) \\
&= \int_{F\times\mathbb{R}_+} z\,\Psi(y)\,\mathbb{P}_{Y,Z}(dy\,dz) \\
&= \mathbb{E}[\Psi(Y)Z].
\end{aligned}$$

which was the desired result. ∎

Remark: The mapping Ψ is measurable thanks to the Fubini theorem. The theorem can be explained informally as follows. If we condition with respect to the σ-field $\mathcal{B}$, the random variable Y, which is $\mathcal{B}$-measurable, behaves like a constant, but on the other hand $\mathcal{B}$ gives no information on the random variable X. Thus the best approximation of $g(X,Y)$ knowing $\mathcal{B}$ is obtained by integrating $g(\cdot,Y)$ with respect to the law of X.

Theorem 1.5.11: Invariance under independent information

Let Z be a random variable in L^1, and let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two sub-σ-fields of $\mathcal{A}$. Assume that $\mathcal{H}_2$ is independent of $\sigma(Z) \vee \mathcal{H}_1$. Then,

$$\mathbb{E}[Z \mid \mathcal{H}_1 \vee \mathcal{H}_2] = \mathbb{E}[Z \mid \mathcal{H}_1].$$

Proof for Theorem.

It suffices to prove that the equality

$$\mathbb{E}[\mathbb{1}_A Z] = \mathbb{E}[\mathbb{1}_A \mathbb{E}[Z \mid \mathcal{H}_1]]$$

holds. ■

1.5.3 Transition Probabilities and Conditional Distributions

Definition 1.5.12: Transition probability

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be two measurable spaces. A transition probability (or transition kernel) from E into F is a mapping

$$\nu : E \times \mathcal{F} \longrightarrow [0, 1]$$

which satisfies the following two properties:

1. for every $x \in E$, $A \mapsto \nu(x, A)$ is a probability measure on $(F, \mathcal{F})$;
2. for every $A \in \mathcal{F}$, the function $x \mapsto \nu(x, A)$ is $\mathcal{E}$-measurable.

At an intuitive level, each time one fixes a "starting point" $x \in E$, the probability measure $\nu(x, \cdot)$ gives a way to choose an "arrival point" $y \in F$ in a random manner (but with a law depending on the starting point x). This notion plays a fundamental role in the theory of Markov chains.

Proposition 1.5.13

Let ν be a transition probability from E into F.

1. If h is a nonnegative (resp. bounded) measurable function on $(F, \mathcal{F})$, then

$$\varphi(x) = \int h(y)\, \nu(x, dy), \qquad x \in E,$$

   is a nonnegative (resp. bounded) measurable function on E.
2. If γ is a probability measure on $(E, \mathcal{E})$, then

$$\mu(A) = \int \gamma(dx)\, \nu(x, A), \qquad A \in \mathcal{F},$$

   is a probability measure on $(F, \mathcal{F})$.

Proof for Proposition.

We omit the easy proof of the proposition. When treating the nonnegative case of (1), we first consider simple functions, and then use an increasing passage to the limit. ■

We now come to the relation between the notion of a transition probability and conditional expectations.

Definition 1.5.14: The conditional distribution

Let X and Y be two random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$ respectively. Any transition probability ν from E into F such that, for every nonnegative measurable function h on F,

$$\mathbb{E}[h(Y) \mid X] = \int h(y)\, \nu(X, dy), \qquad \text{a.s.},$$

is called a conditional distribution of Y knowing X.

Remark The random variable $\int h(y)\, \nu(X, dy)$ is the composition of X with the function

$$x \mapsto \int h(y)\, \nu(x, dy),$$

which is measurable by last Proposition. In particular, this random variable is a function of X, as $\mathbb{E}[h(Y) \mid X]$ should be.

By definition, if ν is a conditional distribution of Y knowing X, we have, for every $A \in \mathcal{F}$,

$$\mathbb{P}(Y \in A \mid X) = \nu(X, A), \qquad \text{a.s.}$$

It is tempting to replace this equality of random variables by an equality of real numbers,

$$\mathbb{P}(Y \in A \mid X = x) = \nu(x, A),$$

for every $x \in E$. Although it gives the intuition behind the notion of conditional distribution, this last equality has no mathematical meaning in general, because one has typically $\mathbb{P}(X = x) = 0$ for every $x \in E$, which makes it impossible to define the conditional probability given the event $\{X = x\}$. The only rigorous formulation is thus the first equality $\mathbb{P}(Y \in A \mid X) = \nu(X, A)$.

Let us briefly discuss the uniqueness of the conditional distribution of Y knowing X. If ν and ν' are two such conditional distributions, we have, for every $A \in \mathcal{F}$,

$$\nu(X, A) = \mathbb{P}(Y \in A \mid X) = \nu'(X, A), \qquad \text{a.s.}$$

which is equivalent to saying that, for every $A \in \mathcal{F}$,

$$\nu(x, A) = \nu'(x, A), \qquad \mathbb{P}_X(dx)\text{-a.s.}$$

Suppose that the measurable space $(F, \mathcal{F})$ is such that any probability measure on $(F, \mathcal{F})$ is characterized by its values on a (given) countable collection of measurable sets (this property holds for $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, by considering the collection of all rectangles with rational coordinates). Then, we conclude from the preceding display that

$$\nu(x, \cdot) = \nu'(x, \cdot), \qquad \mathbb{P}_X(dx)\text{-a.s.}$$

So uniqueness holds in this sense (and clearly we cannot expect more). By abuse of

language, we will nonetheless speak about *the conditional distribution* of Y knowing X.

Let us now consider the problem of the existence of conditional distributions.

Theorem 1.5.15: Existence of conditional distribution

Let X and Y be two random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively. Suppose that $(F, \mathcal{F})$ is a complete separable metric space equipped with its Borel σ-field. Then, there exists a conditional distribution of Y knowing X.

Proof for Theorem.

We will not prove this theorem, see Theorem 6.3 in [1] for the proof of a more general statement. ■

1.5.4 Evaluation of Conditional Expectation

Discret Case

If X is a discrete variable (E is countable), and Y any random variable. then we have the conditional distibution of Y given X, denoted by $\nu(x, A)$, defined by :

$$\nu(x, A) := \begin{cases} \frac{1}{\mathbb{P}(X=x)}\mathbb{P}(Y \in A, X = x) & \text{if } x \in E' := \{y \in E : \mathbb{P}(X = y) > 0\}, \\ \delta_{y_0}(A) & \text{if } x \notin E', \end{cases}$$

where y_0 is an arbitrary fixed point of F. Theorefore we can conclude that :

$$\mathbb{E}[h(Y) \mid X] = \varphi(X)$$

where

$$\varphi(x) = \int h(y)\, \nu(x, dy) = \begin{cases} \frac{1}{\mathbb{P}(X=x)}\mathbb{E}(h(Y)\mathbb{1}_{\{X=x\}}) & \text{if } x \in E', \\ y_0 & \text{if } x \notin E', \end{cases}$$

Where y_0 is an arbitrary fixed point of F.

Continuous Case [Random Variables with a Density]

Suppose that X and Y take values in $\mathbb{R}^m$ and in $\mathbb{R}^n$ respectively, and that the pair (X, Y) has density $p(x, y)$, $(x, y) \in \mathbb{R}^m \times \mathbb{R}^n$. The density of X is then

$$q(x) = \int_{\mathbb{R}^n} p(x, y)\, dy,$$

where we take $q(x) = 0$ if the integral is infinite. We may define the conditional distribution of Y knowing X by :

$$\nu(x, A) = \begin{cases} \frac{1}{q(x)} \int_A p(x, y)\, dy & \text{if } q(x) > 0, \\ \delta_0(A) & \text{if } q(x) = 0. \end{cases}$$

Theorefore we can conclude that :

$$\mathbb{E}[h(Y) \mid X] = \varphi(X)$$

where

$$\varphi(x) = \int h(y)\,\nu(x, dy) = \begin{cases} \dfrac{1}{q(y)} \displaystyle\int_{\mathbb{R}^m} h(x)\,p(x,y)\,dx & \text{if } q(y) > 0, \\ h(0) & \text{if } q(y) = 0. \end{cases}$$

1.6 Gaussian Distribution

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1.7 Convergence of random variables

The study of convergence of random variables is fundamental in probability theory, particularly in the analysis of asymptotic behaviors and limit theorems. Different notions of convergence allow us to formalize how a sequence of random variables approximates another random variable under various probabilistic criteria. These modes of convergence differ in strength and applicability, and understanding their relationships is essential for results such as the law of large numbers and the central limit theorem.

Throughout this section, we consider a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. Let $(\mathbf{X}_n)_{n\in\mathbb{N}}$ be a sequence of random variables that mapsto $\mathbb{R}^d$ and let $\mathbf{X}$ be another random variable taking values in $\mathbb{R}^d$. We introduce below the principal notions of convergence that will be used in the sequel.

Definition 1.7.1: Almost sure convergence

The sequence $(\mathbf{X}_n)$ is said to converge almost surely to $\mathbf{X}$, denoted by

$$\mathbf{X}_n \xrightarrow[a.s.]{n\to\infty} \mathbf{X},$$

if

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} \mathbf{X}_n(\omega) = \mathbf{X}(\omega)\right\}\right) = 1.$$

Proposition 1.7.2

Let $(\mathbf{X}_n)_{n\geq 1}$ and $\mathbf{X}$ be integrable random variables. If

$$\sum_{n=1}^{\infty} \mathbb{E}|\mathbf{X}_n - \mathbf{X}| < \infty,$$

then

$$\mathbf{X}_n \xrightarrow{\text{a.s.}} \mathbf{X}.$$

Proof for Proposition.

Let $\epsilon > 0$ By Markov's inequality,

$$\mathbb{P}\Big(|X_n - X| > \epsilon\Big) \leq \frac{\mathbb{E}|X_n - X|}{\epsilon}.$$

Therefore,

$$\sum_{n=1}^{\infty} \mathbb{P}\Big(|X_n - X| > \epsilon\Big) \leq \frac{1}{\varepsilon} \sum_{n=1}^{\infty} \mathbb{E}|X_n - X| < \infty.$$

By the Borel-Cantelli lemma 1.4.19 :

$$\mathbb{P}\left(\limsup |X_n - X| > \epsilon\right) = 0.$$

Now take $\epsilon = 1/m$ for $m \in \mathbb{N}^*$. Then for every $m \geq 1$,

$$\mathbb{P}\left(\limsup |X_n - X| > \frac{1}{m}\right) = 0.$$

Since a countable union of null sets is null,

$$\mathbb{P}\Big(\bigcup_{m=1}^{\infty} \{\limsup |X_n - X| > \frac{1}{m}\}\Big) = 0.$$

Thus, outside a null set, for every $m \geq 1$,

$$\limsup |X_n - X| \leq \frac{1}{m} \quad a.s.$$

Then :

$$\limsup |X_n - X| = 0 \quad a.s.$$

This implies that $X_n \to X$ almost surely. ∎

Definition 1.7.3: Convergence in probability

We say that the sequence $(\mathbf{X}_n)_{n\in\mathbb{N}}$ converges in probability to $\mathbf{X}$, and write

$$\mathbf{X}_n \xrightarrow[\mathbb{P}]{n\to\infty} \mathbf{X},$$

if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\Big(|\mathbf{X}_n - \mathbf{X}| > \varepsilon\Big) = 0.$$

Proposition 1.7.4

We remind that d is the distance on $L^0(\Omega, \mathcal{A}, \mathbb{P})$ defined by :

$$d(\mathbf{X}, \mathbf{Y}) = \mathbb{E}[|\mathbf{X} - \mathbf{Y}| \wedge 1]$$

We have :

$$\mathbf{X}_n \xrightarrow[\mathbb{P}]{n\to\infty} \mathbf{X} \iff d(\mathbf{X}_n, \mathbf{X}) \xrightarrow{n\to\infty} 0$$

Proof for Proposition.

Left exercice for the reader.
Hint: use the fact that the space $L^0_{\mathbb{R}^d}(\Omega, \mathcal{A}, \mathbb{P})$ is complete for the distance d by proposition 1.3.7.

Definition 1.7.5: Convergence in Lp

Let $p \in [1, \infty)$ and assume that

$$\mathbb{E}[|\mathbf{X}|^p] < \infty \quad \text{and} \quad \mathbb{E}[|\mathbf{X}_n|^p] < \infty \quad \text{for all } n \in \mathbb{N}.$$

The sequence $(\mathbf{X}_n)$ is said to converge to $\mathbf{X}$ in L^p, written

$$\mathbf{X}_n \xrightarrow[\mathcal{L}^p]{n\to\infty} \mathbf{X},$$

if

$$\lim_{n\to\infty} \mathbb{E}\Big[|\mathbf{X}_n - \mathbf{X}|^p\Big] = 0.$$

This is equivalent to saying that each component sequence of $(\mathbf{X}_n)_{n\in\mathbb{N}}$ converges to the corresponding component of $\mathbf{X}$ in the Banach space $L^p(\Omega, \mathcal{A}, \mathbb{P})$. One could also consider the convergence in L^∞, but this convergence is very rarely used in probability theory and we will not discuss it.

1.7.1 Relationship between modes of convergences

Proposition 1.7.6

If :

$$\mathbf{X}_n \xrightarrow[a.s.]{n\to\infty} \mathbf{X}, \quad \text{or} \quad \mathbf{X}_n \xrightarrow[\mathcal{L}^p]{n\to\infty} \mathbf{X}.$$

Then :

$$\mathbf{X}_n \xrightarrow[\mathbb{P}]{n\to\infty} \mathbf{X}.$$

Proof for Proposition.

If $\mathbf{X}_n$ converges almost surely to $\mathbf{X}$, then by dominated convergence :

$$d(\mathbf{X}_n, \mathbf{X}) = \mathbb{E}\Big[\, |\mathbf{X}_n - \mathbf{X}| \wedge 1 \,\Big] \longrightarrow 0 \quad \text{as } n \to \infty,$$

If $\mathbf{X}_n$ converges to $\mathbf{X}$ in L^p, then

$$d(\mathbf{X}_n, \mathbf{X}) \leq \mathbb{E}\Big[|\mathbf{X}_n - \mathbf{X}|\Big] = \|\mathbf{X}_n - \mathbf{X}\|_1 \leq \|\mathbf{X}_n - \mathbf{X}\|_p \longrightarrow 0 \quad \text{as } n \to \infty.$$

Proposition 1.7.7

If :

$$\mathbf{X}_n \xrightarrow[\mathbb{P}]{n\to\infty} \mathbf{X}.$$

Then :

$$\exists, (n_k) \uparrow \infty \quad \text{such that} \quad \mathbf{X}_{n_k} \xrightarrow[a.s.]{k\to\infty} \mathbf{X}$$

Proof for Proposition.

The assertion has been derived in the proof of Proposition **??**.

Proposition 1.7.8

If :

$$\begin{cases} \mathbf{X}_n \xrightarrow[\mathbb{P}]{n\to\infty} \mathbf{X}. \\ \exists r \in (1, \infty) \quad \text{such :} \quad (\mathbb{E}[|\mathbf{X}_n|^r])_{n\in\mathbb{N}} \quad \text{is bounded.} \end{cases}$$

Then :

$$\begin{cases} \mathbb{E}[|\mathbf{X}|^r] < \infty \\ \forall p \in [1, r), \quad \mathbf{X}_n \xrightarrow[\mathcal{L}^p]{n\to\infty} \mathbf{X}. \end{cases}$$

Proof for Proposition.

By assumption, there is a constant C such that $\mathbb{E}[|\mathbf{X}_n|^r] \leq C$ for every n. By the previous proposition we can find a subsequence $(\mathbf{X}_{n_k})$ that converges almost surely to

$\mathbf{X}$. By Fatou's lemma,

$$\begin{aligned}\mathbb{E}[|\mathbf{X}|^r] &= \mathbb{E}\Big[\liminf_{k\to\infty} |\mathbf{X}_{n_k}|^r\Big] \\ &\leq \liminf_{k\to\infty} \mathbb{E}[|\mathbf{X}_{n_k}|^r] \\ &\leq C.\end{aligned}$$

Then, for every $p \in [1, r)$ and every $\varepsilon > 0$,

$$\begin{aligned}\mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^p] &= \mathbb{E}\Big[|\mathbf{X}_n - \mathbf{X}|^p \mathbb{1}_{\{|\mathbf{X}_n - \mathbf{X}| \leq \varepsilon\}}\Big] + \mathbb{E}\Big[|\mathbf{X}_n - \mathbf{X}|^p \mathbb{1}_{\{|\mathbf{X}_n - \mathbf{X}| > \varepsilon\}}\Big] \\ &\leq \varepsilon^p + \mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^r]^{p/r} \mathbb{P}(|\mathbf{X}_n - \mathbf{X}| > \varepsilon)^{1-p/r} \\ &\leq \varepsilon^p + (2C)^{p/r} \mathbb{P}(|\mathbf{X}_n - \mathbf{X}| > \varepsilon)^{1-p/r}.\end{aligned}$$

Since $\mathbf{X}_n$ converges in probability to $\mathbf{X}$,

$$\limsup_{n\to\infty} \mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^p] \leq \varepsilon^p.$$

Letting $\varepsilon \to 0$ yields $\mathbb{E}[|\mathbf{X}_n - \mathbf{X}|^p] \to 0$.

Lemma 1.7.9: Scheffé's lemma

If :

$$\begin{cases} \mathbf{X}_n \geq 0 \quad \text{and} \quad \mathbf{X}_n \xrightarrow[\mathbb{P}]{n\to\infty} \mathbf{X}. \\ \forall n \in \mathbb{N} : \quad \mathbf{X}_n \in L^1 \quad \text{and} \quad \mathbf{X} \in L^1 \\ \mathbb{E}[\mathbf{X}_n] \xrightarrow{n\to\infty} \mathbb{E}[\mathbf{X}] \end{cases}$$

Then :

$$\mathbf{X}_n \xrightarrow[\mathcal{L}^1]{n\to\infty} \mathbf{X}.$$

Proof for Lemma

We write

$$\mathbb{E}[|\mathbf{X}_n - \mathbf{X}|] = \mathbb{E}[\mathbf{X}_n - \mathbf{X}] + 2\mathbb{E}[(\mathbf{X}_n - \mathbf{X})^-].$$

We have $\mathbb{E}[\mathbf{X}_n - \mathbf{X}] \to 0$, and the bound $(\mathbf{X}_n - \mathbf{X})^- \leq \mathbf{X}$ allows us to apply the dominated convergence theorem to obtain $\mathbb{E}[(\mathbf{X}_n - \mathbf{X})^-] \to 0$ (using that almost sure convergence in the dominated convergence theorem can be replaced by convergence in probability).

The next theorem would be seen in the martingale theory (next chapter), but we state it here for the sake of completeness. It gives a equivalent condition for the convergence in L^1 under the assumption of convergence in probability. The condition is expressed in terms of uniform integrability, that we state here for information but it's studied deeply in next chapter.

A collection $(X_i)_{i\in I}$ of random variables in $L^1(\Omega, \mathcal{A}, \mathbb{P})$ is said to be **uniformly integrable**

(u.i.) if

$$\lim_{a\to+\infty} \sup_{i\in I} \mathbb{E}\Big[|X_i|\mathbb{1}_{\{|X_i|>a\}}\Big] = 0.$$

Theorem 1.7.10: Vitali convergence theorem

If

$$\begin{cases} \mathbf{X}_n \xrightarrow[\mathbb{P}]{n\to\infty} \mathbf{X}. \\ \mathbf{X}_n \in L^1 \quad \text{for every } n \in \mathbb{N}. \end{cases}$$

Then the following assertions are equivalent :

$$\begin{cases} 1. \quad \mathbf{X}_n \xrightarrow[L^1]{n\to\infty} \mathbf{X} \\ 2. \quad \text{the sequence } (\mathbf{X}_n)_{n\in\mathbb{N}} \text{ is u.i.} \end{cases}$$

Proof for Theorem.

It would be shown in the section of uniform integrability 2.3.4 (next chapter). ■

It's also relevant to see that if there is $Z \in L^1$ such $|\mathbf{X}_n| \leq Z$ a.s. for every $n \in \mathbb{N}$, then the sequence $(\mathbf{X}_n)_{n\in\mathbb{N}}$ is u.i., which gives a simpler condition to prove L^1-convergence. In fact, we have :

$$\mathbb{E}[|\mathbf{X}_n \mathbb{1}_{|\mathbf{X}_n \geq a}] \leq \mathbb{E}[Z\mathbb{1}_{Z\geq a}]$$

Finaly, taking supremum and limits :

$$\lim_{a\to\infty} \sup_{n\in\mathbb{N}} \mathbb{E}[|\mathbf{X}_n|\mathbb{1}_{|\mathbf{X}_n\geq a}] \leq \lim_{a\to\infty} \mathbb{E}[Z\mathbb{1}_{Z\geq a}] = 0 \quad \text{Because :} \quad Z\mathbb{1}_{Z\geq a} \xrightarrow[a.s.]{n\to\infty} 0$$

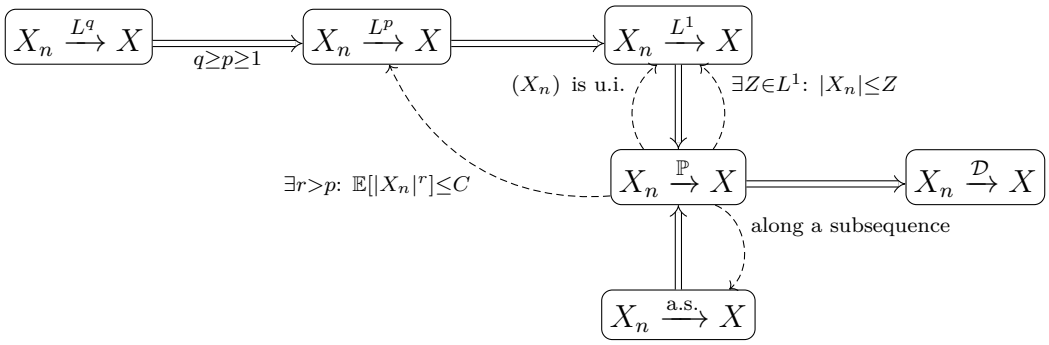

FIGURE 1 – Relations between the different convergences

Figure 1 illustrates the relations between the different types of convergence that we have introduced. The dashed lines give partial converses (under the stated conditions) that correspond to results stated in Propositions 1.7.7, 1.7.8, and to the dominated convergence theorem.

1.7.2 The Strong Law of Large Numbers

Theorem 1.7.11: Kolmogorov's Zero-One Law

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables taking values in arbitrary measurable spaces. For every $n \in \mathbb{N}$, let $\mathcal{B}_n$ be the σ-field

$$\mathcal{B}_n = \sigma(X_k; k \geq n).$$

Define the asymptotic σ-field $\mathcal{B}_\infty$ by

$$\mathcal{B}_\infty = \bigcap_{n=1}^{\infty} \mathcal{B}_n.$$

Then $\mathbb{P}(B) = 0$ or 1, for every $B \in \mathcal{B}_\infty$.

Proof for Theorem.

For every $n \in \mathbb{N}$, set

$$\mathcal{D}_n = \sigma(X_k; k \leq n).$$

Proposition 1.4.17 shows that $\mathcal{D}_n$ is independent of $\mathcal{B}_{n+1}$, hence a fortiori $\mathcal{D}_n$ is independent of $\mathcal{B}_\infty$. Consequently,

$$\forall A \in \bigcup_{n=1}^{\infty} \mathcal{D}_n, \quad \forall B \in \mathcal{B}_\infty, \quad \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Since the class $\bigcup_{n=1}^{\infty} \mathcal{D}_n$ is closed under finite intersections, Proposition 1.4.13 implies that $\mathcal{B}_\infty$ is independent of

$$\sigma\Big(\bigcup_{n=1}^{\infty} \mathcal{D}_n\Big) = \sigma(X_n; n \geq 1).$$

In particular, $\mathcal{B}_\infty$ is independent of itself, but this means that, for every $B \in \mathcal{B}_\infty$,

$$\mathbb{P}(B) = \mathbb{P}(B \cap B) = \mathbb{P}(B)^2,$$

which is only possible if $\mathbb{P}(B) = 0$ or 1. ∎

Let us give a simple illustration of Theorem 1.7.11. In the setting of this theorem, a random variable Y taking values in $[-\infty, \infty]$ which is $\mathcal{B}_\infty$-measurable is necessarily equal to a constant almost surely (this can be verified by observing that the distribution function $x \mapsto \mathbb{P}(Y \leq x)$ takes values in $\{0, 1\}$). Let us then consider a sequence $(X_n)_{n\in\mathbb{N}}$ of independent real random variables. The random variable

$$Z := \limsup_{n\to\infty} \frac{1}{n}(X_1 + \cdots + X_n),$$

which takes values in $[-\infty, \infty]$, is $\mathcal{B}_\infty$-measurable, since we have also, for every $k \in \mathbb{N}$,

$$Z = \limsup_{n\to\infty} \frac{1}{n}(X_k + X_{k+1} + \cdots + X_n),$$

which shows that Z is $\mathcal{B}_k$-measurable for every $k \in \mathbb{N}$. Theorem 1.7.11 thus implies that Z is constant almost surely. In particular, if we know that $\frac{1}{n}(X_1 + \cdots + X_n)$ converges almost surely, the limit must be constant.

Theorem 1.7.12: Strong Law of Large Numbers

Let $(X_n)_{n\geq 1}$ be a sequence of independent and identically distributed real random variables in L^1. Then

$$\frac{1}{n}(X_1 + \cdots + X_n) \xrightarrow[a.s.]{n\to\infty} \mathbb{E}[X_1].$$

Proof for Theorem.

To simplify notation, we set $S_n = X_1 + \cdots + X_n$ for every $n \in \mathbb{N}$, and $S_0 = 0$. Let $a > \mathbb{E}[X_1]$, and

$$M := \sup_{n\in\mathbb{Z}_+} (S_n - na),$$

which is a random variable with values in $[0, \infty]$.
We will show that

$$M < \infty, \quad \text{a.s.} \tag{i}$$

Assume that (1) holds (for any choice of $a > \mathbb{E}[X_1]$).
Since $S_n \leq na + M$ for every $n \in \mathbb{N}$, it immediately follows from (1) that

$$\limsup_{n\to\infty} \frac{1}{n} S_n \leq a, \quad \text{a.s.}$$

By considering a sequence of values of a that decreases to $\mathbb{E}[X_1]$, we find

$$\limsup_{n\to\infty} \frac{1}{n} S_n \leq \mathbb{E}[X_1], \quad \text{a.s.}$$

Replacing X_n by $-X_n$, we get the reverse inequality

$$\liminf_{n\to\infty} \frac{1}{n} S_n \geq \mathbb{E}[X_1], \quad \text{a.s.}$$

and the convergence in the theorem follows.

It remains to prove (1). With the notation of Theorem 1.7.11, we first observe that $\{M < \infty\}$ belongs to $\mathcal{B}_\infty$. Indeed, we have, for every integer $k \in \mathbb{N}$,

$$\{M < \infty\} = \left\{ \sup_{n\in\mathbb{Z}_+} (S_n - na) < \infty \right\} = \left\{ \sup_{n\geq k} (S_n - S_k - na) < \infty \right\},$$

and the event in the right-hand side is $\mathcal{B}_{k+1}$-measurable. It will therefore be enough to

prove that $\mathbb{P}(M < \infty) > 0$, or equivalently that $\mathbb{P}(M = \infty) < 1$. In the remaining part of the proof we verify that $\mathbb{P}(M = \infty) < 1$.

For every $k \in \mathbb{N}$, set

$$M_k := \sup_{0 \le n \le k} (S_n - na), \qquad M'_k := \sup_{0 \le n \le k} (S_{n+1} - S_1 - na).$$

Then, M_k and M'_k are nonnegative random variables with the same distribution, as a consequence of the fact that the random vectors $(X_1, \dots, X_k)$ and $(X_2, \dots, X_{k+1})$ have the same law. It follows that

$$M = \lim_{k \to \infty} \uparrow M_k \qquad \text{and} \qquad M' := \lim_{k \to \infty} \uparrow M'_k$$

also have the same law (just observe that $\mathbb{P}(M' \le x) = \lim \downarrow \mathbb{P}(M'_k \le x) = \lim \downarrow \mathbb{P}(M_k \le x) = \mathbb{P}(M \le x)$ for every $x \in \mathbb{R}$).

On the other hand, it follows from the definitions that, for every integer $k \ge 0$,

$$M_{k+1} = \sup \left(0, \sup_{1 \le n \le k+1} (S_n - na) \right) = \sup(0, M'_k + X_1 - a),$$

which can also be written as

$$M_{k+1} = M'_k - \inf(a - X_1, M'_k).$$

Since M'_k has the same law as M_k (and both these variables are clearly in L^1), we get

$$\mathbb{E}[\inf(a - X_1, M'_k)] = \mathbb{E}[M'_k] - \mathbb{E}[M_{k+1}] = \mathbb{E}[M_k] - \mathbb{E}[M_{k+1}] \le 0$$

thanks to the trivial inequality $M_k \le M_{k+1}$. We may now apply the dominated convergence theorem to the sequence $\inf(a - X_1, M'_k)$, $k \in \mathbb{N}$, noting that these variables are bounded in absolute value by $|a - X_1|$ (recall that $M'_k \ge 0$). It follows that

$$\mathbb{E}[\inf(a - X_1, M')] = \lim_{k \to \infty} \mathbb{E}[\inf(a - X_1, M'_k)] \le 0.$$

We now argue by contradiction to prove that $\mathbb{P}(M = \infty) < 1$. Suppose that $\mathbb{P}(M = \infty) = 1$, then we have also $\mathbb{P}(M' = \infty) = 1$, since M and M' have the same law, and it follows that $\inf(a - X_1, M') = a - X_1$ a.s. But then the inequality in the last display gives $\mathbb{E}[a - X_1] \le 0$, which is a contradiction since we chose $a > \mathbb{E}[X_1]$. This contradiction completes the proof. ■

Remarks.

- The L^1 integrability assumption is optimal in the sense that it is needed for the limit $\mathbb{E}[X_1]$ to be defined (and finite). If we remove the L^1 assumption but assume instead that the random variables X_n are nonnegative and $\mathbb{E}[X_1] = \infty$, it is easy to obtain that
$$\frac{1}{n}(X_1 + \cdots + X_n) \xrightarrow[n \to \infty]{\text{a.s.}} +\infty,$$
by applying the theorem to the variables $X_n \wedge K$, for every $K \in \mathbb{N}$.

- One can show that the convergence of the theorem also holds in L^1. The proof will be given at the end of next chapter as an application of martingale theory. As we already mentioned, the almost sure convergence is the most interesting one from a probabilistic point of view.

1.7.3 Convergence in Distribution

Recall that $C_b(\mathbb{R}^d)$ denote the set of all bounded continuous functions from $\mathbb{R}^d$ into $\mathbb{R}$. We equip $C_b(\mathbb{R}^d)$ with the supremum norm

$$\|\varphi\| = \sup_{x\in\mathbb{R}^d} |\varphi(x)|.$$

We let $M_1(\mathbb{R}^d)$ denote the set of all probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

Definition 1.7.13: Convergence in distribution

A sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ in $M_1(\mathbb{R}^d)$ converges weakly to $\mathbb{Q} \in M_1(\mathbb{R}^d)$ if

$$\forall \varphi \in C_b(\mathbb{R}^d), \qquad \int \varphi \, d\mathbb{Q}_n \xrightarrow{n\to\infty} \int \varphi \, d\mathbb{Q}.$$

A sequence $(X_n)_{n\in\mathbb{N}}$ of random variables with values in $\mathbb{R}^d$ converges in distribution to a random variable X with values in $\mathbb{R}^d$ if the sequence $(P_{X_n})_{n\in\mathbb{N}}$ converges weakly to P_X.
By Proposition 8.5, this is equivalent to saying that

$$\forall \varphi \in C_b(\mathbb{R}^d), \qquad \mathbb{E}[\varphi(X_n)]n \to \infty \mathbb{E}[\varphi(X)].$$

We will write $\mathbb{Q}_n \xrightarrow[(w)]{n\to\infty} \mathbb{Q}$, resp. $X_n \xrightarrow[\mathcal{D}]{n\to\infty} X$, if the sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$, resp. if $(X_n)_{n\in\mathbb{N}}$ converges in distribution to X.

Remark There is some abuse of terminology in saying that the sequence $(X_n)_{n\in\mathbb{N}}$ converges in distribution to X, because the limiting random variable X is not determined uniquely (only its distribution P_X is determined). For this reason, we will sometimes write that a sequence $(X_n)_{n\in\mathbb{N}}$ of random variables converges in distribution to a probability measure $\mathbb{Q}$ — one should of course understand that the laws P_{X_n} converge weakly to $\mathbb{Q}$. We may also note that the convergence in distribution makes sense even if the random variables X_n, $n \in \mathbb{N}$ are defined on different probability spaces (in the present book, we will however always assume that they are defined on the same probability space). This makes the convergence in distribution very different from the other types of convergence studied in this section.

Proposition 1.7.14

If

$$X_n \xrightarrow[\mathbb{P}]{n\to\infty} X$$

Then :

$$X_n \xrightarrow[\mathcal{D}]{n\to\infty} X$$

Proof for Proposition.

Suppose first that X_n converges a.s. to X. Then, for every $\varphi \in C_b(\mathbb{R}^d)$, $\varphi(X_n)$ converges a.s. to $\varphi(X)$ and the dominated convergence theorem gives $\mathbb{E}[\varphi(X_n)] \to \mathbb{E}[\varphi(X)]$.

If X_n only converges in probability to X, then we know from Proposition 10.3 that there is a subsequence of $(X_n)_{n\in\mathbb{N}}$ that converges a.s. to X, and thus, for every $\varphi \in C_b(\mathbb{R}^d)$, $\mathbb{E}[\varphi(X_n)]$ converges to $\mathbb{E}[\varphi(X)]$ along this subsequence. But then, the same argument shows that, from any subsequence of $(X_n)_{n\in\mathbb{N}}$, we can extract a subsequence along which $\mathbb{E}[\varphi(X_n)]$ converges to $\mathbb{E}[\varphi(X)]$. This is only possible if $\mathbb{E}[\varphi(X_n)] \to \mathbb{E}[\varphi(X)]$ as $n \to \infty$.

Remark: The converse to the proposition is (of course) false, because, as we already mentioned, the convergence in distribution of X_n to X does not determine the limiting variable X.

Theorem 1.7.15: Portmanteau Theorem

Let $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ be a sequence in $M_1(\mathbb{R}^d)$ and let $\mathbb{Q} \in M_1(\mathbb{R}^d)$. The following four assertions are equivalent:

1. The sequence $(\mathbb{Q}_n)$ converges weakly to $\mathbb{Q}$.
2. For every open subset G of $\mathbb{R}^d$,
$$\liminf \mathbb{Q}_n(G) \geq \mathbb{Q}(G).$$
3. For every closed subset F of $\mathbb{R}^d$,
$$\limsup \mathbb{Q}_n(F) \leq \mathbb{Q}(F).$$
4. For every Borel subset B of $\mathbb{R}^d$ such that $\mathbb{Q}(\partial B) = 0$,
$$\lim \mathbb{Q}_n(B) = \mathbb{Q}(B).$$

Proof for Theorem.

Let us start by proving that (1)⇒(2). If G is an open subset of $\mathbb{R}^d$, we can find a sequence $(\varphi_p)_{p\in\mathbb{N}}$ of bounded continuous functions such that $0 \leq \varphi_p \leq \mathbb{1}_G$ and $\varphi_p \uparrow \mathbb{1}_G$

(for instance, $\varphi_p(x) = p\,\mathrm{dist}(x, G^c) \wedge 1$). Then

$$\liminf_{n\to\infty} \mathbb{Q}_n(G) \geq \sup_p \left(\liminf_{n\to\infty} \int \varphi_p \, d\mathbb{Q}_n \right) = \sup_p \left(\int \varphi_p \, d\mathbb{Q} \right) = \mathbb{Q}(G).$$

The equivalence (2)⇔(3) is immediate since complements of open sets are closed and conversely.

Let us show that (2) and (3) imply (4). If $B \in \mathcal{B}(\mathbb{R}^d)$, write $\overline{B}$ for the closure of B and B° for the interior of B so that $\partial B = \overline{B} \setminus B^\circ$. By (2) and (3), we have :

$$\begin{cases} \limsup \mathbb{Q}_n(B) \leq & \limsup \mathbb{Q}_n(\overline{B}) \leq \mathbb{Q}(\overline{B}) \\ \liminf \mathbb{Q}_n(B) \geq & \liminf \mathbb{Q}_n(B^\circ) \geq \mathbb{Q}(B^\circ). \end{cases}$$

If $\mathbb{Q}(\partial B) = 0$ then $\mathbb{Q}(\overline{B}) = \mathbb{Q}(B^\circ) = \mathbb{Q}(B)$ and we get (4).

We still have to prove that (4)⇒(1). Let $\varphi \in C_b(\mathbb{R}^d)$. Without loss of generality, we can assume that $\varphi \geq 0$. Then, let $K \geq 0$ such that $0 \leq \varphi \leq K$. By the Fubini theorem,

$$\int \varphi(x)\mathbb{Q}(dx) = \int \left(\int_0^K \mathbb{1}_{\{t \leq \varphi(x)\}} dt \right) \mathbb{Q}(dx) = \int_0^K \mathbb{Q}(E_t^\varphi)\, dt,$$

where E_t^φ is the closed set $E_t^\varphi := \{x \in \mathbb{R}^d : \varphi(x) \geq t\}$. Similarly, for every n,

$$\int \varphi(x)\mathbb{Q}_n(dx) = \int_0^K \mathbb{Q}_n(E_t^\varphi)\, dt.$$

Now note that $\partial E_t^\varphi \subset \{x \in \mathbb{R}^d : \varphi(x) = t\}$, and that there are at most countably many values of t such that

$$\mathbb{Q}(\{x \in \mathbb{R}^d : \varphi(x) = t\}) > 0$$

(indeed, for every $k \in \mathbb{N}$, there are at most k distinct values of t such that $\mathbb{Q}(\{x \in \mathbb{R}^d : \varphi(x) = t\}) \geq \frac{1}{k}$, since the sets $\{x \in \mathbb{R}^d : \varphi(x) = t\}$ are disjoint when t varies, and $\mathbb{Q}$ is a probability measure). Hence (4) implies

$$\mathbb{Q}_n(E_t^\varphi)n \to \infty \mathbb{Q}(E_t^\varphi),$$

for every $t \in [0, K]$, except possibly for a countable set of values of t, which has zero Lebesgue measure. By dominated convergence, we conclude that

$$\int \varphi(x)\mathbb{Q}_n(dx) = \int_0^K \mathbb{Q}_n(E_t^\varphi)\, dtn \to \infty \int_0^K \mathbb{Q}(E_t^\varphi)\, dt = \int \varphi(x)\mathbb{Q}(dx).$$

■

Recall the notation $C_c(\mathbb{R}^d)$ for the space of all real continuous functions with compact support on $\mathbb{R}^d$.

Proposition 1.7.16

Let $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ and $\mathbb{Q}$ be probability measures on $\mathbb{R}^d$. Let H be a subset of $C_b(\mathbb{R}^d)$ whose closure (with respect to the supremum norm) contains $C_c(\mathbb{R}^d)$. The following properties are equivalent.

1. The sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$.
2. We have
$$\forall \varphi \in C_c(\mathbb{R}^d), \qquad \int \varphi \, d\mathbb{Q}_n n \to \infty \int \varphi \, d\mathbb{Q}.$$
3. We have
$$\forall \varphi \in H, \qquad \int \varphi \, d\mathbb{Q}_n n \to \infty \int \varphi \, d\mathbb{Q}.$$

It is obvious that (1)⇒(2) and (1)⇒(3). Then suppose that (2) holds. Let $\varphi \in C_b(\mathbb{R}^d)$ and let $(f_k)_{k\in\mathbb{N}}$ be a sequence in $C_c(\mathbb{R}^d)$ such that $0 \le f_k \le 1$ for every $k \in \mathbb{N}$, and $f_k \uparrow 1$ as $k \to \infty$. Then, for every $k \in \mathbb{N}$, $\varphi f_k \in C_c(\mathbb{R}^d)$ and thus

$$\int \varphi f_k \, d\mathbb{Q}_n n \to \infty \int \varphi f_k \, d\mathbb{Q}. \tag{i}$$

On the other hand,

$$\left| \int \varphi \, d\mathbb{Q}_n - \int \varphi f_k \, d\mathbb{Q}_n \right| \le \|\varphi\| \int (1 - f_k) \, d\mathbb{Q}_n = \|\varphi\| \left(1 - \int f_k \, d\mathbb{Q}_n \right),$$

and similarly

$$\left| \int \varphi \, d\mathbb{Q} - \int \varphi f_k \, d\mathbb{Q} \right| \le \|\varphi\| \left(1 - \int f_k \, d\mathbb{Q} \right).$$

Hence, using (i), we get for every $k \in \mathbb{N}$,

$$\begin{aligned}
\limsup_{n\to\infty} \left| \int \varphi \, d\mathbb{Q}_n - \int \varphi \, d\mathbb{Q} \right| &\le \limsup_{n\to\infty} \left| \int \varphi \, d\mathbb{Q}_n - \int \varphi f_k \, d\mathbb{Q}_n \right| + \left| \int \varphi \, d\mathbb{Q} - \int \varphi f_k \, d\mathbb{Q} \right| \\
&\le \|\varphi\| \left(\limsup_{n\to\infty} \left(1 - \int f_k \, d\mathbb{Q}_n \right) + \left(1 - \int f_k \, d\mathbb{Q} \right) \right) \\
&= 2\|\varphi\| \left(1 - \int f_k \, d\mathbb{Q} \right).
\end{aligned}$$

Now we just have to let k tend to ∞ (noting that $\int f_k \, d\mathbb{Q} \uparrow 1$, by monotone convergence) and we find that $\int \varphi \, d\mathbb{Q}_n$ converges to $\int \varphi \, d\mathbb{Q}$, so that (1) holds.

We complete the proof by verifying that (3)⇒(2). So assume that (3) holds. If $\varphi \in C_c(\mathbb{R}^d)$, then, for every $k \in \mathbb{N}$, we can find a function $\varphi_k \in H$ such that $\|\varphi - \varphi_k\| \le 1/k$, and it follows that

$$\begin{aligned}
&\limsup_{n\to\infty}\left|\int \varphi \, \mathrm{d}\mathbb{Q}_n - \int \varphi \mathrm{d}\mathbb{Q}\right| \\
&\leq \limsup_{n\to\infty}\left(\left|\int (\varphi-\varphi_k)\,\mathrm{d}\mathbb{Q}_n\right| + \left|\int \varphi_k \, \mathrm{d}\mathbb{Q}_n - \int \varphi_k \, \mathrm{d}\mathbb{Q}\right| + \left|\int (\varphi_k-\varphi)\,\mathrm{d}\mathbb{Q}\right|\right) \\
&\leq \frac{2}{k}
\end{aligned}$$

Since this holds for every k, we get that

$$\int \varphi \, d\mathbb{Q}_n \longrightarrow \int \varphi \, d\mathbb{Q}.$$

Recall our notation $\hat{\mathbb{Q}}(\xi) = \int e^{i\xi\cdot x}\,\mathbb{Q}(dx)$ for the Fourier transform of $\mathbb{Q} \in M_1(\mathbb{R}^d)$, and $\Phi_X = \widehat{P_X}$ for the characteristic function of a random variable X with values in $\mathbb{R}^d$.

Theorem 1.7.17: Lévy's Theorem

A sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ in $M_1(\mathbb{R}^d)$ converges weakly to $\mathbb{Q} \in M_1(\mathbb{R}^d)$ if and only if

$$\forall \xi \in \mathbb{R}^d, \qquad \hat{\mathbb{Q}}_n(\xi) \xrightarrow[n\to\infty]{} \hat{\mathbb{Q}}(\xi).$$

Equivalently, a sequence $(X_n)_{n\in\mathbb{N}}$ of random variables with values in $\mathbb{R}^d$ converges in distribution to X if and only if

$$\forall \xi \in \mathbb{R}^d, \qquad \Phi_{X_n}(\xi) \xrightarrow[n\to\infty]{} \Phi_X(\xi).$$

Proof for Theorem.

It is enough to prove the first assertion. First, if we assume that the sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$, the definition of this convergence shows that

$$\forall \xi \in \mathbb{R}^d, \qquad \hat{\mathbb{Q}}_n(\xi) = \int e^{i\xi\cdot x}\,\mathbb{Q}_n(dx) \xrightarrow[n\to\infty]{} \int e^{i\xi\cdot x}\,\mathbb{Q}(dx) = \hat{\mathbb{Q}}(\xi).$$

Conversely, assume that $\hat{\mathbb{Q}}_n(\xi) \to \hat{\mathbb{Q}}(\xi)$ for every $\xi \in \mathbb{R}^d$ and let us show that $(\mathbb{Q}_n)$ converges weakly to $\mathbb{Q}$. To simplify notation, we only treat the case $d = 1$, but the proof in the general case is exactly similar.

Let $f \in C_c(\mathbb{R})$, and, for every $\sigma > 0$, let

$$g_\sigma(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right).$$

be the density of the $\mathbb{N}(0,\sigma^2)$ distribution. We already observed in the proof of Theorem 8.16 that $g_\sigma * f$ converges pointwise to f as $\sigma \to 0$. Since f has compact support here, it is not hard to verify that this convergence is uniform on $\mathbb{R}$ (use Proposition 5.8 (i) and note that, for every $\varepsilon > 0$, we can find a compact set K such that $|g_\sigma * f| \leq \varepsilon$ for every $x \in \mathbb{R}\setminus K$ and $\sigma \in (0,1]$). Thus, if we let H be the subset of

$C_b(\mathbb{R}^d)$ defined by

$$H := \{\varphi = g_\sigma * f : \sigma > 0 \text{ and } f \in C_c(\mathbb{R}^d)\}$$

the closure of H in $C_b(\mathbb{R}^d)$ contains $C_c(\mathbb{R}^d)$.

On the other hand, we also saw in the proof of Theorem 8.16 that, for any $\nu \in M_1(\mathbb{R})$,

$$\begin{aligned}\int g_\sigma * f \, \mathrm{d}v &= \int f(x) g_\sigma * v(x) \mathrm{d}x \\ &= \int f(x) \left((\sigma\sqrt{2\pi})^{-1} \int e^{i\xi x} g_{1/\sigma}(\xi) \widehat{v}(-\xi) \mathrm{d}\xi \right) \mathrm{d}x\end{aligned}$$

We apply this formula to $\nu = \mathbb{Q}_n$ and $\nu = \mathbb{Q}$. From our assumption $\hat{\mathbb{Q}}_n(\xi) \to \hat{\mathbb{Q}}(\xi)$ for every $\xi \in \mathbb{R}$ and to the trivial bound $|\hat{\mathbb{Q}}_n(\xi)| \leq 1$, we can apply the dominated convergence theorem to get that, for every $x \in \mathbb{R}$,

$$\int e^{i\xi x} g_{1/\sigma}(\xi) \hat{\mathbb{Q}}_n(-\xi) \, d\xi \xrightarrow[n\to\infty]{} \int e^{i\xi x} g_{1/\sigma}(\xi) \hat{\mathbb{Q}}(-\xi) \, d\xi.$$

Since the quantities in the last display are bounded by 1 and $f \in C_c(\mathbb{R})$, we can use (10.3) and again the dominated convergence theorem to get

$$\int g_\sigma * f \, d\mathbb{Q}_n \xrightarrow[n\to\infty]{} \int g_\sigma * f \, d\mathbb{Q}.$$

Finally, we have proved that $\int \varphi \, d\mathbb{Q}_n \to \int \varphi \, d\mathbb{Q}$ for every $\varphi \in H$, and we know that the closure of H in $C_b(\mathbb{R}^d)$ contains $C_c(\mathbb{R}^d)$. By Proposition 10.12, this is enough to show that the sequence $(\mathbb{Q}_n)_{n\in\mathbb{N}}$ converges weakly to $\mathbb{Q}$. ■

1.7.4 The Central Limit Theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed real random variables in L^1. The strong law of large numbers asserts that

$$\frac{1}{n}(X_1 + \cdots + X_n) \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X_1].$$

We are now interested in the speed of this convergence, meaning that we want information about the typical size of

$$\frac{1}{n}(X_1 + \cdots + X_n) - \mathbb{E}[X_1]$$

when n is large.

Under the additional assumption that the variables X_n are in L^2, it is easy to guess the answer, because a simple calculation made in the proof of the weak law of large numbers (Theorem 9.13) gives

$$\begin{aligned}\mathbb{E}\left[(X_1 + \cdots + X_n - n\mathbb{E}[X_1])^2\right] &= \operatorname{var}(X_1 + \cdots + X_n) \\ &= n \operatorname{var}(X_1).\end{aligned}$$

This shows that the mean value of $(X_1 + \cdots + X_n - n\mathbb{E}[X_1])^2$ grows linearly with n, and thus the typical size of $X_1 + \cdots + X_n - n\mathbb{E}[X_1]$ is expected to be $\sqrt{n}$, or equivalently the typical size of

$$\frac{1}{n}(X_1 + \cdots + X_n) - \mathbb{E}[X_1]$$

should be $1/\sqrt{n}$. The central limit theorem gives a much more precise statement.

Theorem 1.7.18: Central Limit Theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed real random variables in L^2. Set $\sigma^2 = \mathrm{var}(X_1)$ and assume that $\sigma > 0$. Then

$$\frac{1}{\sqrt{n}}(X_1 + \cdots + X_n - n\mathbb{E}[X_1]) \xrightarrow[n\to\infty]{(d)} \mathbb{N}(0, \sigma^2).$$

where $\mathbb{N}(0, \sigma^2)$ is the Gaussian distribution with mean 0 and variance σ^2. Hence, for every $a, b \in \mathbb{R}$ with $a < b$,

$$\lim_{n\to\infty} \mathbb{P}(X_1+\cdots+X_n \in [n\mathbb{E}[X_1]+a\sqrt{n},\, n\mathbb{E}[X_1]+b\sqrt{n}]) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b \exp\left(-\frac{x^2}{2\sigma^2}\right) dx.$$

Proof for Theorem.

The second part of the statement follows from the first one, thanks to Proposition 10.11. To prove the first assertion, note that we can assume $\mathbb{E}[X_1] = 0$ (just replace X_n by $X_n - \mathbb{E}[X_n]$). Then set

$$Z_n = \frac{1}{\sqrt{n}}(X_1 + \cdots + X_n).$$

The characteristic function of Z_n is

$$\begin{aligned}
\Phi_{Z_n}(\xi) &= \mathbb{E}\left[\exp\left(i\xi\frac{X_1 + \cdots + X_n}{\sqrt{n}}\right)\right] \\
&= \mathbb{E}\left[\exp\left(i\frac{\xi}{\sqrt{n}}X_1\right)\right]^n \\
&= \Phi_{X_1}\left(\frac{\xi}{\sqrt{n}}\right)^n.
\end{aligned}$$

where, in the second equality, we used the fact that the random variables X_i are independent and identically distributed. On the other hand, by Proposition 8.17, we have

$$\Phi_{X_1}(\xi) = 1 + i\xi\mathbb{E}[X_1] - \frac{1}{2}\xi^2\mathbb{E}[X_1^2] + o(\xi^2) = 1 - \frac{\sigma^2\xi^2}{2} + o(\xi^2)$$

as $\xi \to 0$. For every fixed $\xi \in \mathbb{R}$, we have thus

$$\Phi_{X_1}\left(\frac{\xi}{\sqrt{n}}\right) = 1 - \frac{\sigma^2\xi^2}{2n} + o\left(\frac{1}{n}\right)$$

as $n \to \infty$. Thanks to proposition 1.7.8 and the last display, we have, for every $\xi \in \mathbb{R}$,

$$\lim_{n\to\infty} \Phi_{Z_n}(\xi) = \lim_{n\to\infty} \left(1 - \frac{\sigma^2\xi^2}{2n} + o\left(\frac{1}{n}\right)\right)^n = \exp\left(-\frac{\sigma^2\xi^2}{2}\right),$$

and, by Lemma 8.15, the limit is the characteristic function of the Gaussian $\mathbb{N}(0, \sigma^2)$ distribution. An application of Lévy's theorem 1.7.17 now completes the proof. ■

Remark When $\sigma = 0$ there is not much to say since the variables X_n are all equal a.s. to the constant $\mathbb{E}[X_1]$.

1.7.5 The Multidimensional Central Limit Theorem

We now assume that $(X_n)_{n\in\mathbb{N}}$ is a sequence of independent and identically distributed random variables with values in $\mathbb{R}^d$, such that $\mathbb{E}[\|X_1\|] < \infty$. We can apply the strong law of large numbers to each component of X_n and we have

$$\frac{1}{n}(X_1 + \cdots + X_n) \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X_1],$$

as in the real case. If we assume furthermore that $\mathbb{E}[\|X_1\|^2] < \infty$, then we expect to have also an analog of the central limit theorem. However, in contrast with the law of large numbers, this is not immediate because the convergence in distribution of a sequence of random vectors does not follow from the convergence in distribution of each component sequence (in fact, the law of the limiting vector is not determined by its marginal distributions, as we already noticed in Chapter 8).

To extend the central limit theorem to the case of random variables with values in $\mathbb{R}^d$, we must first generalize the Gaussian distribution in higher dimensions.

Definition 10.16 A random variable X with values in $\mathbb{R}^d$ is called a Gaussian vector if any linear combination of the components of X is a (real) Gaussian variable. This is equivalent to the existence of a vector $m \in \mathbb{R}^d$ and a $d \times d$ symmetric nonnegative definite matrix C such that, for every $\xi \in \mathbb{R}^d$,

$$\mathbb{E}[\exp(i\xi \cdot X)] = \exp\left(i\xi \cdot m - \frac{1}{2}\,{}^t\xi C\xi\right). \tag{10.5}$$

Moreover, we have then $\mathbb{E}[X] = m$ and $K_X = C$, and we say that X follows the Gaussian $\mathbb{N}_d(m, C)$ distribution.

To establish the equivalence of the two forms of the definition, first assume that any linear combination of the components of X is Gaussian. In particular, these components are in L^2 and thus the expectation $\mathbb{E}[X]$ and the covariance matrix K_X are well-defined. Then, for every $\xi \in \mathbb{R}^d$, we have $\mathbb{E}[\xi \cdot X] = \xi \cdot \mathbb{E}[X]$ and $\mathrm{var}(\xi \cdot X) = {}^t\xi K_X \xi$ (by (8.3)), so that $\xi \cdot X$ has the Gaussian $\mathbb{N}(\xi \cdot \mathbb{E}[X], {}^t\xi K_X \xi)$ distribution. Formula (10.5), with $m = \mathbb{E}[X]$ and $C = K_X$, now follows from Lemma 8.15 giving the characteristic function of a real Gaussian variable. Conversely, if (10.5) holds, then by replacing ξ with $\lambda\xi$ for $\lambda \in \mathbb{R}$, and using again Lemma 8.15, we get that the characteristic function of $\xi \cdot X$ coincides with the characteristic function of the Gaussian $\mathbb{N}(\xi \cdot m, {}^t\xi C\xi)$ distribution, so

that, in particular, $\xi \cdot X$ is Gaussian.

Remark In order for X to be a Gaussian vector, it is not sufficient that each component of X is a Gaussian (real) random variable. Let us give a simple example with $d = 2$. Let X_1 be Gaussian $\mathbb{N}(0,1)$ and let U be a random variable with law $\mathbb{P}(U = 1) = \mathbb{P}(U = -1) = 1/2$, which is independent of X_1. If we set $X_2 = UX_1$, then one immediately checks that X_2 is also Gaussian $\mathbb{N}(0,1)$. On the other hand, (X_1, X_2) is certainly not a Gaussian vector, because $\mathbb{P}(X_1 + X_2 = 0) = \mathbb{P}(U = -1) = 1/2$, which makes it impossible for $X_1 + X_2$ to be Gaussian.

Let $Y_1, \ldots, Y_d$ be independent Gaussian (real) random variables. Then $(Y_1, \ldots, Y_d)$ is a Gaussian vector. Indeed we noticed at the end of Section 9.5 that any linear combination of independent Gaussian random variables is Gaussian. Alternatively, we can also notice that formula (10.5) holds, with a diagonal matrix C. Conversely, if the covariance matrix of a Gaussian vector is diagonal, then its components are independent. This is an immediate application of Proposition 9.8 (iv), and we will come back to this result in the next chapter.

Proposition 1.7.19

Let $m \in \mathbb{R}^d$ and let C be a symmetric nonnegative definite $d \times d$ matrix. One can construct a Gaussian $\mathbb{N}_d(m, C)$ random vector.

Proof for Proposition.

It is enough to treat the case $m = 0$ (if X is Gaussian $\mathbb{N}_d(0, C)$, $X + m$ is Gaussian $\mathbb{N}_d(m, C)$). Set $A = \sqrt{C}$ in such a way that A is a symmetric nonnegative definite matrix and $A^2 = C$. Let $Y_1, \ldots, Y_d$ be independent Gaussian $\mathbb{N}(0,1)$ variables, and $Y = (Y_1, \ldots, Y_d)$, so that the covariance matrix K_Y is the identity matrix. Then $X = AY$ follows the Gaussian $\mathbb{N}_d(0, C)$ distribution. Indeed any linear combination of the components of X is also a linear combination of $Y_1, \ldots, Y_d$ and is therefore Gaussian. Moreover, by (8.2),

$$K_X = AK_Y{}^tA = A^2 = C,$$

which shows that X is Gaussian $\mathbb{N}_d(0, C)$. ∎

Theorem 1.7.20: Multidimensional Central Limit Theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent and identically distributed random variables with values in $\mathbb{R}^d$, such that $\mathbb{E}[\|X_1\|^2] < \infty$. Then,

$$\frac{1}{\sqrt{n}}(X_1 + \cdots + X_n - n\mathbb{E}[X_1]) \xrightarrow[n\to\infty]{(d)} \mathbb{N}_d(0, K_{X_1}).$$

Proof for Theorem.

Replacing X_n by $X_n - \mathbb{E}[X_n]$ allows us to assume that $\mathbb{E}[X_1] = 0$. For every $\xi \in \mathbb{R}^d$,

the central limit theorem in the real case (Theorem 10.15) shows that

$$\frac{1}{\sqrt{n}}(\xi \cdot X_1 + \cdots + \xi \cdot X_n) \xrightarrow[n\to\infty]{(d)} \mathbb{N}(0, \sigma^2),$$

where $\sigma^2 = \mathbb{E}[(\xi \cdot X_1)^2] = {}^t\xi K_{X_1}\xi$. This implies that

$$\mathbb{E}\left[\exp\left(i\xi \cdot \frac{X_1 + \cdots + X_n}{\sqrt{n}}\right)\right] \longrightarrow \exp\left(-\frac{1}{2}\,{}^t\xi K_{X_1}\xi\right)$$

and Lévy's theorem (Theorem 10.13) gives the desired result. ■

CHAPITRE 2

MARTINGALE AND CONTINUOUS SEMIMARTINGALES THEORY

2.1 Filtrations and information flow

2.1.1 Filtration

Definition 2.1.1: Filtration

A filtration on $(\Omega, \mathcal{A}, \mathbb{P})$ is an increasing family of sub-σ-fields of $\mathcal{A}$. The family in the most applications of this book is indexed by time, which is either discrete ($\mathbb{N}$) or continuous ($\mathbb{R}_+$).

- **The discrete-time case:** A discrete-time filtration is a family $(\mathcal{F}_n)_{n\in\mathbb{N}}$ of σ-fields such that :
$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots \subset \mathcal{A}$$
- **The continuous-time case:** A continuous-time filtration is a family $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$ of σ-fields such that :
$$\mathcal{F}_0 \subset \mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$$

We also say that $(\Omega, \mathcal{A}, (\mathcal{F}_n)_{n\in\mathbb{N}}, \mathbb{P})$ and $(\Omega, \mathcal{A}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ are filtered probability spaces.

The σ-field $\mathcal{F}_n$ (resp. $\mathcal{F}_t$) gathers the information available at time n (resp. t), events that are $\mathcal{F}_n$-measurable (resp. $\mathcal{F}_t$-measurable) are interpreted as those that depend only on what has happened up to time n (resp. t). We will write :

$$\mathcal{F}_\infty = \begin{cases} \bigvee_{n\in\mathbb{N}} \mathcal{F}_n := \sigma\left(\bigcup_{n\in\mathbb{N}} \mathcal{F}_n\right) & \text{in the discrete-time case} \\ \bigvee_{t\geq 0} \mathcal{F}_t := \sigma\left(\bigcup_{t\geq 0} \mathcal{F}_t\right) & \text{in the continuous-time case} \end{cases}$$

If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a random process, then, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), we may define $\mathcal{F}_n^X$ (resp. $\mathcal{F}_t^X$) as the smallest σ-field on Ω for which $X_0, X_1, \ldots, X_n$ (resp.

$X_s \quad s \leq t)$ are measurables. In other words, we have :

$$\begin{cases} \mathcal{F}_n^X = \sigma(X_0, X_1, \ldots, X_n) & \text{in the discrete-time case} \\ \mathcal{F}_t^X = \sigma(X_s : s \leq t) & \text{in the continuous-time case} \end{cases}$$

Then $(\mathcal{F}_n^X)_{n\in\mathbb{N}}$ (resp. $(\mathcal{F}_t^X)_{t\geq 0}$) is a filtration called *the canonical filtration* of $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$).

Remark: Let $(\mathcal{F}_t)_{0\leq t\leq\infty}$ be a filtration on $(\Omega, \mathcal{F}, P)$. We set, for every $t \geq 0$

$$\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s,$$

and $\mathcal{F}_{\infty+} = \mathcal{F}_\infty$. Note that $\mathcal{F}_t \subset \mathcal{F}_{t+}$ for every $t \in [0, \infty]$. The collection $(\mathcal{F}_{t+})_{0\leq t\leq\infty}$ is also a filtration. We say that the filtration $(\mathcal{F}_t)$ is right-continuous if

$$\mathcal{F}_{t+} = \mathcal{F}_t, \qquad \forall t \geq 0.$$

Let $(\mathcal{F}_t)$ be a filtration and let $\mathcal{N}$ be the class of all $(\mathcal{F}_\infty, P)$-negligible sets (i.e. $A \in \mathcal{N}$ if there exists an $A' \in \mathcal{F}_\infty$ such $A \subset A'$ and $P(A') = 0$). The filtration is said to be complete if $\mathcal{N} \subset \mathcal{F}_0$ (and thus $\mathcal{N} \subset \mathcal{F}_t$ for every t).

If $(\mathcal{F}_t)$ is not complete, it can be completed by setting $\mathcal{F}'_t = \mathcal{F}_t \vee \sigma(\mathcal{N})$, for every $t \in [0, \infty]$, using the notation $\mathcal{F}_t \vee \sigma(\mathcal{N})$ for the smallest σ-field that contains both $\mathcal{F}_t$ and $\sigma(\mathcal{N})$ (recall that $\sigma(\mathcal{N})$ is the σ-field generated by $\mathcal{N}$). We will often apply this completion procedure to the canonical filtration of a random process $(X_t)_{t\geq 0}$, and call the resulting filtration the *completed canonical filtration* of X.

Assumption: *In this book we will consider only filtrations that are right-continuous and complete.*

2.1.2 Adapted processes

For the discrete-time case, we have only one definition for measurablity with respect to a filtration, but for the continuous-time case, we have 3 types/definitions of measurability. We start with the discrete-time case.

Definition 2.1.2: Adapted Process [discrete]

A random process $(X_n)_{n\in\mathbb{N}}$ is said to be adapted to the filtration $(\mathcal{F}_n)_{n\in\mathbb{N}}$ if, for every $n \in \mathbb{N}$, X_n is $\mathcal{F}_n$-measurable.

In the continuous-time case, we have the following definitions.

Definition 2.1.3: Measurable Process [continuous]

A process $X = (X_t)_{t\geq 0}$ with values in a measurable space $(E, \mathcal{E})$ is said to be measurable if the mapping

$$(\omega, t) \mapsto X_t(\omega)$$

defined on $\Omega \times \mathbb{R}_+$ equipped with the product σ-field $\mathcal{F} \otimes \mathcal{B}(\mathbb{R}_+)$ is measurable. (We recall that $\mathcal{B}(\mathbb{R}_+)$ stands for the Borel σ-field of $\mathbb{R}$.)

This is stronger than saying that, for every $t \geq 0$, X_t is $\mathcal{F}$-measurable. On the other hand, considering for instance the case where $E = \mathbb{R}$, it is easy to see that if the sample paths of X are continuous, or only right-continuous, the fact that X_t is $\mathcal{F}$-measurable for every t implies that the process is measurable in the previous sense – see the argument in the proof of Proposition xxx below.

Definition 2.1.4: Adapted and Progressive Process

A random process $(X_t)_{t\geq 0}$ with values in a measurable space $(E, \mathcal{E})$ is called **adapted** if, for every $t \geq 0$, X_t is $\mathcal{F}_t$-measurable. This process is said to be **progressive** if, for every $t \geq 0$, the mapping

$$(\omega, s) \mapsto X_s(\omega)$$

defined on $\Omega \times [0, t]$ is measurable for the σ-field $\mathcal{F}_t \otimes \mathcal{B}([0, t])$.

Proposition 2.1.5

Let $(X_t)_{t\geq 0}$ be a random process with values in a metric space (E, d) (equipped with its Borel σ-field). Suppose that X is adapted and that the sample paths of X are right-continuous (i.e. for every $\omega \in \Omega$, $t \mapsto X_t(\omega)$ is right-continuous). Then X is progressive. The same conclusion holds if one replaces right-continuous by left-continuous.

Proof for Proposition.

We treat only the case of right-continuous sample paths, as the other case is similar. Fix $t > 0$. For every $n \geq 1$ and $s \in [0, t]$, define a random variable X_s^n by setting

$$X_s^n = X_{kt/n} \quad \text{if } s \in [(k-1)t/n, kt/n), k \in \{1, \dots, n\},$$

and $X_t^n = X_t$. The right-continuity of sample paths ensures that, for every $s \in [0, t]$ and $\omega \in \Omega$,

$$X_s(\omega) = \lim_{n\to\infty} X_s^n(\omega).$$

On the other hand, for every Borel subset A of E,

$$\begin{aligned}\{(\omega, s) \in \Omega \times [0, t] : X_s^n(\omega) \in A\} &= (\{X_t \in A\} \times \{t\}) \\ &\quad \cup \left(\bigcup_{k=1}^{n} \left(\{X_{kt/n} \in A\} \times \left[\frac{(k-1)t}{n}, \frac{kt}{n} \right) \right) \right)\end{aligned}$$

which belongs to the σ-field $\mathcal{F}_t \otimes \mathcal{B}([0, t])$. Hence, for every $n \geq 1$, the mapping

$(\omega, s) \mapsto X_s^n(\omega)$, defined on $\Omega \times [0,t]$, is measurable for $\mathcal{F}_t \otimes \mathcal{B}([0,t])$. Since a pointwise limit of measurable functions is also measurable, the same measurability property holds for the mapping $(\omega, s) \mapsto X_s(\omega)$ defined on $\Omega \times [0,t]$. It follows that the process X is progressive. ∎

2.2 Martingales and stopping times

2.2.1 Martingales

Definition 2.2.1: Martingale

An adapted real-valued process $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) such that $X_n \in L^1$ for every $n \in \mathbb{N}$ (resp. $X_t \in L^1$ for every $t \geq 0$) is called

- a **martingale** if :

$$\begin{cases} \forall n \in \mathbb{N} \quad X_n = \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 \quad X_t = \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

- a **supermartingale** if :

$$\begin{cases} \forall n \in \mathbb{N} \quad X_n \geq \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 \quad X_t \geq \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

- a **submartingale** if :

$$\begin{cases} \forall n \in \mathbb{N} \quad X_n \leq \mathbb{E}[X_{n+1}|F_n] & \text{For the discret time} \\ \forall s \geq t \geq 0 \quad X_t \leq \mathbb{E}[X_s|F_t] & \text{For the continuous time} \end{cases}$$

If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a submartingale, $(-X_n)_{n\in\mathbb{N}}$ (resp. $(-X_t)_{t\geq 0}$) is a supermartingale. For this reason, some of the results below are stated for supermartingales only, but the analogous results for submartingales immediately follow.

Proposition 2.2.2

Let $\varphi : \mathbb{R} \longrightarrow \mathbb{R}_+$ be a convex function, and let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be an adapted process, such that $\mathbb{E}[\varphi(X_n)] < \infty$ for every $n \in \mathbb{N}$ (resp. $\mathbb{E}[\varphi(X_t)] < \infty$ for every $t \geq 0$).

- If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a martingale, then $(\varphi(X_n))_{n\in\mathbb{N}}$ (resp. $(\varphi(X_t))_{t\geq 0}$) is a submartingale.
- If $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is a submartingale, and if we assume in addition that φ is increasing, $(\varphi(X_n))_{n\in\mathbb{N}}$ (resp. $(\varphi(X_t))_{t\geq 0}$) is a submartingale.

In particular, if X_n (resp. X_t) is a martingale, $|X_n|$ (resp. $|X_t|$) is a submartingale (and

so is X_n^2, provided that $\mathbb{E}[X_n^2] < \infty$ for every n) and, if X_n (resp. X_t) is a submartingale, $(X_n)^+$ (resp. $(X_t)^+$) is also a submartingale.

Proposition 2.2.3

Let $(M_n)_{n\in\mathbb{N}}$ (resp. $(M_t)_{t\geq 0}$) be a square integrable martingale ($\in L^2$). Then :

- **Discrete-time case:** If $n \geq m \geq 1$. Then,

$$\mathbb{E}\left[\sum_{k=m}^{n-1}(M_{k+1}-M_k)^2 \mid \mathcal{F}_m\right] = \mathbb{E}[M_n^2 - M_m^2 \mid \mathcal{F}_m] = \mathbb{E}[(M_n - M_m)^2 \mid \mathcal{F}_m].$$

  In particular,

$$\mathbb{E}\left[\sum_{i=m}^{n-1}(M_{k+1}-M_{t_k})^2\right] = \mathbb{E}[M_n^2 - M_m^2] = \mathbb{E}[(M_n - M_m)^2].$$

- **Continuous-time case:** If $0 \leq s < t$ and let $s = t_0 < t_1 < \cdots < t_p = t$ be a subdivision of the interval $[s,t]$. Then,

$$\mathbb{E}\left[\sum_{i=1}^{p}(M_{t_i}-M_{t_{i-1}})^2 \mid \mathcal{F}_s\right] = \mathbb{E}[M_t^2 - M_s^2 \mid \mathcal{F}_s] = \mathbb{E}[(M_t - M_s)^2 \mid \mathcal{F}_s].$$

  In particular,

$$\mathbb{E}\left[\sum_{i=1}^{p}(M_{t_i}-M_{t_{i-1}})^2\right] = \mathbb{E}[M_t^2 - M_s^2] = \mathbb{E}[(M_t - M_s)^2].$$

For the discrete-time martingale we can define the following operation with predictable processes that conserve martingality.

Definition 2.2.4: Predictable Process

A sequence $(H_n)_{n\in\mathbb{N}}$ of real random variables is called predictable if H_n is bounded and $\mathcal{F}_{n-1}$-measurable, for every $n \in \mathbb{N}$.

Proposition 2.2.5

Let $(X_n)_{n\in\mathbb{N}}$ be an adapted process, and let $(H_n)_{n\in\mathbb{N}}$ be a predictable sequence. We set $(H \bullet X)_0 = 0$ and, for every integer $n \geq 1$:

$$(H \bullet X)_n = H_1(X_1 - X_0) + H_2(X_2 - X_1) + \cdots + H_n(X_n - X_{n-1})$$

Then,

- If $(X_n)_{n\in\mathbb{N}}$ is a martingale, $((H \bullet X)_n)_{n\in\mathbb{N}}$ is also a martingale.
- If $(X_n)_{n\in\mathbb{N}}$ is a supermartingale (resp. a submartingale), and $H_n \geq 0$ for every $n \in \mathbb{N}$, then $((H \bullet X)_n)_{n\in\mathbb{N}}$ is a supermartingale (resp. a submartingale).

Proof for Proposition.

Let us prove first assertion. Since the random variables H_n are bounded, one immediately checks that the random variables $(H \bullet X)_n$ are integrable. It is also straightforward to verify that the process $((H \bullet X)_n)_{n\in\mathbb{N}}$ is adapted (all random variables entering the definition of $(H \bullet X)_n$ are $\mathcal{F}_n$-measurable). Then it suffices to verify that, for every $n \in \mathbb{N}$,

$$\mathbb{E}[(H \bullet X)_{n+1} - (H \bullet X)_n \mid \mathcal{F}_n] = 0.$$

However, $(H\bullet X)_{n+1}-(H\bullet X)_n = H_{n+1}(X_{n+1}-X_n)$ and since H_{n+1} is $\mathcal{F}_n$-measurable, We can pull the measurable term out of the expectation, that

$$\begin{aligned}\mathbb{E}[H_{n+1}(X_{n+1} - X_n) \mid \mathcal{F}_n] &= H_{n+1}\mathbb{E}[X_{n+1} - X_n \mid \mathcal{F}_n] \\ &= H_{n+1}(\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] - X_n) = 0.\end{aligned}$$

The proof of the second assertion is similar. ∎

2.2.2 Stopping times

Definition 2.2.6: Stopping time

A random variable $T : \Omega \longrightarrow \mathbb{N} \cup \{+\infty\}$ (resp. $T : \Omega \longrightarrow [0, \infty]$) is a **stopping time** of the filtration $(\mathcal{F}_n)$ (resp. $(\mathcal{F}_t)$) if $\{T \leq n\} \in \mathcal{F}_n \quad \forall n \in \mathbb{N}$ (resp. $\{T \leq t\} \in \mathcal{F}_t \quad \forall t \geq 0$) The σ-field of the past before T is then defined by

$$\begin{cases}\mathcal{F}_T = \{A \in \mathcal{F}_\infty : \forall n \in \mathbb{N}, A \cap \{T \leq n\} \in \mathcal{F}_n\}. & \text{For the discrete time} \\ \mathcal{F}_T = \{A \in \mathcal{F}_\infty : \forall t \geq 0, A \cap \{T \leq t\} \in \mathcal{F}_t\}. & \text{For the continuous time}\end{cases}$$

The reader will verify that $\mathcal{F}_T$ is indeed a σ-field.

In what follows, discrete (resp. continuous) "stopping time" will mean stopping time of the filtration $(\mathcal{F}_n)$ (resp. $(\mathcal{F}_t)$) unless otherwise specified. If T is a continuous stopping time, we also have $\{T < t\} \in \mathcal{F}_t$ for every $t > 0$, by considering $\{T < t\} = \bigcup_{n=1}^{\infty} \left\{T \leq t - \frac{1}{n}\right\}$, and moreover

$$\{T = \infty\} = \left(\bigcup_{n=1}^{\infty} \{T \leq n\}\right)^c \in \mathcal{F}_\infty.$$

A good advantage of assuming the right continuity of the filtration is that, in this case, the definition of stopping times in the continuous time case can be equivalently stated by replacing $\{T \leq t\} \in \mathcal{F}_t$ by $\{T < t\} \in \mathcal{F}_t$ for every $t \geq 0$. In fact :

$$\{T \leq t\} = \bigcap_{n=1}^{\infty} \{T < t + \frac{1}{n}\}. \in \mathcal{F}_{t+} = \mathcal{F}_t$$

Proposition 2.2.7

Let S and T be two stopping times

- If $S \leq T$. Then, $\mathcal{F}_S \subset \mathcal{F}_T$.
- We have in general, $S \wedge T$ and $S \vee T$ are also stopping times. Moreover $\mathcal{F}_{S\wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$.

Proof for Proposition.

Can be easily proven by using the definition of stopping times and σ-field of stopping times and manipulation of sets.

Lemma 2.2.8: Measurability w.r.t. stopped filtration

Let T be a stopping time. A function $\omega \mapsto Y(\omega)$ defined on the set $\{T < \infty\}$ and taking values in the measurable set $(E, \mathcal{E})$ is $\mathcal{F}_T$-measurable if and only if, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), the restriction of Y to the set $\{T = n\}$ is $\mathcal{F}_n$ measurable (resp. $\{T \leq t\}$ $\mathcal{F}_t$-measurable).

Proof for Lemma

For the discrete-time case, first assume that, for every $n \in \mathbb{N}$, the restriction of Y to $\{T = n\}$ is $\mathcal{F}_n$-measurable. Then, for every measurable subset A of E,

$$\{Y \in A\} \cap \{T \leq n\} = \bigcup_{k=0}^{n} (\{Y \in A\} \cap \{T = k\}) \in \mathcal{F}_n.$$

Letting $n \to \infty$, we first obtain that $\{Y \in A\} \in \mathcal{F}_\infty$, and then we deduce from the previous display that $\{Y \in A\} \in \mathcal{F}_T$. Conversely, if Y is $\mathcal{F}_T$-measurable, $\{Y \in A\} \in \mathcal{F}_T$ and thus $\{Y \in A\} \cap \{T = n\} = \{Y \in A\} \cap \{T \leq n\} \setminus \{Y \in A\} \cap \{T \leq n-1\} \in \mathcal{F}_n$, giving the desired result.

For the continuous-time case, first assume that, for every $t \geq 0$, the restriction of Y to $\{T \leq t\}$ is $\mathcal{F}_t$-measurable. Then, for every measurable subset A of E,

$$\{Y \in A\} \cap \{T \leq t\} \in \mathcal{F}_t.$$

Letting $t \to \infty$, we first obtain that $\{Y \in A\} \in \mathcal{F}_\infty$, and then we deduce from the previous display that $\{Y \in A\} \in \mathcal{F}_T$.

Conversely, if Y is $\mathcal{F}_T$-measurable, $\{Y \in A\} \in \mathcal{F}_T$ and thus $\{Y \in A\} \cap \{T \leq t\} \in \mathcal{F}_t$, giving the desired result.

Theorem 2.2.9: Stopped process Measurability

Let $(Y_n)_{n\in\mathbb{N}}$ (resp. $(Y_t)_{t\geq 0}$) be an adapted (resp. a progressive) process, and let T be a stopping time. Then the random variable Y_T, which is defined on the event $\{T < \infty\}$ by $Y_T(\omega) = Y_{T(\omega)}(\omega)$, is $\mathcal{F}_T$-measurable.

Proof for Theorem.

Let B be a Borel subset of $\mathbb{R}$. Let's begin with the dicrete-time case, consider the restriction of Y_T to the set $\{T = n\}$, which is given by Y_n on this set. Since Y_n is $\mathcal{F}_n$-measurable, the restriction of Y_T to $\{T = n\}$ is also $\mathcal{F}_n$-measurable. Hence, by Lemma 2.2.8, Y_T is $\mathcal{F}_T$-measurable. For the continuous-time case. Let $t \geq 0$. The restriction to $\{T \leq t\}$ of the function $\omega \mapsto X_T(\omega)$ is the composition of the two mappings

$$\begin{array}{ccc} \{T \leq t\} \ni \omega & \mapsto & (\omega, T(\omega) \wedge t) \\ \mathcal{F}_t & & \mathcal{F}_t \otimes \mathcal{B}([0,t]) \end{array}$$

and

$$\begin{array}{ccc} \Omega \times [0,t] \ni (\omega, s) & \mapsto & X_s(\omega) \\ \mathcal{F}_t \otimes \mathcal{B}([0,t]) & & \mathcal{E} \end{array}$$

which are both measurable (the first one since $T \wedge t$ is $\mathcal{F}_t$-measurable, by Proposition 2.2.7, and the second one by the definition of a progressive process). It follows that the restriction to $\{T \leq t\}$ of the function $\omega \mapsto X_T(\omega)$ is $\mathcal{F}_t$-measurable, which gives the desired result by the lemma 2.2.8. ■

If T is a stopping time, then, for every $n \in \mathbb{N}$ (resp. $t \geq 0$), $n \wedge T$ (resp. $t \wedge T$) is also a stopping time (Proposition 2.2.7) and it follows from the preceding theorem that $Y_{n\wedge T}$ (resp. $Y_{t\wedge T}$) is $\mathcal{F}_{n\wedge T}$-measurable (resp. $\mathcal{F}_{t\wedge T}$-measurable), hence also $\mathcal{F}_n$-measurable (resp. $\mathcal{F}_t$-measurable) by Proposition 2.2.7.

Proposition 2.2.10

If (S_n) is a sequence of stopping times then :

- if (S_n) is monotone increasing, then $S = \lim \uparrow S_n$ is a stopping time.
- If (S_n) is monotone decreasing, then $S = \lim \downarrow S_n$ is a stopping time and

$$\mathcal{F}_S = \bigcap_n \mathcal{F}_{S_n}$$

Proof for Proposition.

- If (S_n) is monotone increasing, then for every $t \geq 0$, $\{S \leq t\} = \bigcap_{n=1}^{\infty} \{S_n \leq t\}$, which belongs to the σ-field $\mathcal{F}_t$ since S_n is a stopping time for every n. Hence, S is a stopping time. same argument for the discrete time case.
- If (S_n) is monotone decreasing, then for every $t \geq 0$, $\{S < t\} = \bigcup_{n=1}^{\infty} \{S_n < t\}$, which belongs to the σ-field $\mathcal{F}_t$ since S_n is a stopping time for every n. Hence, S is a stopping time by the fact that $\mathcal{F}_t$ is right-continuous [check remarks after the definition 2.2.6].
- For the last assertion we first verify that $\mathcal{F}_S \subset \bigcap_n \mathcal{F}_{S_n}$ thanks to last proposition

2.2.7. Conversely, if $A \in \bigcap_n \mathcal{F}_{S_n}$, then

$$A \cap \{S < t\} = \bigcup_{n=1}^{\infty} A \cap \{S_n < t\} \in \mathcal{F}_t,$$

which gives $A \in \mathcal{F}_S$ again this is allowed thanks to the right-continuity of the filtration.

Proposition 2.2.11

Let T be a stopping time and let S be an $\mathcal{F}_T$-measurable random variable with values in $[0, \infty]$, such that $S \geq T$. Then S is also a stopping time. In particular, if T is a continuous stopping time,

$$T_n = \frac{\lceil 2^n T \rceil}{2^n} \cdot \mathbb{1}_{\{T<\infty\}} + \infty \cdot \mathbb{1}_{\{T=\infty\}}, \quad n = 0, 1, 2, \ldots$$

defines a sequence of stopping times that decreases to T.

Proof for Proposition.

For the first assertion, we write, for every $n \in \mathbb{N}$,

$$\{S \leq n\} = \{S \leq n\} \cap \{T \leq n\} \in \mathcal{F}_n$$

since $\{S \leq n\}$ is $\mathcal{F}_T$-measurable. And similar argument holds for the continuous time case. Further more, the second assertion follows since $T_n \geq T$, and T_n is a function of T, hence $\mathcal{F}_T$-measurable, and $T_n \downarrow T$ as $n \uparrow \infty$ by construction.

Finally lets defining hitting time.

Definition 2.2.12: Hitting time

Let $(X_t)_{t \geq 0}$ be an adapted process with values in a metric space (E, d).

1. Assume that the sample paths of X are right-continuous, and let O be an open subset of E. Then

$$T_O = \inf\{t \geq 0 : X_t \in O\}$$

is a stopping time of the filtration $(\mathcal{F}_t)$.

2. Assume that the sample paths of X are continuous, and let F be a closed subset of E. Then

$$T_F = \inf\{t \geq 0 : X_t \in F\}$$

is a stopping time (of the filtration $(\mathcal{F}_t)$).

In fact the stopping time property can be proven by considering the two sets :

$$\begin{cases} \{T_O < t\} & = \bigcup_{s\in[0,t)\cap\mathbb{Q}} \{X_s \in O\} \in \mathcal{F}_t, \\ \{T_F \leq t\} & = \left\{\inf_{0\leq s\leq t} d(X_s, F) = 0\right\} = \left\{\inf_{s\in[0,t]\cap\mathbb{Q}} d(X_s, F) = 0\right\} \in \mathcal{F}_t. \end{cases}$$

2.3 Martingale convergence theory

2.3.1 Upcrossing numbers

Discret time

Consider first a deterministic real sequence $y = (y_n)_{n\in\mathbb{N}}$, and $a < b$, the **upcrossing number** of this sequence along $[a, b]$ before time n, denoted by $M^y_{ab}(n)$, is the largest integer k such that there exists a strictly increasing finite sequence

$$m_1 < n_1 < m_2 < n_2 < \cdots < m_k < n_k$$

of nonnegative integers smaller than or equal to n with the properties $y_{m_i} \leq a$ and $y_{n_i} \geq b$, for every $i \in \{1, \ldots, k\}$. In what follows we consider a sequence $Y = (Y_n)_{n\in\mathbb{N}}$ of real random variables, and the associated upcrossing number $M^Y_{ab}(n)$ is then an integer-valued random variable. where $M^y_{ab}(0) = 0$ if such a sequence does not exist. We will use the following simple analytic lemma, whose proof is left to the reader.

Lemma 2.3.1

The sequence $(y_n)_{n\in\mathbb{N}}$ converges in $\overline{\mathbb{R}}$ if and only if, for every choice of the rationals a and b with $a < b$, we have $M^y_{ab}(n) < \infty$.

Lemma 2.3.2: Doob's Upcrossing Inequality

Let $X = (X_n)_{n\in\mathbb{N}}$ be a supermartingale. Then, for every reals $a < b$ and every $n \in \mathbb{N}$,

$$E[M^Y_{ab}(n)] \leq \frac{1}{b-a} E[(Y_n - a)^-]$$

Continuous time

Let $f : I \longrightarrow \mathbb{R}$ be a function defined on a subset I of $\mathbb{R}_+$. If $a < b$, the upcrossing number of f along $[a, b]$, denoted by $M^f_{ab}(I)$, is the maximal integer $k \geq 1$ such that there exists a finite increasing sequence $s_1 < t_1 < \cdots < s_k < t_k$ of elements of I such that $f(s_i) \leq a$ and $f(t_i) \geq b$ for every $i \in \{1, \ldots, k\}$ (if, even for $k = 1$, there is no such subsequence, we take $M^f_{ab}(I) = 0$, and if such a subsequence exists for every $k \geq 1$, we take $M^f_{ab}(I) = \infty$). Upcrossing numbers are a convenient tool to study the regularity of functions.

In the next lemma, the notation

$$\lim_{s\downarrow\downarrow t} f(s) \quad (\text{resp. } \lim_{s\uparrow\uparrow t} f(s))$$

means

$$\lim_{s\downarrow t, s>t} f(s) \quad (\text{resp. } \lim_{s\uparrow t, s<t} f(s)).$$

We say that $g : \mathbb{R}_+ \to \mathbb{R}$ is **càdlàg** (for the French "continue à droite avec des limites à gauche") if g is right-continuous and has left-limits at every $t > 0$.

Lemma 2.3.3

Let D be a countable dense subset of $\mathbb{R}_+$ and let f be a real function defined on D. We assume that, for every $T \in D$,

- the function f is bounded on $D \cap [0, T]$;
- for all rationals a and b such that $a < b$,

$$M^f_{ab}(D \cap [0, T]) < \infty.$$

Then, the right-limit

$$f(t+) := \lim_{s\downarrow\downarrow t, s\in D} f(s)$$

exists for every real $t \geq 0$, and similarly the left-limit

$$f(t-) := \lim_{s\uparrow\uparrow t, s\in D} f(s)$$

exists for every real $t > 0$. Furthermore, the function $g : \mathbb{R}_+ \to \mathbb{R}$ defined by $g(t) = f(t+)$ is càdlàg.

We omit the proof of this analytic lemma. It is important to note that the right and left-limits $f(t+)$ and $f(t-)$ are defined for every $t \geq 0$ ($t > 0$ in the case of $f(t-)$) and not only for $t \in D$.

Lemma 2.3.4: Doob's Upcrossing Inequality

Let D be a countable dense subset of $\mathbb{R}_+$ and let $X = (X_t)_{t\geq 0}$ be a supermartingale. Then, for every reals $a < b$ and every $t \geq 0$,

$$E[M^X_{ab}(D \cap [0, T])] \leq \frac{1}{b-a} E[(X_T - a)^-]$$

Can be obtained by monotone convergence applied on the discret version Lemma 2.3.2 by considering a sequence of finite subsets of D.

2.3.2 Lp Convergence and Doob's inequalities

Lemma 2.3.5

Let $(X_n)_{n\in\mathbb{N}}$ be a submartingale, and let S and T be two bounded stopping times such that $S \leq T$. Then

$$\mathbb{E}[X_S] \leq \mathbb{E}[X_T].$$

Proof for Lemma

to be proven. ■

Theorem 2.3.6: Doob's supermartingale inequalities

1. (Maximal inequality) Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a supermartingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $\lambda > 0$,

$$\begin{cases} \lambda\mathbb{P}\left(\sup_{0\leq m\leq n} |X_m| > \lambda\right) \leq \mathbb{E}[|X_0|] + 2\mathbb{E}[|X_n|] & \text{For the discrete case} \\ \lambda\mathbb{P}\left(\sup_{0\leq s\leq t} |X_s| > \lambda\right) \leq \mathbb{E}[|X_0|] + 2\mathbb{E}[|X_t|] & \text{For the continuous case} \end{cases}$$

2. (Doob's inequality in L^p) Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a martingale (with right-continuous sample paths in the continuous case). Then, for every $n \in \mathbb{N}$ (resp. $t > 0$) and every $p > 1$,

$$\begin{cases} \mathbb{E}\left[\sup_{0\leq m\leq n} |X_m|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_n|^p] & \text{For the discrete case} \\ \mathbb{E}\left[\sup_{0\leq s\leq t} |X_s|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_t|^p] & \text{For the continuous case} \end{cases}$$

Note that part (2) of the inequality is useful only if $\mathbb{E}[|X_n|^p] < \infty$(resp. $\mathbb{E}[|X_t|^p] < \infty$).

Proof for Theorem.

to be proven. ■

Corollary 2.3.7

Let $p \in (1, \infty)$ and let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a nonnegative submartingale. For every $n \in \mathbb{N}$ (resp. $t \geq 0$), set

$$\begin{cases} \widetilde{X}_n = \sup_{0\leq k\leq n} X_k & \text{For discret time} \\ \widetilde{X}_t = \sup_{0\leq s\leq t} X_s & \text{For continuous time} \end{cases}$$

Then,

$$\begin{cases} \mathbb{E}[(\widetilde{X}_n)^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[(X_n)^p] & \text{For discret time} \\ \mathbb{E}[(\widetilde{X}_t)^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[(X_t)^p] & \text{For continuous time} \end{cases}$$

Consequently, if $(Y_n)_{n\in\mathbb{N}}$ (resp. $(Y_t)_{t\geq 0}$) is a martingale, and

$$\begin{cases} Y_n^* = \sup_{0\leq k\leq n} |Y_k| & \text{For discret time} \\ Y_t^* = \sup_{0\leq s\leq t} |Y_s| & \text{For continuous time} \end{cases}$$

we have, for every $n \in \mathbb{N}$ (resp. $t \geq 0$),

$$\begin{cases} \mathbb{E}[(Y_n^*)^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|Y_n|^p]. & \text{For discret time} \\ \mathbb{E}[(Y_t^*)^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|Y_t|^p]. & \text{For continuous time} \end{cases}$$

Proof for Corollary.

to be proven.

In what follows, you will find that most of the results stated in the continuous time case requiring the right-continuity of the sample paths of the process, which is not that hard to verify in most of the cases thanks to the following theorem, which gives a sufficient condition for such a regularity of the sample paths.

Theorem 2.3.8: Sufficient condition for càdlàg modification

Assume that the filtration $(\mathcal{F}_t)$ is right-continuous and complete. Let $X = (X_t)_{t\geq 0}$ be a supermartingale, such that the function $t \mapsto \mathbb{E}[X_t]$ is right-continuous. Then X has a modification with càdlàg [a] sample paths, which is also an $(\mathcal{F}_t)$-supermartingale.

a. refers to the french sentence "continue à droite et limite à gauche" which means "right-continuous and have left-limits"

Proof for Theorem.

The proof of this theorem could be found in [2, thm. 3.18]

Theorem 2.3.9: Doob's Lp convergence theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a martingale and $p \in (1, \infty)$. Suppose that

$$\sup_{n\in\mathbb{N}} \mathbb{E}[|X_n|^p] < \infty,$$

Then, X_n converges a.s. and in L^p to a random variable X_∞ such that

$$\mathbb{E}[|X_\infty|^p] = \sup_{n\in\mathbb{N}} \mathbb{E}[|X_n|^p]$$

and we have

$$\mathbb{E}[(X_\infty^*)^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_\infty|^p].$$

Proof for Theorem.

to be proven. ■

Theorem 2.3.10: Kolmogorov's criterion for a.s. convergence

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent real random variables in L^2. Assume that $\mathbb{E}[X_n] = 0$ for every $n \in \mathbb{N}$. Then the following two conditions are equivalent:

1. $\sum_{n=0}^{\infty} \mathbb{E}[(X_n)^2] < \infty$.
2. The series $\sum_{n=0}^{\infty} X_n$ converges a.s. and in L^2.

Proof for Theorem.

to be proven. ■

2.3.3 Almost sure convergence

Theorem 2.3.11: Almost convergence of supermartingales

Let X be a supermartingale (with right-continuous sample paths if time is continuous). Assume that the collection $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is bounded in L^1. Then, it converge almost surely to a random variable $X_\infty \in L^1$.

Proof for Theorem.

We will prove it in the continuous time case, the discrete time case can be proven by similar arguments and it is left exercice for the reader. Let D be a countable dense subset of $\mathbb{R}_+$. Using lemma 2.3.4, we have, for every $a < b$ and every $T \in D$,

$$\mathbb{E}\left[M_{ab}^X(D \cap [0,T])\right] \leq \frac{1}{b-a}\mathbb{E}\left[(X_T - a)^-\right]$$

By Monotone convergence theorem, we get, for every $a < b$,

$$\mathbb{E}\left[M_{ab}^X(D)\right] \leq \frac{1}{b-a}\sup_{t\geq 0}\mathbb{E}\left[(X_t - a)^-\right] < +\infty.$$

Since the collection $(X_t)_{t\geq 0}$ is bounded in L^1, the right-hand side of the previous display is finite. Hence, a.s. for all rationals $a < b$, we have $M_{ab}^X(D) < \infty$. It follows from Lemma 2.3.3 that the limit

$$X_\infty := \lim_{D\ni t\to\infty} X_t$$

exists a.s. in $[-\infty, +\infty]$. We can in fact exclude the values $-\infty$ and $+\infty$ since, by Fatou's lemma,

$$\mathbb{E}[|X_\infty|] \leq \liminf_{D \ni t \to \infty} \mathbb{E}[|X_t|] < +\infty$$

and we get that $X_\infty \in L^1$. The right-continuity of X samples (which we have not used yet) allows us to remove the restriction to D in the limit. ■

Corollary 2.3.12

Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a nonnegative supermartingale. Then, X_n (resp. X_t) converges a.s. as $n \to \infty$ (resp. $t \to \infty$). Its limit X_∞ is in L^1 and we have $X_n \geq \mathbb{E}[X_\infty \mid \mathcal{F}_n]$ for every $n \in \mathbb{N}$. (resp. $X_t \geq \mathbb{E}[X_\infty \mid \mathcal{F}_t]$ for every $t \geq 0$).

Backward supermartingales A useful variant of the previous theorem is the following one, which is stated in a backward form in discrete time setup, we will not prove it however proof could be found in [2, thm. 12.30]. Since it will be used only in a proof step of supermartingale optional theorem in the continuous time case.

A backward filtration is a sequence $(\mathcal{F}_n)_{n\in\mathbb{Z}_-}$ of sub-σ-fields of $\mathcal{A}$, which is indexed by the set $\mathbb{Z}_- = \{0, -1, -2, \dots\}$ of all nonpositive integers, and is such that $\mathcal{F}_n \subset \mathcal{F}_m$ whenever $n, m \in \mathbb{Z}_-$ and $n \leq m$. We then set

$$\mathcal{F}_{-\infty} = \bigcap_{n\in\mathbb{Z}_-} \mathcal{F}_n$$

which is also a sub-σ-field of $\mathcal{A}$. It is important to observe that, in contrast with the "forward case" studied in the previous sections, the σ-field $\mathcal{F}_n$ becomes smaller and smaller when $n \downarrow -\infty$ (the larger $|n|$ is, the smaller the σ-field $\mathcal{F}_n$).

In what follows, we fix a backward filtration $(\mathcal{F}_n)_{n\in\mathbb{Z}_-}$. Let $X = (X_n)_{n\in\mathbb{Z}_-}$ be a sequence of real random variables indexed by $\mathbb{Z}_-$. We say that X is a backward martingale (resp. a backward supermartingale, a backward submartingale) if X_n is $\mathcal{F}_n$-measurable and $\mathbb{E}[|X_n|] < \infty$ for every $n \in \mathbb{Z}_-$, and if, for every $m, n \in \mathbb{Z}_-$ such that $n \leq m$,

$$X_n = \mathbb{E}[X_m|\mathcal{F}_n] \quad (\text{resp. } X_n \geq \mathbb{E}[X_m|\mathcal{F}_n], X_n \leq \mathbb{E}[X_m|\mathcal{F}_n]).$$

Theorem 2.3.13: Backward supermartingale convergence

Let $(X_n)_{n\in\mathbb{Z}_-}$ be a backward supermartingale. Suppose that

$$\sup_{n\in\mathbb{Z}_-} \mathbb{E}[|X_n|] < \infty.$$

Then the sequence $(X_n)_{n\in\mathbb{Z}_-}$ is uniformly integrable and converges a.s. and in L^1 to a random variable X_∞ as $n \to -\infty$. Moreover for every $n \in \mathbb{Z}_-$,

$$\mathbb{E}[X_n|\mathcal{F}_{-\infty}] \leq X_\infty,$$

and equality holds in the last display if $(X_n)_{n\in\mathbb{Z}_-}$ is a backward martingale.

2.3.4 Martingale Closure

Definition 2.3.14: Closed Martingale

A martingale $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is said to be *closed* if there exists a random variable $Z \in L^1$ such that, for every $n \in \mathbb{N}$ (resp. $t \geq 0$),

$$\begin{cases} X_n = E[Z \mid \mathcal{F}_n] & \text{For the discrete time} \\ X_t = E[Z \mid \mathcal{F}_t] & \text{For the continuous time} \end{cases}$$

Definition 2.3.15: Uniform integrability

A collection $(X_i)_{i\in I}$ of random variables in $L^1(\Omega, \mathcal{A}, \mathbb{P})$ is said to be *uniformly integrable* (u.i. in short) if

$$\lim_{a\to+\infty} \left(\sup_{i\in I} E[|X_i| \mathbb{1}_{\{|X_i|>a\}}] \right) = 0.$$

A uniformly integrable collection $(X_i)_{i\in I}$ is bounded in L^1. To verify this, take a large enough so that

$$\sup_{i\in I} E[|X_i| \mathbb{1}_{\{|X_i|>a\}}] \leq 1$$

and write $E[|X_i|] \leq E[|X_i| \mathbb{1}_{\{|X_i|\leq a\}}] + E[|X_i| \mathbb{1}_{\{|X_i|>a\}}] \leq a + 1$. The converse is false: a collection $(X_i)_{i\in I}$ which is bounded in L^1 may not be uniformly integrable.

Proposition 2.3.16

Let $(X_i)_{i\in I}$ be a bounded subset of L^1. The following two conditions are equivalent:

1. The collection $(X_i)_{i\in I}$ is u.i.
2. For every $\varepsilon > 0$, one can find $\delta > 0$ such that, for every event $A \in \mathcal{A}$ of probability $\mathbb{P}(A) < \delta$, one has

$$\forall i \in I, \quad \mathbb{E}[|X_i| \mathbb{1}_A] < \varepsilon.$$

Proof for Proposition.

(1)⇒(2) Let $\varepsilon > 0$. We can first fix $a > 0$ such that

$$\sup_{i\in I} \mathbb{E}[|X_i| \mathbb{1}_{\{|X_i|>a\}}] < \frac{\varepsilon}{2}.$$

If we set $\delta = \varepsilon/(2a)$, then the condition $\mathbb{P}(A) < \delta$ implies, that, for every $i \in I$,

$$\mathbb{E}[|X_i|\mathbb{1}_A] \leq \mathbb{E}[|X_i|\mathbb{1}_{A\cap\{|X_i|\leq a\}}] + \mathbb{E}[|X_i|\mathbb{1}_{\{|X_i|>a\}}] \leq a\mathbb{P}(A) + \frac{\varepsilon}{2} < \varepsilon.$$

(2)⇒(1) Set $C = \sup_{i\in I} \mathbb{E}[|X_i|]$. By the Markov inequality, we have for every $a > 0$,

$$\forall i \in I, \quad \mathbb{P}(|X_i| > a) \leq \frac{C}{a}.$$

Let $\varepsilon > 0$ and choose $\delta > 0$ so that the property stated in (2) holds. Then, if a is large enough so that $C/a < \delta$, we have $\mathbb{P}(|X_i| > a) < \delta$ for every $i \in I$, and thus, by (2),

$$\forall i \in I, \quad \mathbb{E}[|X_i|\mathbb{1}_{\{|X_i|>a\}}] < \varepsilon.$$

We conclude that the collection $(X_i)_{i\in I}$ is u.i.

Theorem 2.3.17: Vitali convergence theorem

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables in L^1 that converges to X_∞ in probability. The following two conditions are equivalent:

1. *The sequence $(X_n)_{n\in\mathbb{N}}$ converges to X_∞ in L^1.*
2. *The sequence $(X_n)_{n\in\mathbb{N}}$ is uniformly integrable.*

Proof for Theorem.

- (1)⇒(2) If the sequence $(X_n)_{n\in\mathbb{N}}$ converges in L^1, then it is bounded in L^1. Then, let $\varepsilon > 0$. We can choose N large enough so that, for every $n \geq N$,

  $$\mathbb{E}[|X_n - X_N|] < \frac{\varepsilon}{2}.$$

  Since the finite set $\{X_1, \ldots, X_N\}$ is u.i., Proposition 2.3.16 allows us to choose $\delta > 0$ such that, for every event A of probability $\mathbb{P}(A) < \delta$, we have

  $$\forall n \in \{1, \ldots, N\}, \quad \mathbb{E}[|X_n|\mathbb{1}_A] < \frac{\varepsilon}{2}.$$

  Then, if $n > N$, we have also

  $$\mathbb{E}[|X_n|\mathbb{1}_A] \leq \mathbb{E}[|X_N|\mathbb{1}_A] + \mathbb{E}[|X_n - X_N|] < \varepsilon.$$

  By Proposition 2.3.16, the sequence $(X_n)_{n\in\mathbb{N}}$ is uniformly integrable.
- (2)⇒(1) Using the characterization of uniform integrability in Proposition 2.3.16(ii), it is straightforward to verify that the collection $(X_n - X_m)_{n,m\in\mathbb{N}}$ is also u.i. Hence, if $\varepsilon > 0$ is fixed, we can choose $a > 0$ large enough so that, for every $m, n \in \mathbb{N}$,

  $$\mathbb{E}[|X_n - X_m|\mathbb{1}_{\{|X_n - X_m|>a\}}] < \varepsilon.$$

Then, for every $m, n \in \mathbb{N}$,

$$\begin{aligned}\mathbb{E}[|X_n - X_m|] &\leq \mathbb{E}[|X_n - X_m|\mathbb{1}_{\{|X_n - X_m| \leq \varepsilon\}}] + \mathbb{E}[|X_n - X_m|\mathbb{1}_{\{\varepsilon < |X_n - X_m| \leq a\}}] \\ &\quad + \mathbb{E}[|X_n - X_m|\mathbb{1}_{\{|X_n - X_m| > a\}}] \\ &\leq 2\varepsilon + a\,\mathbb{P}(|X_n - X_m| > \varepsilon).\end{aligned}$$

The convergence in probability of the sequence $(X_n)_{n\in\mathbb{N}}$ implies that

$$\mathbb{P}(|X_n - X_m| > \varepsilon) \leq \mathbb{P}\left(|X_n - X_\infty| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(|X_m - X_\infty| > \frac{\varepsilon}{2}\right) \xrightarrow[n,m\to\infty]{} 0.$$

It follows that

$$\limsup_{m,n\to\infty} \mathbb{E}[|X_n - X_m|] \leq 2\varepsilon$$

and, since ε was arbitrary, this shows that the sequence $(X_n)_{n\in\mathbb{N}}$ is Cauchy in L^1, hence converges in L^1.

■

Theorem 2.3.18: Doob's Uniform Integrability convergence theorem

Let X be a martingale (with right-continuous sample paths if time is continuous). Then the following properties are equivalent:

1. X is closed;
2. the collection $(X_n)_{n\in\mathbb{N}}$ ($(X_t)_{t\geq 0}$ if time is continuous) is uniformly integrable;
3. X_n (resp. X_t) converges a.s. and in L^1 as $n \to \infty$ (resp. $t \to \infty$).

Moreover, if these properties hold, we have $X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n]$ (resp. $X_t = E[X_\infty \mid \mathcal{F}_t]$) for every $n \in \mathbb{N}$ (resp. $t \geq 0$), where $X_\infty \in L^1$ is the a.s. limit of X_n (resp. X_t) as $n \to \infty$ (resp. $t \to \infty$).

Proof for Theorem.

- (1)⇒(2): if $Z \in L^1$, the collection of all random variable $\mathbb{E}[Z|\mathcal{G}]$ when $\mathcal{G}$ varies among all sub-σ-fields of $\mathcal{F}_\infty$ is u.i.
- (2)⇒(3): If (2) holds, then $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is bounded in L^1, and thus converges a.s. to some random variable X_∞ by Theorem **??**. By uniform integrability, the latter convergence also holds in L^1.
- (3)⇒(1): If (3) holds, for every $m \in \mathbb{N}$ (resp. $s \geq 0$), we can pass to the limit $n \to +\infty$ (resp. $t \to +\infty$) in the equality, , $X_m = \mathbb{E}(X_n \mid \mathcal{F}_m)$ (resp. $X_s = \mathbb{E}(X_t \mid \mathcal{F}_s)$) and using the theorem 1.5.4 we get $X_m = \mathbb{E}(X_\infty \mid \mathcal{F}_m)$ (resp. $X_s = \mathbb{E}(X_\infty \mid \mathcal{F}_s)$), which gives the desired result.

■

Corollary 2.3.19

Let $Z \in L^1(\Omega, \mathcal{A}, \mathbb{P})$. The martingale $X_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ (resp. $X_t = \mathbb{E}[Z \mid \mathcal{F}_t]$) converges a.s. and in L^1 to $X_\infty = \mathbb{E}[Z \mid \mathcal{F}_\infty]$.

2.4 Optional stopping theorems

2.4.1 The theorem for martingales

Theorem 2.4.1: Optional stopping theorem – martingale

Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a uniformly integrable martingale (with right-continuous sample paths if time is continnuous). Let S and T be two stopping times with $S \leq T$. Then X_S and X_T are in L^1 and

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof for Theorem.

We begins with the discrete time case. From the preceding section, we know that the martingale $(X_n)_{n\in\mathbb{N}}$ is closed and $X_n = \mathbb{E}[X_\infty|\mathcal{F}_n]$ for every n. Let us verify that $X_T \in L^1$:

$$\begin{aligned}
\mathbb{E}[|X_T|] &= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}|X_n|] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}|\mathbb{E}[X_\infty|\mathcal{F}_n]|] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&\leq \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}\mathbb{E}[|X_\infty||\mathcal{F}_n]] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[\mathbb{1}_{\{T=n\}}|X_\infty|] + \mathbb{E}[\mathbb{1}_{\{T=\infty\}}|X_\infty|] \\
&= \mathbb{E}[|X_\infty|] < \infty.
\end{aligned}$$

Then, if $A \in \mathcal{F}_T$,

$$\begin{aligned}
\mathbb{E}[\mathbb{1}_A X_T] &= \sum_{n\in\mathbb{N}\cup\{\infty\}} \mathbb{E}[\mathbb{1}_{A\cap\{T=n\}} X_T] \\
&= \sum_{n\in\mathbb{N}\cup\{\infty\}} \mathbb{E}[\mathbb{1}_{A\cap\{T=n\}} X_n] \\
&= \sum_{n\in\mathbb{N}\cup\{\infty\}} \mathbb{E}[\mathbb{1}_{A\cap\{T=n\}} X_\infty] \\
&= \mathbb{E}[\mathbb{1}_A X_\infty].
\end{aligned}$$

In the first equality, we use the fact that $X_T \in L^1$ to justify the interchange of sum and expectation via the Fubini theorem. In the third equality, we use the properties $X_n = \mathbb{E}[X_\infty|\mathcal{F}_n]$ and $A \cap \{T = n\} \in \mathcal{F}_n$ for $A \in \mathcal{F}_T$. Since X_T is $\mathcal{F}_T$-measurable and in L^1, the preceding display shows that X_T satisfies the characteristic property of the conditional expectation $\mathbb{E}[X_\infty|\mathcal{F}_T]$.

Finally, if S and T are two stopping times such that $S \leq T$, the property $\mathcal{F}_S \subset \mathcal{F}_T$ (Proposition 2.2.7) and Theorem of nested σ-fields in conditional expectation 1.5.8 imply that

$$X_S = \mathbb{E}[X_\infty|\mathcal{F}_S] = \mathbb{E}[\mathbb{E}[X_\infty|\mathcal{F}_T]|\mathcal{F}_S] = \mathbb{E}[X_T|\mathcal{F}_S].$$

For the continuous time case, Set, for every integer $n \geq 0$,

$$T_n = \frac{\lceil 2^n T \rceil}{2^n} \cdot \mathbb{1}_{\{T<\infty\}} + \infty \cdot \mathbb{1}_{\{T=\infty\}}, \quad n = 0, 1, 2, \ldots$$

and similarly

$$S_n = \frac{\lceil 2^n S \rceil}{2^n} \cdot \mathbb{1}_{\{S<\infty\}} + \infty \cdot \mathbb{1}_{\{S=\infty\}}, \quad n = 0, 1, 2, \ldots$$

By Proposition 2.2.11, (T_n) and (S_n) are two sequences of stopping times that decrease respectively to T and to S. Moreover, we have $S_n \leq T_n$ for every $n \geq 0$.

Now observe that, for every fixed n, $2^n S_n$ and $2^n T_n$ are stopping times of the discrete filtration $\mathcal{H}_k^{(n)} := \mathcal{F}_{k/2^n}$, and $Y_k^{(n)} := X_{k/2^n}$ is a discrete martingale with respect to this filtration. From the discrete case we proven before, we get that $Y^{(n)}_{2^n S_n}$ and $Y^{(n)}_{2^n T_n}$ are in L^1, and

$$X_{S_n} = Y^{(n)}_{2^n S_n} = E[Y^{(n)}_{2^n T_n} \mid \mathcal{H}^{(n)}_{2^n S_n}] = E[X_{T_n} \mid \mathcal{F}_{S_n}]$$

(here we need to verify that $\mathcal{H}^{(n)}_{2^n S_n} = \mathcal{F}_{S_n}$, but this is straightforward).

Let $A \in \mathcal{F}_S$. Since $\mathcal{F}_S \subset \mathcal{F}_{S_n}$, we have $A \in \mathcal{F}_{S_n}$ and thus

$$\mathbb{E}[\mathbb{1}_A X_{S_n}] = \mathbb{E}[\mathbb{1}_A X_{T_n}].$$

By the right-continuity of sample paths, we get a.s.

$$X_S = \lim_{n\to\infty} X_{S_n}, \quad X_T = \lim_{n\to\infty} X_{T_n}.$$

These limits also hold in L^1. Indeed, thanks again to the optional stopping theorem for uniformly integrable discrete martingales, we have $X_{S_n} = \mathbb{E}[X_\infty \mid \mathcal{F}_{S_n}]$ for every n, and thus the sequence (X_{S_n}) is uniformly integrable (and the same holds for the sequence (X_{T_n})).

The L^1-convergence implies that X_S and X_T belong to L^1, and also allows us to pass to the limit $n \to \infty$ in the equality $\mathbb{E}[\mathbb{1}_A X_{S_n}] = \mathbb{E}[\mathbb{1}_A X_{T_n}]$ in order to get

$$\mathbb{E}[\mathbb{1}_A X_S] = \mathbb{E}[\mathbb{1}_A X_T].$$

Since this holds for every $A \in \mathcal{F}_S$, and since the variable X_S is $\mathcal{F}_S$-measurable (by the

remarks before the theorem), we conclude that

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S],$$

which completes the proof. ■

In particular, for every stopping time S, we have

$$X_S = \mathbb{E}[X_\infty \mid \mathcal{F}_S],$$

and

$$\mathbb{E}[X_S] = \mathbb{E}[X_\infty] = \mathbb{E}[X_0].$$

Corollary 2.4.2

If T and S are stopping time such that $S \leq T$, and both S and T are bounded, then X_S and X_T are in L^1 and

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof for Corollary.

Let $a \geq 0$ such that $S \leq T \leq a$. We apply Optional Stopping Theorem 2.4.1 to the martingale $\{X_{n\wedge a}\}_{n\geq 0}$ (resp. $\{X_{t\wedge a}\}_{t\geq 0}$) which is closed by X_a.

Corollary 2.4.3

haha Let $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) be a martingale (with right-continuous sample paths if time is continuous), and let T be a stopping time.

1. The process $(X_{n\wedge T})_{n\in\mathbb{N}}$ (resp. $(X_{t\wedge T})_{t\geq 0}$) is still a martingale.
2. Suppose in addition that the martingale $(X_n)_{n\in\mathbb{N}}$ (resp. $(X_t)_{t\geq 0}$) is uniformly integrable. Then the process $(X_{n\wedge T})_{n\in\mathbb{N}}$ (resp. $(X_{t\wedge T})_{t\geq 0}$) is also a uniformly integrable martingale, and more precisely we have,

$$\begin{cases} X_{n\wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_{n\wedge T}], & n \in \mathbb{N} \quad \text{for the discrete case,} \\ X_{t\wedge T} = \mathbb{E}[X_T \mid \mathcal{F}_{t\wedge T}], & t \geq 0 \quad \text{for the continuous case.} \end{cases}$$

Proof for Corollary.

We will prove it for the continuous time case, the discrete time case can be proven by similar arguments and it is left exercice for the reader. We start with the proof of (2). Note that $t \wedge T$ is a stopping time by property (f) of stopping times. By Theorem 2.4.1, $X_{t\wedge T}$ and X_T are in L^1, and we also know that $X_{t\wedge T}$ is $\mathcal{F}_{t\wedge T}$-measurable, hence $\mathcal{F}_t$-measurable since $\mathcal{F}_{t\wedge T} \subset \mathcal{F}_t$. So in order to get formula in (2), it is enough to prove that, for every $A \in \mathcal{F}_t$,

$$E[\mathbb{1}_A X_T] = E[\mathbb{1}_A X_{t\wedge T}].$$

Let us fix $A \in \mathcal{F}_t$. First, we have trivially

$$E[\mathbb{1}_{A\cap\{T\leq t\}}X_T] = E[\mathbb{1}_{A\cap\{T\leq t\}}X_{t\wedge T}]. \tag{*}$$

On the other hand, by Theorem 2.4.1, we have

$$X_{t\wedge T} = E[X_T \mid \mathcal{F}_{t\wedge T}],$$

and we notice that we have both $A \cap \{T > t\} \in \mathcal{F}_t$ and $A \cap \{T > t\} \in \mathcal{F}_T$ (the latter as a straightforward consequence of the definition of $\mathcal{F}_T$), so that $A \cap \{T > t\} \in \mathcal{F}_t \cap \mathcal{F}_T = \mathcal{F}_{t\wedge T}$. Using the preceding display, we obtain

$$E[\mathbb{1}_{A\cap\{T>t\}}X_T] = E[\mathbb{1}_{A\cap\{T>t\}}X_{t\wedge T}].$$

By adding this equality to (*), we get the desired result.

To prove (1), we just need to apply (2) to the (uniformly integrable) martingale $(X_{t\wedge a})_{a\geq 0}$, for any choice of $a \geq 0$.

2.4.2 The theorem for supermartingales

Theorem 2.4.4: Optional stopping theorem - supermartingale

Let $(Z_n)_{n\in\mathbb{N}}$ (resp. $(Z_t)_{t\geq 0}$) be a nonnegative supermartingale (with right-continuous sample paths if time is continnuous). Let U and V be two stopping times such that $U \leq V$. Then Z_U and Z_V are in L^1, and

$$Z_U \geq \mathbb{E}[Z_V \mid \mathcal{F}_U].$$

Proof for Theorem.

Let's begin with the discrete time case. We first note that $(X_n)_{n\in\mathbb{N}}$ is bounded in L^1. Theorem 2.3.11 then shows that X_n converges a.s. to X_∞, so that the definition of X_T makes sense for any stopping time, even on the event $\{T = \infty\}$. If T is a bounded stopping time, we know that $\mathbb{E}[X_T] \leq \mathbb{E}[X_0]$ (Theorem 12.12). Fatou's lemma then shows that, for any stopping time T,

$$\mathbb{E}[X_T] \leq \liminf_{k\to\infty} \mathbb{E}[X_{T\wedge k}] \leq \mathbb{E}[X_0]$$

and thus $X_T \in L^1$.

Then let S and T be two stopping times such that $S \leq T$. Let us first assume that $T \leq N$ for some fixed integer N. According to Lemma 2.3.5, we have then $\mathbb{E}[X_S] \geq \mathbb{E}[X_T]$. More generally, let $A \in \mathcal{F}_S$, and set

$$S^A(\omega) = \begin{cases} S(\omega) & \text{if } \omega \in A, \\ N & \text{if } \omega \notin A. \end{cases}$$

It is straightforward to verify that S^A is a stopping time (just note that $\{S^A \leq n\} = \Omega$

if $n \geq N$, and $\{S^A \leq n\} = A \cap \{S \leq n\} \in \mathcal{F}_n$ if $n < N$, using the definition of $\mathcal{F}_S$). We define T^A in a similar manner, and note that T^A is also a stopping time since $A \in \mathcal{F}_S \subset \mathcal{F}_T$. Since $S^A \leq T^A$, we have $\mathbb{E}[X_{S^A}] \geq \mathbb{E}[X_{T^A}]$, and it follows that

$$\mathbb{E}[X_S \mathbb{1}_A] \geq \mathbb{E}[X_T \mathbb{1}_A].$$

Let us come back to the general case where S and T satisfy $S \leq T$ but are no longer assumed to be bounded. Let $A \in \mathcal{F}_S$, and $k \in \mathbb{N}$. Then $A \cap \{S \leq k\}$ belongs both to $\mathcal{F}_k$ (by the definition of $\mathcal{F}_S$) and to $\mathcal{F}_S$ (recall that S is $\mathcal{F}_S$-measurable). From Proposition 2.2.7, $A \cap \{S \leq k\}$ belongs to $\mathcal{F}_{S \wedge k}$. Hence, by applying the bounded case to the stopping times $S \wedge k$ and $T \wedge k$, we get

$$\mathbb{E}[X_S \mathbb{1}_{A \cap \{S \leq k\}}] = \mathbb{E}[X_{S \wedge k} \mathbb{1}_{A \cap \{S \leq k\}}] \geq \mathbb{E}[X_{T \wedge k} \mathbb{1}_{A \cap \{S \leq k\}}].$$

By monotone convergence,

$$\lim_{k \to \infty} \uparrow \mathbb{E}[X_S \mathbb{1}_{A \cap \{S \leq k\}}] = \mathbb{E}[X_S \mathbb{1}_{A \cap \{S < \infty\}}].$$

On the other hand, Fatou's lemma gives

$$\liminf_{k \to \infty} \mathbb{E}[X_{T \wedge k} \mathbb{1}_{A \cap \{S \leq k\}}] \geq \mathbb{E}[X_T \mathbb{1}_{A \cap \{S < \infty\}}].$$

So we have obtained that

$$\mathbb{E}[X_S \mathbb{1}_{A \cap \{S < \infty\}}] \geq \mathbb{E}[X_T \mathbb{1}_{A \cap \{S < \infty\}}].$$

Since $X_S = X_T = X_\infty$ on the event $\{S = \infty\}$, it follows that

$$\mathbb{E}[X_S \mathbb{1}_A] \geq \mathbb{E}[X_T \mathbb{1}_A] = \mathbb{E}[\mathbb{E}[X_T \mid \mathcal{F}_S] \mathbb{1}_A].$$

Since this holds for every $A \in \mathcal{F}_S$, and both X_S and $\mathbb{E}[X_T \mid \mathcal{F}_S]$ are $\mathcal{F}_S$-measurable, it follows that $X_S \geq \mathbb{E}[X_T \mid \mathcal{F}_S]$ a.s. (apply the last display with the set $A = \{X_S < \mathbb{E}[X_T \mid \mathcal{F}_S]\}$). This completes the proof of the theorem in the discrete case.

Now the continuous time case, Firstly, we make the extra assumption that U and V are bounded and we verify that we then have $E[Z_U] \geq E[Z_V]$. Let $p \geq 1$ be an integer such that $U \leq p$ and $V \leq p$. For every integer $n \geq 0$, set

$$U_n = \sum_{k=0}^{p2^n - 1} \frac{k+1}{2^n} \mathbb{1}_{\{k2^{-n} < U \leq (k+1)2^{-n}\}}, \quad V_n = \sum_{k=0}^{p2^n - 1} \frac{k+1}{2^n} \mathbb{1}_{\{k2^{-n} < V \leq (k+1)2^{-n}\}}$$

in such a way (by Proposition 2.2.11) that (U_n) and (V_n) are two sequences of bounded stopping times that decrease respectively to U and V, and additionally we have $U_n \leq V_n$ for every $n \geq 0$. The right-continuity of sample paths ensures that $Z_{U_n} \to Z_U$ and $Z_{V_n} \to Z_V$ a.s. as $n \to \infty$. Then, by the optional stopping theorem for discrete supermartingales in the case of bounded stopping times (which we just proved), with

respect to the filtration $(\mathcal{F}_{k/2^{n+1}})_{k\geq 0}$, we have for every $n \geq 0$,

$$Z_{U_{n+1}} \geq E[Z_{U_n} \mid \mathcal{F}_{U_{n+1}}].$$

Setting $Y_n = Z_{U_{-n}}$ and $\mathcal{H}_n = \mathcal{F}_{U_{-n}}$, for every integer $n \leq 0$, we get that the sequence $(Y_n)_{n\leq 0}$ is a backward supermartingale with respect to the filtration $(\mathcal{H}_n)_{n\leq 0}$. Since, for every $n \geq 0$, $E[Z_{U_n}] \leq E[Z_0]$ (by another application of the discrete optional stopping theorem), the sequence $(Y_n)_{n\leq 0}$ is bounded in L^1, and by the convergence theorem for backward supermartingales in Theorem 2.3.13, it converges in L^1. Hence the convergence of Z_{U_n} to Z_U also holds in L^1 and similarly the convergence of Z_{V_n} to Z_V holds in L^1. Since $U_n \leq V_n$, by yet another application of the discrete optional stopping theorem, we have $E[Z_{U_n}] \geq E[Z_{V_n}]$. Using the L^1-convergence of Z_{U_n} and Z_{V_n} we can pass to the limit $n \to \infty$ and obtain that $E[Z_U] \geq E[Z_V]$ as claimed.

Let us prove the statement of the theorem (no longer assuming that U and V are bounded). By the first step of the proof applied to the stopping times 0 and $U \wedge p$, we have $E[Z_{U\wedge p}] \leq E[Z_0]$ for every $p \geq 1$, and Fatou's lemma gives $E[Z_U] \leq E[Z_0] < \infty$ and similarly $E[Z_V] < \infty$. Fix $A \in \mathcal{F}_U \subset \mathcal{F}_V$ and recall our notation U^A for the stopping time defined by $U^A(\omega) = U(\omega)$ if $\omega \in A$ and $U^A(\omega) = \infty$ otherwise (cf. property (d) of stopping times). By the first part of the proof, we have, for every $p \geq 1$,

$$E[Z_{U^A\wedge p}] \geq E[Z_{V^A\wedge p}].$$

By writing each of these two expectations as a sum of expectations over the sets A^c, $A \cap \{U \leq p\}$ and $A \cap \{U > p\}$, and noting that $U > p$ implies $V > p$, we get

$$E[Z_U \mathbb{1}_{A\cap\{U\leq p\}}] \geq E[Z_V \mathbb{1}_{A\cap\{U\leq p\}}].$$

Letting $p \to \infty$ then gives

$$E[Z_U \mathbb{1}_{A\cap\{U<\infty\}}] \geq E[Z_V \mathbb{1}_{A\cap\{U<\infty\}}].$$

On the other hand, the equality

$$E[Z_U \mathbb{1}_{A\cap\{U=\infty\}}] = E[Z_V \mathbb{1}_{A\cap\{U=\infty\}}]$$

is trivial and by adding it to the preceding display, we obtain

$$E[Z_U \mathbb{1}_A] \geq E[Z_V \mathbb{1}_A] = E[E[Z_V \mid \mathcal{F}_U]\mathbb{1}_A].$$

Since this holds for every $A \in \mathcal{F}_U$ and Z_U is $\mathcal{F}_U$-measurable, the desired result follows. ∎

Remark This implies that $\mathbb{E}[Z_U] \geq \mathbb{E}[Z_V]$, and since $Z_U = Z_V = Z_\infty$ on the event $\{U = \infty\}$, it also follows that

$$\mathbb{E}[\mathbb{1}_{\{U<\infty\}} Z_U] \geq \mathbb{E}[\mathbb{1}_{\{U<\infty\}} Z_V] \geq \mathbb{E}[\mathbb{1}_{\{V<\infty\}} Z_V].$$

2.4.3 Some applications of the optional stopping theorem

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